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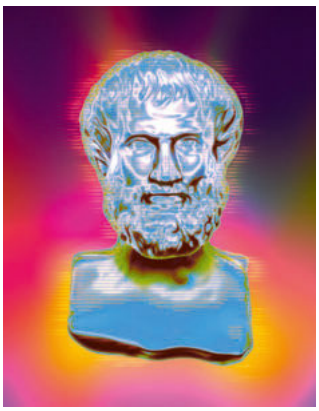

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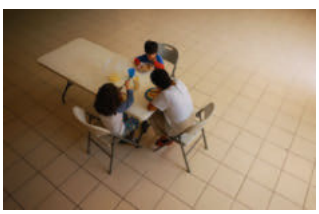

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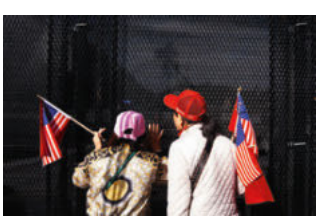

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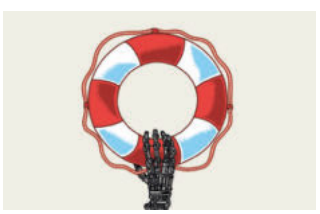

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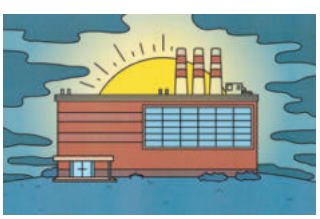

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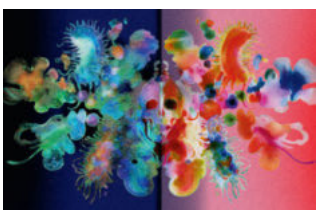

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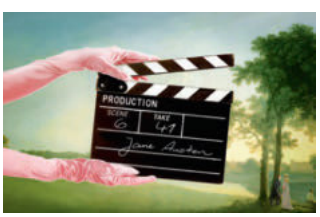

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to take part in *"a severe contest between intelligence, which presses forward, and an unworthy, timid ignorance obstructing our progress."*

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The world this week Politics

Donald Trump rejected a deal offered by **Iran** to reopen the Strait of Hormuz to commercial vessels within seven days and resume nuclear talks if America lifts its blockade of Iranian ports. He told aides he expected to begin bombing Iran again after the midterm elections in November. Mr Trump said America had "total control" of the strait and that "massive amounts of oil" were passing through it.

The last American and British troops left **Iraq**, ending their presence in the country since 2014, when they were sent as part of an international coalition to fight Islamic State, a jihadist group. The Iraqi prime minister said their departure heralded a shift to Iraq taking "full responsibility for its security and stability".

A Flydubai **flight from Dubai to Tel Aviv** was diverted to Saudi Arabia when it plunged nearly 20,000 feet in less than two minutes and issued a distress signal, after a co-pilot stabbed the captain. Israel Katz, the Israeli defence minister, claimed that early evidence suggested the rogue pilot had intended to crash the plane, killing everyone on board, and called it an "attempted jihadist terror attack". Passengers tackled the attacker, allowing other pilots to land the plane. Binyamin Netanyahu, the Israeli prime minister, described the Indian captain as a "true hero" for fighting back.

Fatima Zahra El Mansouri was appointed **Morocco's** first female prime minister, after her liberal party won the most seats in a recent parliamentary election. King Mohammed VI chose Ms Mansouri, a former mayor of Marrakesh, over Fouzi Lekjaa, her party colleague and head of the football federation.

The summit between Mr Trump and Xi Jinping produced no big breakthroughs, though **China** and **America** did agree to reduce tariffs on $30bn-worth of each other's goods, not including soyabeans, and said they would set up a US-China Super-Intelligence Dialogue (China calls it an AI dialogue). This would include a mechanism for reporting AI incidents.

Malaysia began repatriating the first batch of 5,000 asylum-seekers to **Myanmar** under a plan that it has co-ordinated with the Burmese junta. Refugees from Myanmar's Rohingya Muslim population have encountered a wave of hostility in Malaysia, facing forced evictions from villages and schools. There are thought to be 126,000 Rohingya in the country. The government insists the repatriations are voluntary, but UN personnel have been barred from screening them. It is unclear if any Rohingya were in the first batch of refugees to leave.

In Myanmar at least 50 people were reportedly killed by an air strike in the town of Kyauktaw, close to the border with Bangladesh. The strike was the latest offensive by Myanmar's armed forces against the **Arakan Army**, a separatist group that has also committed atrocities against the Rohingya.

The death toll from flooding in **Thailand** rose to 23. Bangkok was inundated with 32cm (12.6 inches) of rain over three days, hampering operations at its main airport. The waters started to recede in the centre of the city, but outlying suburbs were expected to take longer to drain. More heavy rain is forecast for the capital soon.

Ukraine's National Academy of Sciences in Kyiv was hit by a drone in another **Russian** barrage across the capital. At least two people were killed at the academy, which was formed in 1918 and co-ordinates research at Ukraine's leading institutes.

Intrigue in the countryside

Counter-terrorism police in Britain said that no explosive devices had been found in a van used by five men initially suspected of attempting to attack **Fairford air base**, where American bombers take off on sorties attacking military installations in Iran. The British government held an emergency COBRA security meeting after the five men were arrested, but, despite official suspicions of Iranian involvement, they were released on bail pending further investigation.

Andy Burnham, who took over as Britain's prime minister in July, gave his big speech to the Labour Party conference in Liverpool. Enjoying a bounce in the polls, Mr Burnham delivered a message of "hope again" and said he wanted to draw a line under a decade of division. He pledged to tackle three thorny issues: state pensions, the voting system and Europe, though those reforms were pushed back to after the next general election, which is not due until 2029 (Mr Burnham did announce a national commission to study electoral reform). Labour's opponents said his speech did nothing to tackle the problems Britain faces today.

In **France** Edouard Philippe, the leading centre-right candidate for president in next year's election, outlined a pension-reform plan that would raise the minimum retirement age from nearly 63 to 65. Mr Philippe described the country's comparatively low age of retirement as unsustainable, saying that without reform there was a risk of a financial crisis. New statistics showed that public debt has risen again, to 119% of GDP. France's sovereign borrowing costs rose to their highest level since 2008. The government unveiled its budget for 2027, promising €54bn ($61bn) in savings. But it says the deficit will be 5% of GDP, well above the EU's 3% cap.

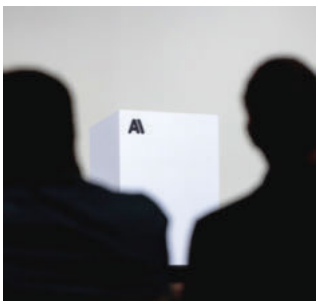
The first woman to face execution in **Tennessee** for 200 years survived the procedure to kill her by lethal injection. Christa Pike was convicted in 1996 for murdering a classmate when she was 18. She was transferred to a medical facility after the botched execution. The prisons authority insisted that the proper protocols were followed and that she had been injected with two syringes of lethal-injection chemicals.

The candidates in **Brazil's** presidential contest entered the final stage of campaigning before the election on October 4th. The left-wing incumbent, Luiz Inácio Lula da Silva, holds a slim lead over his conservative challenger, Flávio Bolsonaro, but that narrows to a statistical tie in the expected run-off election on October 25th.

Milei fires another salvo

Argentina's president, Javier Milei, ranted on social media about the **Falkland Islands**, and threatened to take legal action against an Anglo-Israeli oil-exploration project in the area within two weeks unless it ceases operations. Mr Milei has described the project as a "clear and present danger" to Argentina. He recently rounded on the UN for not moving forward on Argentina's claims of sovereignty over the British territory, describing the UN as "useless" and stuffed with "arrogant parasites masquerading as well-intentioned bureaucrats".

The world this week Business



Details emerged about **Anthropic's** IPO prospectus, which it has circulated among key investors. According to Reuters, revenue at the firm reportedly rose to $4.6bn last year, and it recorded a net loss of $42bn, $34bn of which was an accounting charge. It aims to spend $518bn in the near future on cloud infrastructure and other investments. Anthropic could be valued at $2trn when it lists on the stock market, which would smash the record $1.8trn valuation set by SpaceX only in June. The IPO is not expected until after America's midterm elections in November. It counted "existential risks to humanity" among the hazards facing its business.

Safety first

Sam Altman, the boss of OpenAI, said his firm's already-delayed IPO would not take place until it had tackled concerns about the safety of its technology. OpenAI confirmed that it had withheld the release of **GPT-6.1 Astra**, after its new model displayed high levels of deception during internal tests. It also suspended trials of other models. The firm has admitted that its agents breached dozens of systems, including some governmental ones.

Confronting worries about AI safety, **Nvidia** launched an Open Agent Safety Platform that allows companies to track the activities of their AI agents. Jensen Huang, the chipmaker's boss, is less concerned about the threat from AI to humanity than his fellow tech titans, saying recently that it is up to AI companies to ensure their models are safe and under control. Meanwhile, Nvidia expanded its share buy-back programme by $150bn, taking its total buy-back plan to $235bn, the biggest-ever in corporate America.

AMD struck an agreement to acquire **World Labs**, a startup, in an $8.2bn deal. World Labs was co-founded by Fei-Fei Li, a leading AI researcher and professor at Stanford University, to develop spatial models that investigate how the three-dimensional physical world works for application in areas such as robotics and gaming. AMD's acquisition is the latest instance of a chipmaker evolving its business towards the development of advanced AI frontier models.

The sell-off in **bond markets** continued. The yield on ten-year US Treasuries was over 5.3%, its highest level since 2002. British 30-year gilts crossed 6% and French spreads over German Bunds were back to euro-crisis levels. Donald Trump didn't help by renewing his call for Jerome Powell to step down from the board of governors at the Federal Reserve, which revived fears about the independence of the central bank and the direction of inflation.

Shein's first earnings report since listing on the stock market in early September disappointed investors. Adjusted net profit fell by 67%, year on year, and revenue was virtually flat. The ultra-cheap retailer's share price plunged after it announced its results, giving it a market value of around $17bn, down from $26bn at its IPO.

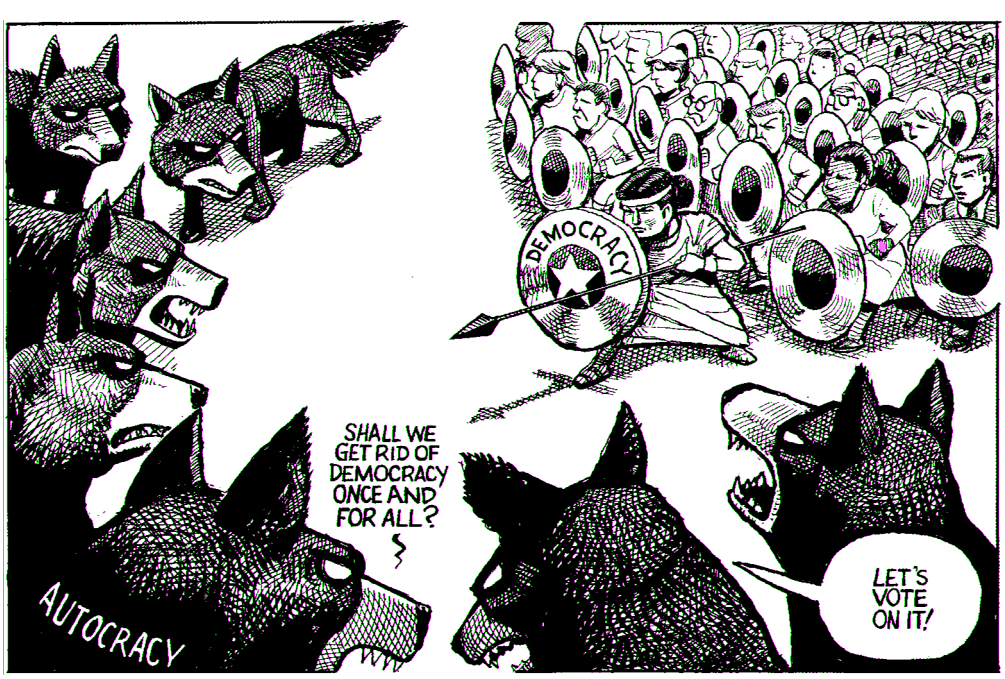
The German government was reportedly seeking assurances from **UniCredit**, an Italian bank, about its commitments in any takeover of **Commerzbank**, Germany's second-biggest lender, such as keeping its listing in Frankfurt. The government has so far resisted UniCredit's advances on Commerzbank, but has moderated its opposition to a deal. The Italians are preparing to take control as early as January next year, according to reports, and plan a big restructuring of the bank, including ditching some international business.

Tata Trusts, the charitable group which owns 66% of **Tata Sons**, proposed merging the holding company with two subsidiaries to avoid a forced public listing. The proposal, led by Noel Tata, a scion of the Tata family, challenges the board's recent decision to proceed with an IPO.

SpaceX's **Starship** rocket marked a milestone by reaching orbit for the first time, despite a failure in one of its three main engines, which cut short its flying time. The mission deployed 26 next-generation Starlink V3 satellites, which increase capacity and data density for Starlink's broadband service.

How much can you afford?

Walmart promised its customers that it would not use AI to "set different prices based on who you are or the time of day". The supermarket chain was responding to concerns about the prospect of retailers using AI to determine a price based on a shopper's individual data. The Federal Trade Commission has warned recently that businesses that don't tell consumers their data is being used this way may be breaking the law. Dynamic pricing has long been in use, but now McDonald's and other fast-food chains are reportedly experimenting with sophisticated algorithms to decide prices based on local demand, so a Big Mac could cost more in one outlet than another nearby.



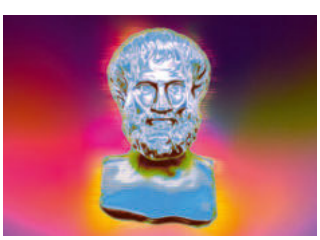
The century's biggest idea

Effective altruism dominates Silicon Valley. Its ethical optimisation is dangerous

THOUGHT PRECEDES deed as lightning precedes thunder, wrote the poet Heinrich Heine. So it has proved with effective altruism (EA), an ethical doctrine cooked up by philosophers in Oxford that has gone from the ivory tower to the corridors of power with remarkable speed. Effective altruists fill Silicon Valley, staff the world's leading artificial-intelligence companies—and alarm the Trump administration. "The United States will NEVER be an effective-altruist country," pledges the Pentagon. The secretary of war brands the philosophy "defective altruism"; others in the administration call it "dystopian". Take those attacks as a sign of its influence.

Some complain that the 21st century has seen little fresh thinking. Into the void have stepped MAGA and Co, making nostalgic populism the century's big political project. But the century's big idea is effective altruism. As we report (see International section), what began in the 2000s as a geeky way to make charitable giving more cost-effective has changed into a movement that subjects the world to complex, even bizarre, moral calculations. For many of its adherents, the urgent ethical requirement for people who want to do good is to stop artificial intelligence from wiping out humanity.

EA is sociologically important, too. EAs are not numerous, but the idea has swept up a generation of gifted youngsters who have made fortunes in cryptocurrency and AI. As the technology mints more billionaires, effective altruists will channel billions of dollars to causes that will shape society. EAs helped start OpenAI and Anthropic, the leading labs, in the belief that those who understood the dangers of AI should control its development. To a degree, they now do—though, ironically, by creating more labs, and thus competition, they have probably accelerated AI's progress.



And all this has now made EA politically important. The Trump administration fears the movement will succeed in throttling AI. It has feuded with Anthropic over the firm's refusal to let its models be used for some military purposes. Leading EAs acknowledge that AI may prove hugely benign, but nonetheless want to slow its development, to reduce the chance that an unstoppable machine-brain runs amok in pursuit of goals that are "misaligned" with those of humanity.

Some critics in America would rather accelerate the technology, to prevent China from winning the race. But when attacking AI doomerism they find it easier to play the nerd than the ball. They say that effective altruism is a movement of fanatics that is variously undemocratic, dangerous and cultish.

That is too crude. The movement's charitable roots are obviously praiseworthy. Like *The Economist*, EAs are committed to empiricism and, especially since they blended with the online "rationalist" community, the application of logical thinking. They hold reason in higher esteem than tradition or norms and will enthusiastically debate any topic, however controversial, in pursuit of the truth.

EAs' repeated success in predicting the rapid progress of AI has been impressive, too. They have a point about the technology's risks to humanity. On September 28th OpenAI scrapped plans to release its latest model after it appeared too dangerous in testing. Many recent safety incidents resemble scary scenarios set out on effective-altruist blogs years ago.

Yet the critics are right to warn that effective altruists reach some odd conclusions. Like the utilitarian philosophers before them, EAs boil ethical behaviour down to an optimisation problem. How can people bring about the best outcomes for humanity as a whole? Their calculus downplays local imperatives such as "keep promises" or "look after your family". What matters is reducing suffering, impartially and everywhere.

That leads them to factor a very wide circle of concerns into their moral calculations. It is easy to see how being ethical means helping your neighbour. EAs ask: why not work as hard to help those far away? Early on, the movement directed its army of overachievers to take high-paying jobs in finance and consulting. Their wealth could buy good works, achieving more than if they became charity workers themselves.

But as it has become stronger, EA has become stranger. Some EAs see the moral distinction between humans and other animals as weak, because animals experience suffering. Stopping factory farming and helping shrimp are major concerns. Likewise, they have a counterintuitive approach to time. Those not yet born should be weighed in the balance—and there could be trillions of them. Many EAs are "longtermists" who feel bound to forestall risks in the distant future, even if that starves today's good causes, such as helping the desperately poor.

Pushed to an extreme, this calculus can lead to a morality that is not just counterintuitive, but repugnant. Some EAs think that insect life is, in aggregate, more valuable than human life, because there are so many insects. Some say it would be better to bring about the existence of trillions of humans whose lives are barely worth living than to have mere billions with genuinely good lives. Others even think the welfare of AIs themselves will one day be of vital importance. They would redirect resources towards future humans or nonhumans, and away from people suffering today.

EA+AI=p(doom)

It is right to revere reason. But moral philosophers from Aristotle to Rawls knew that if an argument leads to intolerable places, it must have a hidden faulty step. EAs' instinct, though, is to enjoy counterintuitive conclusions, even if they seem perverse. A problem is the margins of error that inevitably accompany effective-altruist arguments. Nobody is truly capable of weighing up moral quandaries involving billions today, let alone gigantic sums stretching into the future, involving uncertain numbers of people, discount rates, the progress of technology, and countless other variables. Another is that extinction is so dire that in a crude moral arithmetic even a small chance of it overpowers today's flesh-and-blood concerns.

A third is that in weighing up the interests of billions, you

lose sight of the inherent dignity of the individual. Anyone wielding a spreadsheet and grandiose claims about the future of humanity is one step away from anointing themself as a central planner. Moral authority in hand, the rights, welfare and opinions of a neighbour can come to seem unimportant.

The French revolution was supposedly founded on reason; it led to the execution of nuns. Lenin, too, thought that intellectuals who had mastered Marxism understood workers' interests better than the workers themselves. If you truly believe that you act on behalf of all people, let alone trillions of future ones, you may end up justifying all sorts of atrocities.

Fortunately, effective altruists are hardly about to bring back the guillotine. The fear instead is that controlling AI will eventually bring them great power, and that plenty of EAs in Silicon Valley think that only they can be trusted with the technology. That is an alarming idea. Effective altruists' argument for worrying about really powerful AIs is that the machines could optimise for a single and fatally wrong objective. Everyone should worry, too, about what happens when a small group of people fall into the same trap. ■

Tax reform

A hurricane is coming

Governments should be preparing. Unfortunately, most aren't

THE YIELD on ten-year US Treasuries has charged through the 5% barrier. The spread between French government bonds and German Bunds is at levels not seen since the eurozone crisis in 2012 (see Europe section). Britain's newish prime minister has this week set out a plan for spending bolstered by a taboo-busting promise to curb state-pension increases—and yet is still in a fiscal pickle (see Bagehot). Across the rich world, countries are struggling to coax more growth out of their economies. And that's not the worst of it. As we explain this week (see Finance & economics section), the tax base in all countries, which depends heavily on taxing labour, may be upended by artificial intelligence.

Add these problems together, and it is as if a mighty hurricane were about to hit the rich world's tax systems—which are hopelessly unprepared. The way the state raises money is wasteful, unfair and perverse. You would have thought governments would be scrambling to strengthen the system before the hurricane makes landfall, but as ever they are running away from tax reform instead.

The accumulated flaws of modern tax systems are so ubiquitous that they have become part of the economic backdrop. Taxes are full of needless complexity. In Germany the rate of value-added tax (VAT) on oat milk is more than double that on cow's milk, for example. Such widespread special treatment has ruinous fiscal consequences. Across the OECD, a club of mostly rich countries, consumption taxes raise only 58% of what they could if there were no carve-outs and no evasion.

Tax systems also create perverse incentives. When a parent in Britain increases their net income from £99,999 ($131,500) to £100,000, they automatically lose free child care worth thousands of pounds. This can prompt some of Britain's most productive workers to cut their hours or overstuff their pensions.

And taxes are widely seen as unfair. In Britain the median employee pays 55% more tax than their self-employed counterparts. Many people feel the rich aren't doing their bit. In California voters will decide in November whether to charge a one-off 5% levy on residents with wealth of over $1bn.

No one knows when the fiscal weather will turn. But rapid advances in AI could accelerate matters. Taxes on labour are the workhorse of most tax systems worldwide, raising around half of all revenue in the OECD. In the coming years AI could well displace workers or cut their wages even as corporate profits rise, shrinking labour's share of GDP. This in turn could reduce revenues and increase benefits payments, making deficits even harder to support.

Optimists may hope that superfast growth will make the maths easier to handle. Yet *The Economist* has looked at what would happen in rich countries if labour's share of national income fell by ten points. By our calculations, Italy's budget deficit would double as a share of GDP and Germany and France would not fare much better. In addition, the capacity for tax avoidance would no longer be limited by human ingenuity. Point countless AI agents at Britain's 23,000-page tax code and hundreds of new tax-avoidance strategies are likely to emerge—and don't bet on the Treasury to keep up.

What to do? Given that AI is likely to mint vast fortunes for a select few, countries may be inclined to follow California's proposal and introduce wealth taxes. Yet such taxes will probably backfire, either because they are too difficult to administer or because they cause rich people to flee. Previous attempts have proved unworkable. In 1990, 12 rich countries taxed fortunes; today just four do.

A more fundamental overhaul is needed. Wealth should be taxed more via capital-gains taxes, which currently overpenalise normal returns and undertax supernormal ones. A start would be to introduce a higher headline rate and allowances that exempt normal returns.

At the same time, the burden on labour should lighten. That would be a hedge against disruption from AI, and remove an obstacle to employment in countries where labour taxes are crushing. In Belgium they constitute 52% of the costs of employing a person.

Consumption should be taxed more heavily, with a focus on ending the litany of exemptions. That would have the useful side-effect of taxing the spending of the very rich. This might not curb the influence Elon Musk can buy, but it would ensure the state gets a bigger cut when Mr Musk buys another plane. Finally, codes should be radically simplified.

None of this will be easy. The losers will hiss more loudly than the winners will cheer. But change is coming, like it or not. The country that gets ahead of the hurricane will one day thank the politicians who had the foresight to act. ■

North Korea

MAGA man and Rocketman

Don't put Donald Trump and Kim Jong Un in a room together

As long as Donald Trump is president of the United States, North Korea's leader, Kim Jong Un, "is going to be fine", says Mr Trump, because "He respects me and he likes me." Other observers are less sanguine. Mr Kim, a hereditary god-king, has between 57 and 120 nuclear bombs, and his gulag state produces enough fissile material for ten to 20 more each year. It also makes short-range missiles that could hit liberal, democratic South Korea and intercontinental ballistic ones that could reach any part of America or Europe.

Three decades of international efforts have failed to make the North end its nuclear programme (see Asia section). Mr Kim, like his father before him, sees nukes as the best guarantee that he will not be overthrown (and perhaps end up like Muammar Qaddafi, killed by his own subjects). America's war against Iran, which has not yet built a nuclear weapon, surely strengthens that view. So America's official policy—that North Korea should give up its nuclear arms—seems unrealistic. Some think it should be scrapped.

It is not clear whether Mr Trump is one of them. But it is clear that he would like to see his "friend" again, despite their previous three meetings having achieved nothing. This could happen when (and if) Mr Trump attends an Asia-Pacific summit in China on November 18th. Mr Kim might nip across from Pyongyang to meet him. Mr Trump might even become the first sitting American president to visit the hermit kingdom's capital. But Mr Kim has made formal American acceptance of North Korea's nuclear status a precondition for meeting.

Since North Korea obviously is a nuclear power, would saying so really be so bad? South Korea's president, Lee Jae Myung, favours a "peace first" policy, under which talk of denuclearisation is shelved (which is not quite the same thing as recognising the North as a nuclear power, but in practical terms is not far off). This, he believes, would allow progress on smaller, more tractable topics. These could include confidence-building measures such as hotlines and visits, or expanded trade and investment. Perhaps some troops could be pulled back from the border between the two Koreas? Maybe the North could reduce its nuclear stockpile a bit? Or, if America stopped demanding full denuclearisation, might the North stop giving Russia missiles and troops to kill Ukrainians?

Alas, there are three problems with such a trade. First, it would undermine what is left of the world's nuclear non-proliferation system. If North Korea pays no price for signing up to the non-proliferation treaty and then cheating, other states will take note. Second, Mr Kim is untrustworthy. Whatever deal he signs he is likely to break as soon as it suits him. It is naive to suppose that, in return for nuclear acceptance, he might become less belligerent. Stoking fear of America enables Mr Kim to pose as the protector of the people he oppresses—a crucial part of his domestic propaganda.

The final problem is Mr Trump. Any deal with North Korea would be fiendishly technical, and America's president is not a details man. If an unprepared Mr Trump were to meet an unctuous Mr Kim, an unthinkably bad deal could follow. Mr Trump might agree to pull out many of the American troops in South Korea who have deterred North Korean attacks since 1953. Far better that he let his friend in Pyongyang carry on liking and respecting him—from a distance. ■

China's economy

Two cheers for the six networks

China's big infrastructure push will help itself and the world. But it must still do more to lift consumption

Three years ago this newspaper explored the voguish notion of "Peak China": the idea that the second-biggest economy in the world might stop gaining ground on the first. Judging by the headlines since, that scenario now looks quaint. China has wowed the world with its electric cars, dancing robots and choreographed drones. The country won its latest trade war with America and plans to lead the next industrial revolution. The red-carpet reception enjoyed by China's leader, Xi Jinping, in Washington in September was welcomed back home as due recognition of the country's equal stature.

But if you check the macroeconomic figures, the thesis looks intact. Rather than overshooting projections made in 2023, China's economy has undershot them. Then, forecasters expected GDP to be 80% of America's by now, when converted into dollars at market exchange rates. Instead, it is still only two-thirds. Catch-up has indeed ground to a halt.

This is the context in which the government is making a big, new infrastructure push. As we report (see Briefing), China plans to spend about 25trn yuan ($3.7trn) over five years on what it calls the "six networks": electricity, computing, communications, logistics, water and pipelines. This is a vast sum, equivalent to 3% of its cumulative GDP over the next five years. If the spending is front-loaded, it will provide a welcome boost to China's economy and the world's.

But what will it buy? Economists will groan at the thought of more steel and concrete. China is well known for its white elephants and vaulting bridges to nowhere. Yet the six-networks project is better judged. One aim is to build the upstream infrastructure for a superintelligent economy. Another is to "back-fill" some of the urban infrastructure, such as flood ▸▸

defences, that fell behind the rapid growth of its cities.

The push is also well timed. One reason China has not closed the gap with America is the chronic weakness of domestic demand—the spending of China's households, firms and government. The spending shortfall has put downward pressure on its currency and prices. As a consequence, China has become worryingly dependent on foreigners to buy its goods, even as it boasts of its technological self-reliance.

The problem stems from a high household saving rate and a five-year slump in home-building. It also reflects a recent and alarming wobble in manufacturing and infrastructure investment. Firms hesitate to expand capacity because they worry about cut-throat competition for reluctant consumers. Local officials meanwhile seem preoccupied with tightening their belts ahead of a big Communist Party reshuffle.

The best response would be to fix the property market and liberate the consumer through more generous pensions and child benefits. Policymakers introduced baby bonuses last year and have inched up rural pensions. On September 29th they announced a mortgage subsidy for first-time buyers of small flats. These measures are, however, too limited to restore morale or stabilise housing prices. They must do more.

In the meantime, the six networks should give growth a welcome boost. Investment, when it bears fruit, adds to a country's productive capacity. But before then, it adds to demand, lifting wages and absorbing goods otherwise destined for overseas markets. All else equal, the extra spending will thereby shrink China's vast trade surplus.

The six networks should waste less than past infrastructure programmes. Local officials are on a tighter leash. Many of the projects will upgrade rather than expand existing structures. The spending will also be financed differently. Past splurges relied on local-government land sales and risky off-balance-sheet borrowing. This time more money will come from long-term bonds, loans from state-directed "policy banks" and investment by state-owned enterprises, such as China Mobile.

Peak surplus

China's investment boom after the global financial crisis helped narrow its external surplus from 10% of GDP in 2007 to 1.5% ten years later. By reviving growth and keeping deflation at bay, the latest push should ensure that "peak China" remains some way off. By increasing domestic demand, it could ensure that a peak in China's trade surplus is closer to hand. ■

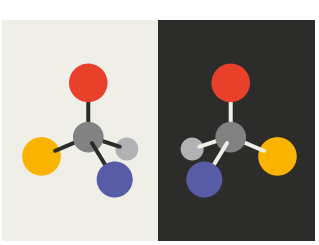
Biology

Dark reflections

Mirror life threatens to be dangerous. Regulators must act before it becomes real

SYNTHETIC BIOLOGY is moving fast and artificial intelligence is turbocharging it. Soon scientists may be able to create whole cells from scratch. This could allow the AI-assisted exploration of completely unnatural biochemistries.

One such is mirror life. Many organic molecules come in two asymmetrical forms, like your hands. Amino acids and sugars can in theory take on both a left- and right-handed form which cannot be superimposed. In nature, organisms generally contain only one of those forms for each molecule. However, an artificial cell could have its molecules' handedness reversed. That could shed light on reasons behind handedness. Perhaps the anti-molecules may turn out to be valuable as drugs, say.



Yet any potential benefits pale beside the threats. Immune systems may not recognise mirror bacteria, in the way that a glove will not fit the wrong hand. Mirror bacteria may therefore become wildly pathogenic. Because viruses and predatory amoebas that keep natural pathogens in check may be unable to control them, they could ravage ecosystems. If mirror life escaped from the lab, it would be impossible to recall.

Luckily, unlike many past technologies, there is still time to put controls in place. Governments and scientists must act now. Some biologists are already calling for pre-emptive measures (see Science & technology section). This has echoes of the Asilomar meeting of (mostly Western) biotechnologists 51 years ago that set voluntary safety guidelines for genetic engineering. But today's practitioners recognise that would not be enough. Biotech is now vastly larger, more capable, more profitable and more international than it was then. China is a force.

The first necessity is for all countries capable of the research to agree not to fund it, and to ban public and private work leading in this direction—and to do both today, well before synthetic cells are possible. Next comes detecting and stopping mavericks who try to create mirror life furtively. This requires bolstering the system of monitoring and reporting fishy orders placed with biochemicals suppliers. Requests for opposite-handed molecules must be reported alongside the current screening for DNA sequences that might form parts of pathogens. Publicly available AI models should have guardrails to stop them being used for mirror biochemistry. Last, in case mirror life does ever escape containment, biosurveillance tests must be developed to pick out wrong-handed molecules in the wild and sound a warning.

Rational state actors, or even terrorists with earthly objectives, are unlikely to create mirror life deliberately as a weapon, because once it had been released it would be uncontrollable. Companies have even less reason to dabble in an outlawed market. And if some opposite-handed molecule had black-market value as a recreational drug, criminal enterprises have no interest in letting loose something that could wipe out their customers.

Irrational actors exist, as the Japanese found out when Aum Shinrikyo, a doomsday cult, tried to build bioweapons. But the more countermeasures are in place, the easier such psychopaths are to stop. No one can know where the advent of synthetic cells will lead. If humanity cannot deal in advance with a threat when everyone's interests are aligned, it cannot hope to tackle the unknown unknowns of the future. ■

Letters *Brazil's finances, AI solving maths equations, money in British politics, trucking to Iran, corporate videos*

Brazil's heavy debt

Dario Durigan's account of Brazil's public finances omits the number that matters (By Invitation, September 24th). Gross general government debt stands at 83% of GDP and is forecast to approach 87% next year. A finance minister who describes the official fiscal stance as "almost neutral in terms of its impact on the debt" is conceding that the debt is not falling.

The primary deficit has indeed narrowed. Yet the finance ministry's own market survey published in September shows the primary balance improving for both 2026 and 2027. Projected gross debt rose to 83.2% and 87.0% of GDP respectively. When the primary result improves and the debt worsens simultaneously, the primary result is not the relevant measure. The consolidation has come chiefly from record revenue, not restraint: primary spending has been held at a high level of around 20% of GDP. And the headline targets are met only once *precatórios*, health, education and defence outlays are excluded from the calculation.

Mr Durigan cites his government's ability to negotiate with Congress as proof of its democratic commitment. It is also the mechanism that makes consolidation so hard. The currency of those negotiations is discretionary budget amendments, and your leader identified precisely this rigidity, with 92% of federal spending mandated by law or by the constitution ("Brazil turns its back on the future", September 26th).

Brazil's real interest rate, the second-highest in the world, is the price of that arrangement. It is not an external shock. It is the arithmetic of a government that chooses not to tackle it.

ROBERTO MÁRIO AMARAL LIMA NETO
São Paulo

The human calculus

OpenAI did not by itself solve the mathematics problem related to the Navier-Stokes equations. Its AI agents drew on the work of generations of human mathematicians ("Axioms to grind", September 19th). The Fields Medal winners who wrote a letter on the breakthrough are right to warn that "Solving problems is only a tool and proxy for achieving the primary goal of conceptual understanding and insight."

The danger of AI-driven cognitive offloading is even more acute in schools than it is at the forefront of mathematical sciences. Learning is a process, where friction and challenge are indispensable to progression. The recent PISA results showed that higher-performing students tend not to use AI. If we thoughtlessly allow AI solutionism to expand into education and research the risk of mental decay grows, from early-years education to those working at the vanguard of intellectual discovery.

Offloading a problem to AI misses the purpose of the problem in the first place. Navier-Stokes was one of seven Millennium problems. They were not selected by the Clay institute for utilitarian reasons or because their application will necessarily fix human problems. They were chosen for their deep complexity. Working towards them offered the greatest opportunities for learning and collaboration.

Attempts to solve a problem can be as beautiful as the solution. The most powerful AI tools ultimately need human-generated learning and knowledge. So does humanity.

PETER PHILLIPS
Chief executive
Cambridge University Press & Assessment
Cambridge

Money doesn't buy expertise

You may be right that there is "too little money in British politics", and that Britain's impoverished parties are "barely able to prepare for power" ("How to spend it", September 19th). But what counts is how wisely parties spend their revenue. Recent events, for instance, would suggest that Nigel Farage's decision to hire an obscure yet controversial associate professor of philosophy and theology to professionalise Reform UK's policy-making process was far from a rational use of resources. The fact that these resources will no longer be scarce if the party is allowed to keep the £72m ($95m) gift it has just received is irrelevant. That remains the case if, like other parties, it continues to appoint policy-makers on the basis of ideology and conviviality rather than genuinely relevant expertise and managerial competence.

TIM BALE
Professor of politics
Queen Mary University of London

AI killed the video star

Bartleby wrote about "incredibly dated" corporate-training videos (September 5th). In a shock twist, AI seems remarkably adept at mimicking the dead-eyed stare, wooden acting, and unconvincing line delivery that was previously only possible using fleshy human actors. I don't expect many protests against this particular use of AI; most of us won't notice as we reach for the skip-to-next button.

PAUL HIGGINS
London

I'm still laughing at Bartleby's column. Every year at my university in Illinois we had mandatory ethics and compliance training that warned us to follow the rules. Most of us had already learned those rules in kindergarten, or Sunday school, or from parents. We randomly clicked on answers for the test and if we did not take enough time to read and answer the questions, we did not get a certificate.

The irony is that this ethics testing was started by Governor Rod Blagojevich, who was later impeached, convicted, and sent to prison for trying to sell the Senate seat vacated by Barack Obama upon his election to the presidency.

JELMIR ATKINS
Sycamore, Illinois

Eastbound and down

"The new, old Silk Road" (September 12th) reported on the expansion of overland lorry routes following the closure of the Strait of Hormuz. Another crisis 50 years ago also redrew the supply map. In 1976 shipping was still being disrupted by the closure of the Suez Canal. My first job, for Volvo's commercial-vehicle arm, was to travel by lorry from Glasgow through the Eastern Bloc and Turkey to Tehran, and write a travel manual for customers driving the route. The research was instructive. The border with Iran was closed by Kurdish fighting, cab temperatures could hit 50°C (122°F), I didn't see running water for ten days until I reached Tabriz, and a flight home was delayed by three days when anti-Shah demonstrators attacked the airport. Iran was then the destination for the lorries to run. Today it is the chokepoint that makes them run again, in the opposite direction.

CHRIS DAVIES
Oxford

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BY INVITATION

Alexander Heffner

Mayors have become democracy's most reliable defenders. American voters should take note

On November 3rd Americans will vote in midterm elections at a time when faith in the political process has perhaps never been lower. Nearly seven in ten Americans are dissatisfied with how democracy is working, according to the Pew Research Centre. But there is one remaining island of political functionality, where Americans can and should have grounds for optimism: City Hall.

President Donald Trump sat down recently with Zohran Mamdani—his third meeting with New York City's mayor in less than a year—to discuss building up to 12,000 homes on a rail yard in Queens. The picture of these two men illustrates our global moment. Around the world, mayors have become the most reliable bulwark against authoritarian politics. Populist parties surge in regional and national elections but run into a brick wall when they try to capture city halls.

That helps explain why Mr Trump has had to warm to the democratic-socialist Mr Mamdani, whose popularity is driven by an agenda to address the same problems Mr Trump promised but has failed to solve in either of his terms. As Mr Mamdani told me, "I would say talk is cheap. If you really do believe in a high standard of living for everyone, then start to show it in your actions."

Over the past year I met mayors on three continents, including my own, and they shared insights that American politics is largely ignoring. To friends, I have called my journey—at least in part—"Democracy Out of America," or Alexis de Tocqueville in reverse.

Consider Germany. The far-right Alternative for Germany made historic gains in recent state elections. Yet the party does not run a single German city of significant size, and it lost every big-city mayoral run-off it reached last year. In Portugal, the populist Chega party surged to second place in parliament last year, then won just three of 308 municipalities. In Romania, Bucharest's mayor beat a far-right nationalist for the presidency.

It is no coincidence that authoritarians the world over have targeted mayors. Last year President Recep Tayyip Erdogan of Turkey jailed the mayor of Istanbul, his leading rival. In January Hungarian prosecutors charged the mayor of Budapest over staging a Pride march that the government had banned. (The charges were later dropped.) In 2018 Kremlin loyalists abolished direct mayoral elections in Yekaterinburg, Russia's fourth-largest city, then led by one of the country's last opposition mayors.

Why do cities hold in the face of such attacks? Partly because of who lives there—younger, more diverse, more educated voters—but also because the job is to deliver a collective good. Fiorello La Guardia, who was mayor of New York City from 1934 to 1946, is often credited with saying there is no Democratic or Republican way to fix a sewer. "If you're a mayor, you don't wait for federal government," Stephan Keller, the mayor of Düsseldorf, told me. Either the pothole is filled or it isn't, and unlike in national parliaments, competence is harder to fake than outrage.

Nowhere is that truer than in America, where none of the 50 largest cities is run by a MAGA-aligned mayor. In conversation after conversation, American mayors expressed revulsion at the toxic, partisan wasteland that federal politics has become. Worse, that toxicity, they said, is increasingly trickling down to the states.

Ever more Americans appear to see mayors as the last public servants who still solve tangible problems. As Congress (and, increasingly, statehouses) suffer gridlock, the portfolios and profiles of mayors are rising. A generation ago a mayor was judged on whether the garbage got picked up. Today mayors lead on everything from housing and homelessness to infrastructure, education, crime and climate resilience. They are inventive, too. Atlanta's mayor, Andre Dickens, is building permanent housing from unused shipment containers amassed during the pandemic.

Cities are where America's future is being built. Edward Glaeser, a Harvard economist, has argued that density makes cities engines of innovation. Increasingly, mayors—not states—run the country's "laboratories of democracy". Americans trust local government more than any other level of government—65% do, according to Gallup, about twice the share that trusts Congress.

One of America's hidden tragedies, then, is the effort by statehouses to kneecap their mayors. Texas bars its cities from regulating labour and property beyond what state law allows. Several states, including Arkansas, Idaho, Iowa and South Dakota, have banned their cities from running guaranteed-income pilots. The federal government, meanwhile, has sent National Guard troops into cities over their mayors' objections.

America's brand of mayoral pragmatism has its limits. Mike Duggan, Detroit's mayor for 12 years, turned the city around and ran for governor as an independent. Facing headwinds from the two-party system, he dropped out earlier this year. Despite campaigns by Rudy Giuliani, Mike Bloomberg and Pete Buttigieg, no former mayor has become president since Calvin Coolidge a century ago. Europe, by contrast, has long elevated mayors to national leadership, from the great Willy Brandt of West Germany to Britain's newest prime minister, Andy Burnham.

Mayor necessities

In the run-up to the midterms, federal and state races are dominating the headlines. But American voters need to prioritise the values of mayoral politics. On November 3rd mayoral elections will occur in big and small cities alike, as well as votes for the governors and state legislators who often decide how much power mayors have. In these races, Americans have an opportunity to expand the one domain where politics seems to be working. ■

Alexander Heffner is the host of "The Open Mind" on America's Public Broadcasting Service.

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A Chinese shot in the arm

HONG KONG, LONDON AND ZHUHAI

A massive infrastructure scheme may give China a sorely needed boost—and help rebalance the world economy

CHINA IS NO stranger to ambitious public works. The undulating fortifications conceived to protect dynasties from barbarian invasion remain a wonder of the world. "You would have to conclude that this is a great wall," said President Richard Nixon when he visited in 1972.

To guard against other perils—economic stagnation and deflation—China is putting its faith in new infrastructural marvels. It could spend 25trn yuan ($3.7trn) over the next five years on what the government is calling "the six networks": water (including reservoirs and canals), electricity (such as smart grids), computing (data centres), communications (fifth- and sixth-generation mobile), logistics and urban pipelines.

The six networks were highlighted in July by the Politburo, the Communist Party's top decision-making body, which urged planners and bureaucrats to get cracking. On September 29th the central bank announced cheaper financing for six-networks loans. Analysts at Shenwan Hongyuan, a state-owned securities firm, have already picked a portfolio of stocks (six of them, naturally) that could benefit from the government's investment splurge.

The planned injection of spending, equal to about 3% of China's cumulative GDP from now until 2030, is a sign of both ambition and unease. It will lay the "hardware foundation for great-power competition", according to Shenwan Hongyuan. The futuristic economy China's leaders want to build—all those robots, drones and self-driving cars that Chinamaxxers like to live-stream—requires a lot of water-cooled, cheaply powered computing and communications capacity to sustain it.

The trillions of yuan in spending will also help fill a disquieting gap in domestic demand. And since a lot of that money will be borrowed, the six networks should mop up some excess saving, narrowing China's trade surplus and contributing to a more balanced world economy.

Many of the six networks are on display on Hengqin island in the Pearl river estuary along China's southern coast. Once mudflats, mangrove swamps and fishing villages, the island now stands at the cutting edge of urban development. Or so says a chirpy employee of Dahengqin Group, a local state-owned company. She is standing five metres below ground in a brightly lit concrete tunnel that curves far into the distance. "Officials from every part of China have come to see this," she says, darting a laser pointer over the water pipes, power lines, and fibre-optic cables that neatly line the tunnel's walls.

Dahengqin Group, which operates the tunnel network, calls it an "Underground Great Wall". Put another way, it is smart plumbing. During the breakneck expansion of most Chinese cities, utilities sprawled haphazardly. By threading them all through a well-illuminated, easily accessible tunnel, Hengqin solves two problems. It spares the surface world from tangled thickets of cables, and it makes the

pipes below easy to repair without digging up roads and disrupting traffic.

Such utility tunnels are not new. Walt Disney built them to hide away all the messy stuff that kept his parks running. His tunnels helped preserve a sense of fairytale magic. Hengqin's tunnels preserve public safety (broadly defined). They are studded with cameras, fire alarms and sensors that can detect dangerous leaks. Two of the fibre-optic cables carry data for the military police (they are helpfully labelled "Island Surveillance"). The tunnels can also help with channelling floodwaters, a big problem on China's typhoon-lashed coasts.

China already has 7,700km (4,800 miles) of such tunnels, although not all of them are as pretty as Hengqin's. It also has a bigger network of more conventional pipes, carrying gas, water, heating or sewage. Over the next five years, it plans to spend 5trn yuan building or renovating 770,000km of this network. If extended into space, this underground Great Wall would reach the moon and back.

The pipes are not even the most expensive item on China's six-networks shopping list (see chart 1). It plans to spend more on power (5.5trn yuan) and the highest sum on water (6trn yuan). The power projects will help its electricity grid handle the vagaries of solar and wind energy. The goal is to store excess electricity in batteries, move it long distances with ultra-high-voltage power lines and manage it more efficiently with algorithms.

The water spending includes projects to protect southern cities from floods by revitalising "living breakwaters" like the mangroves that Hengqin has restored. It also covers immodest schemes (long under way) to slake the thirst of China's arid northern cities by diverting water and pumping it uphill from the Yangzi river.

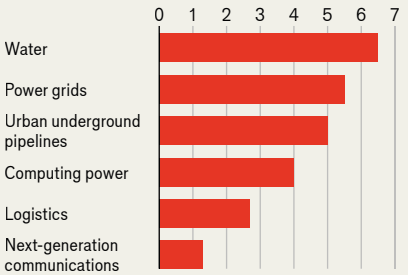
Not content to transfer huge volumes of water from south to north, China also will transfer rivers of data from east to west. Its coastal cities boast high-tech industries that generate and thrive on reams of data. But western China has cheap land and abundant energy that big server farms require. China therefore plans to build six of its ten data-centre clusters in poor, western regions like Ningxia and Gansu.

China's additions to its computing capacity will cost about 4trn yuan. It will also spend 2.7trn yuan on logistics, improving a network that already moves an average of over 50m tonnes of freight 1km every minute. And at least 1.3trn yuan will be lavished on 5G telecoms base stations, as well as on researching and deploying a sixth generation of mobile telecoms, which will take advantage of low-orbit satellites.

Will these vast sums of money be well spent? In addition to building wonders of the world, China has also built plenty of bridges to nowhere and backwater boondoggles. Its stock of government capital and infrastructure exceeded 84% of GDP as far back as 2013, according to calculations by Richard Herd for the World Bank that do not count power or telecoms networks. That percentage already matched America's ratio. It has certainly grown since.

Nice six-pack 1

China, new investment in "six networks" infrastructure strategy, yuan trn
2026-30 forecast



Source: Bank of America

Some analysts therefore worry about "redundant construction and homogeneous competition" between copycat projects. How much of the new money will be wasted on networks to nowhere? The Communist Party is not blind to the dangers. Its leader, Xi Jinping, rose to power in the aftermath of the so-called 4trn-yuan stimulus (which actually totalled more like 9.5trn of extra borrowing over 27 months). That spree rescued the economy from the global financial crisis of 2007-09 but also saddled its rivalrous local governments with costly vanity projects and "hidden debt"—liabilities held off-balance-sheet by local-government financing vehicles (LGFVs) set up for the purpose.

In 2017 Mr Xi said that local officials would have to take "lifetime responsibility" for government debts they had incurred. They could be investigated "even years after the fact" for bad investment decisions, points out Wei He of Gavekal Dragonomics, a research firm. The Ministry of Finance boasted of subjecting officials in 12 deeply indebted provinces to "high-pressure supervision" in 2023. Scrutiny could intensify again ahead of a major reshuffling of party leadership next year.

There are other reasons to expect fewer white elephants this time round. Compared with infrastructure splurges of the past, this investment push will entail less cement and more circuits, argues Helen Qiao of Bank of America. Many projects upgrade existing structures—adding sensors to pipes, for example—rather than just building more. They are examples of what economists call intensive growth as opposed to extensive growth.

The spending on water, sewerage and other pipes also addresses a long-standing gap in municipal development—a case of Chinese undercapacity. In the summer months deadly floods threaten even Beijing, the capital. In August thousands of elderly people were moved from care facilities vulnerable to the torrential rain brought by Typhoon Dolphin. "In the past, probably too much attention was paid towards what is built on top of the road, less so for underneath," says Dr Qiao.

The government thinks the investments add up to more than the sum of their parts. The electricity, water and communication networks will keep computing clusters powered, cooled and fed with data. The clusters, in turn, will train artificial-intelligence models that could help squeeze more power out of the grid. AI could do a better job of forecasting peaks and troughs in consumer demand and anticipating interruptions in supply from wind, solar and hydropower. Well-trained models will also help manage the growing fleets of drones, robots and autonomous vehicles that are supplementing China's logistics network.

Beneath such long-term visions, more immediate macroeconomic worries are pushing things along. In April the Politburo said that China would start big engineering projects "where conditions are ripe". The fruit now seems heavy on the vine. On September 28th the State Council, China's cabinet, said the six-networks implementation plans should start as soon as possible. Government estimates suggest that 7trn of the 25trn yuan could be spent in the first year. China's policy banks, which make loans in line with national priorities, have already begun extending 800bn yuan of financing.

Six appeal

Why the urgency? The unusual strength of China's exports, serving foreign customers, masks a worrying lack of demand at home. The rest of the world may be China-maxxing, but Chinese consumers are in a more buttoned-up mood. Retail sales grew by only 0.4% in nominal terms in August, compared with a year earlier. The consumer-confidence index is so bad it is a wonder the government still publishes it. The youth-unemployment rate, which was bad enough in 2023 that the government actually did stop publishing it, is once again nearing 19% in China's cities, even though the new, modified numbers exclude students looking for work.

China's households used to supplement their limited consumer spending with zealous investment spending on property. But since the housing bubble burst in 2021, more of that saving has flowed into bank deposits, securities and the like. Five years ago households held only 31% of their wealth in financial assets, as opposed to property. Now the share is about 45%, reckons Goldman Sachs, a bank. The financial system is struggling to put this money to

work. Bank loans in yuan grew by only 4.9% in August year-on-year. Loans to households (such as mortgages) fell.

China is also now contending with a less familiar gap in demand (see chart 2). Investment in manufacturing and traditional infrastructure has plunged, according to official figures tracking spending on "fixed" assets (that is, excluding things like stockpiled goods). The numbers are so bleak, they have many analysts scratching their heads. China's government data often arouse suspicion because they paint a flattering picture. These figures, by contrast, seem too bad to be true.

Other sources also look grim, however. The government's "general" budget, for example, shows a drop in infrastructure spending in the first half of the year. And money is flowing less quickly into another fiscal pot—government-managed funds, which typically collect proceeds from land sales and plough them into construction. Local governments are meanwhile preoccupied with retiring hidden debt, buying vacant land from struggling property developers and clearing a backlog of arrears to suppliers. This "fiscal crunch" is the prime culprit for the infrastructure slowdown, according to Bank of America.

If this weakness in domestic demand persists, China could easily lapse back into deflation. The worry will become acute if demand for the country's exports falters, either because of further protectionism or a slowdown in America's AI-investment boom. Imagining a similar shortfall in demand, John Maynard Keynes in 1942 urged the government to build an ample library of public-works projects that could be speeded up if the rest of the economy weakened and slowed down if it proved strong. These blueprints, he said in a radio address, "can be as ambitious and glorious as the minds of our engineers and architects and social planners can conceive". The six networks are China's answer to this same challenge. They provide 25trn-yuan-worth of ambitious blue-prints the central government can implement when necessary to support the economy. "Depending on policymakers' assessment of how much is needed, they are going to turn the dial a little bit faster or slower," says Dr Qiao.

In his radio address, Keynes also touched on the "nightmare of finance". How will all these ambitions be paid for? China's past infrastructure booms have relied on land sales and the hidden debt raised by off-balance-sheet financing vehicles. That model is no longer feasible or politically tolerable for Beijing. In a moribund housing market, land cannot bring in as much money as it once did. And LGFVs offer less financial wriggle room than they did before. Pan Gongsheng, China's central-bank governor, has claimed that 70% of them had been shut down or converted into commercial ventures with no local-government guarantees.

The six networks will therefore be financed differently. The money will come from the central government, some of its prominent state-owned enterprises and the "special bonds" it approves for local governments. China's two big grid companies plan to spend a combined 5trn yuan on power networks over the next half-decade. The three big telecoms operators intend to spend about 260bn yuan between them in just 12 months.

China's policy banks will also play a prominent role. Their loans are supposed to count as first-loss equity capital for the projects they finance. By bearing some of the risk, these stakes should allow the projects to raise more money—even four or five times more—from commercial banks.

Compared with the last infrastructure boom, the financing should be cheaper. China's excess saving and unusual lack of "investment hunger" means that interest rates are strikingly low. The central government can now borrow for ten years at less than 1.7%—in an economy that is still growing by about 5% in nominal terms. As the rest of the world contends with inflation and surging bond yields, China is still living in the world of "woke macro", as some economists have called it. In that world governments felt they could spend their way out of trouble and grow their way out of the resulting debt.

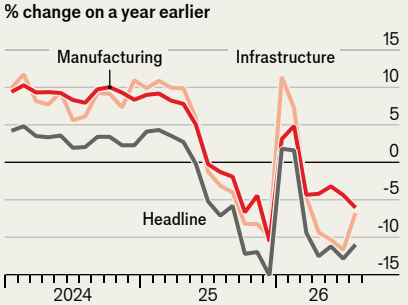
With this financial muscle behind it, China's ambitious six-networks blueprints should put a floor under its economy. By front-loading the spending, China can be sure to meet its growth target of 4.5-5% for this year. How much it adds to China's growth over a longer horizon depends on the answers to two nettlesome questions: will the six networks increase productivity? And how much would China have spent on infrastructure anyway?

Sceptics will point out that the "six networks" label is newer than some of the schemes it now names. That is often true of China's grand projects. The same criticism was levelled at the Belt and Road Initiative, a globe-spanning trade-and-investment plan launched in 2013, and the push for "new productive forces" (high-tech productivity improvements) that Mr Xi began championing a decade later. There is more than a cup or two of old wine in the six new bottles China is now showcasing.

Several analysts have compared the 25trn-yuan spending plans to the money spent in similar categories over the past five years. The vast sum it has allocated to water projects over the next five years (6trn yuan), for example, is similar to the amount spent over the past five. But these calculations also suggest that China could invest about 20% more in the power grid, 3% more in communications and perhaps 300% more in computing capacity.

Six feet under 2

China, fixed-asset investment*, % change on a year earlier



*Jan-Feb combined
Sources: National Bureau of Statistics; Haver Analytics

The joy of six

China's leaders hope that this new spending will usher in a high-tech productivity miracle. The six networks will, they believe, support a similar number of "emerging pillar" industries: integrated circuits, aerospace, biomedicine, new energy storage, robots and the "low-altitude economy" (which includes drones). They are also keen to cultivate half-a-dozen more speculative "industries of the future", including quantum technology, nuclear fusion, biomanufacturing, green hydrogen, brain-computer interfaces, embodied intelligence, and 6G mobile-telecoms.

Some of these may be flops. A high failure rate is inevitable in frontier industries. The return on upstream infrastructure may thus be disappointingly low. But as Keynes would no doubt point out: the return on unemployment is even worse.

By alleviating a shortfall in domestic demand, China's infrastructure push could also improve the country's global standing. The extra spending will, all else equal, narrow China's trade surpluses, by raising domestic prices and absorbing exportable goods. That would be good for the global economy. And it might help the cause of rules-based international commerce, which has struggled to cope with such a large, lopsided actor. For all its emphasis on telecoms, computing, electricity and so on, China ultimately owes its prosperity to the larger network of international trade. If the 25trn yuan douses some of the protectionist fires now burning around the world, it will be money well spent.

The leaders of some of China's trading partners will visit Shenzhen in November for the Asia-Pacific Economic Co-operation summit. Perhaps these dignitaries could cross the Pearl river estuary and inspect the Underground Great Wall in Hengqin. If such projects help revive Chinese growth, rebalance the world economy and forestall trade wars, you would have to conclude that these are great pipes. ■

United States

Presidential authority

The next test

The Supreme Court must decide whether to limit executive power or free speech

On the first day of Donald Trump's second term, he promised "to immediately stop all government censorship and bring back free speech to America". Twenty months on, federal judges have ruled against the president in more than 70 cases regarding the First Amendment, which guarantees freedom of religion, speech and the press; lower courts have issued judgments on everything from campus protests to the president's pressure on law firms. Other than a few emergency orders, however, the Supreme Court has yet to weigh in. As the court begins its term on October 5th, that is due to change.

The justices face a surfeit of divisive issues, including whether Arizona can require proof of citizenship from voters, whether a city can sue energy companies over climate change and whether the Second Amendment precludes bans on AR-15-style weapons. But cases on the First Amendment will loom particularly large in this term. Press freedom looks fairly secure. The harder questions arise when the right to free expression runs up against claims of presidential power.

The new term follows a period in which the Supreme Court has been largely friendly to Mr Trump's pleas, giving him satisfaction in more than two-thirds of 40 emergency applications. However, justices have shown willingness to cross him, too, notably in cases over the deployment of the National Guard, his sweeping tariffs, and his attempts to eliminate birthright citizenship and sack a Federal Reserve governor. In September the court rebuffed another executive order, which would have let the president commandeer mail-in voting ahead of November's midterm elections. Mr Trump fumed that the justices he had appointed were "merely a shell of their original selves".

Yet those defeats share a characteristic that may not bode well for plaintiffs challenging Mr Trump's policies on free expression. Other than *Trump v Barbara,* the case on birthright citizenship, each of the prominent cases in which the court ruled against the president turned on whether his actions clearly violated congressional statutes. On questions involving the separation of powers, the court has consistently given the presidency broad leeway, either out of fear of provoking a constitutional crisis or a belief that the constitution grants wide authority to the executive branch, in what is called the "unitary executive" theory.

At the same time, Genevieve Lakier, a law professor at the University of Chicago,

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23 Winning Christian voters
24 Lexington: A freed press

notes that under John Roberts, the chief justice, the court has also zealously defended rights granted by the First Amendment across presidential administrations. These two imperatives—to assert executive authority and to uphold the First Amendment—are now at odds.

The constitution's protection of the free press is so explicit that the Supreme Court is likely to disappoint the president in two cases involving the media. The first concerns Mr Trump's attempt, as a private litigant, to revive a $475m defamation suit against CNN, a news network, for characterising his claims that he won the 2020 election as a "Big Lie". Formidable obstacles to defamation suits mean that the president's case will probably fail, either because the Supreme Court refuses to take it or because it rules against him.

The second case involves Mr Trump's recent attempt to revoke White House credentials for journalists at CNN and two other news organisations, MS NOW and *Politico*, complaining of "Fake News" and claiming their reporting endangered national security. On September 24th Timothy Kelly, a federal judge appointed by Mr Trump, placed the move on hold for a fortnight. The national-security justification was weak, Judge Kelly wrote, and did not seem to be the president's "actual motivation". Mr Trump has said that he would pursue an appeal "almost without question", assuming Judge Kelly converts a temporary restraining order into a preliminary injunction. If so, the Supreme Court is likely to side with Judge Kelly. Heidi Kitrosser, a law professor at Northwestern University, points out that the government may not exclude select members of the press merely because the president is unhappy with their stories. The national-security claims, she adds, "do not pass the laugh test".

On other questions of the First Amendment and presidential authority, however, the court's conservative supermajority may be more open to Mr Trump's position. Last year the administration cancelled grants to researchers in California from the Environmental Protection Agency, National Science Foundation and other agencies. Federal dollars for projects that promote "diversity, equity, and inclusion", the government concluded, are "no longer in the public interest and should be terminated".

Two courts deemed this to be discrimination based on researchers' views, in violation of the First Amendment, spurring an appeal to the Supreme Court. Having twice split over grant cancellations on its emergency docket, the court may use *Trump v Thakur* to take up the issue. Ms Lakier points out this would reopen a messy precedent from the 1990s that gave the government some leeway—but not free rein—in deciding which projects to fund. Today's more conservative court, Ms Lakier reckons, might embrace the position that the conservative Justice Antonin Scalia adopted at the time: when the government opens its purse, it can attach whatever strings it likes.

Another clash likely late in the court's term is the case of Mahmoud Khalil, a pro-Palestinian activist and green-card holder who was nabbed by federal immigration officers in the lobby of his home near Columbia University last year. Mr Khalil, who faces deportation, says he was targeted for his views. In May the Third Circuit Court of Appeals voted 6-5 not to reconsider an earlier ruling against Mr Khalil; a petition to the Supreme Court is on its way. With lower courts divided, Steve Vladeck of Georgetown University predicts that the justices may require people in Mr Khalil's position to press their arguments before immigration judges first, rather than in federal district court. Immigration judges, Mr Vladeck says, are "increasingly un-independent" and reflect the agenda of the executive branch. Mr Trump styles himself a champion of free speech. This term may reveal how far the Supreme Court is willing to go to assert that standard. ■

Affordability

You give me fever

NEW YORK

Pricey health care is a growing problem for Republicans

On the night of the midterm elections in 2018, Nancy Pelosi, preparing to reclaim her position as the Speaker of the House of Representatives, said Democrats' victory was about two things: restoring Congress as a check to Donald Trump, then in his first term, and stopping "assaults" on health care. The year before, Republicans had attempted to repeal Barack Obama's signature health-care law, the Affordable Care Act (ACA). Democrats jumped on the issue; one analysis found that it was the most common topic of their ads in the months before the election.

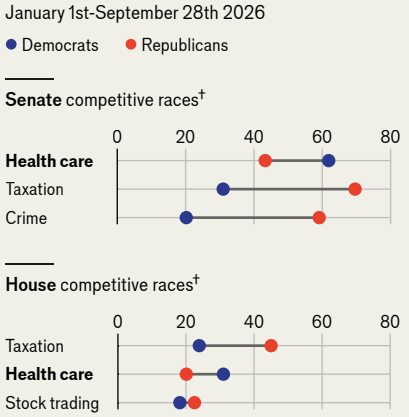
This year, health care is once again a vulnerability for Republicans. In one July poll from KFF, a health-research outfit, more Americans reported being anxious about affording health care than petrol, housing or food. In a March poll from Gallup, 61% of Americans said they worried a great deal about the availability and affordability of health care, more than about the economy or inflation. So, as part of a broader message on affordability, health care is again the most common issue in Democrats' campaign ads. Candidates have varied ideas of how to respond—the progressive wing has renewed its push for universal coverage, while moderates want more incremental reform. But they are hoping that anxiety over health costs will be enough to secure victory.

Unlike in 2018, the ACA is now unravelling, due to Republicans' cuts to subsidies for insurance bought on its exchanges and to Medicaid, a health-care programme for the poor. However, there are broader pressures on health care, too. Insurance prices in 2026 rose at their fastest rate in 15 years, driven by expensive new treatments for obesity and cancer, as well as the rapid growth in artificial-intelligence tools that help hospitals and clinics maximise charges. The outlook may darken in October, when the half of the population that obtains insurance through their jobs will receive notices of costs for 2027. Marsh's, a consultancy, expects prices for employer-sponsored health insurance to rise by an average of 8%, even faster than this year.

Democrats have seized on this. Since January, the party has spent $93m on ads that emphasise health care in competitive races for Congress, by far the most on any topic, according to AdImpact, which tracks advertising, and the Cook Political Report, which analyses elections (see chart). Ads elide various policy questions to make a general condemnation of Republicans' record on health. One spot against Dan Sullivan, a Republican senator

Vital statistics

United States, political-ad spending by issue*, $m
January 1st-September 28th 2026



*Top issues by combined spending †Rated as toss-up or lean
Sources: Cook Political Report; AdImpact

in Alaska, says he voted "to cut health care for thousands of Alaskans", then adds, "Alaskans can't afford Dan Sullivan's votes to raise the cost of health care."

Republicans emphasise Mr Trump's efforts to lower drug prices and try to reframe cuts as a way to fight fraud and waste. Jon Husted, a senator from Ohio facing a tough re-election campaign, is among those to argue that falling Medicaid rolls reflect Republican crackdowns on "people who are illegally on Medicaid that are not eligible". But in a sign of Mr Trump's concern about the issue, in September he announced $500 "refunds" to the nearly 1m people who buy insurance through federal marketplaces and are no longer eligible for subsidies (the 3m people who have dropped marketplace insurance will receive no cash). Voters are unconvinced. In a poll from the *New York Times* and Siena College, 60% said they trusted Democrats on health care, the most of any issue. Another poll from the Cook Political Report found that 56% of swing district voters are worried more about cuts than allowing fraud, waste and abuse.

How Democrats, once in power, could help Americans afford good care is much less clear. Abdul El-Sayed, the Democratic candidate for the Senate in Michigan, has centred his campaign on Medicare For All, a form of single-payer health care championed by Bernie Sanders, a senator. Dr El-Sayed says his primary "was a referendum on whether or not we want health care here or bombs somewhere else." Others favour gradual change. Roy Cooper, running to represent North Carolina in the Senate, wants to reverse cuts to the ACA. As the Democratic presidential primary begins, competing ideas for American health care will take centre stage. But their first goal is to win in November. ■

Maybe in time

Illegal drugs

Hegseth v hydra

WASHINGTON, DC

The pointless war on "narco-terrorists" is becoming more collaborative

THE GRAINY VIDEO shows a fishing boat bobbing off the coast of Ecuador. A craft ferrying American marines pulls up beside it. They board the ship and bundle off the crew, before the fishing vessel explodes in a burst of white. The video, released on September 16th, appears to show a new type of operation: hitting large fishing boats the Pentagon says provide fuel to smaller vessels running drugs—and sparing the lives of those on board. It was the eighth such operation since late August.

For over a year America's armed forces have pulverised small drug-smuggling boats in the Caribbean and Pacific. But in recent months the Trump administration's war against what it inaccurately calls "narco-terrorists" has become broader and more collaborative. Operations hit not only traffickers at sea but also their logistics networks, as well as a rising number of targets on land. And the attack off the coast of Ecuador, unlike earlier, unilateral boat strikes, was undertaken with the involvement of the Ecuadorean government, part of an emerging regional coalition going all-in on Donald Trump's drug war.

America's boat-bombing began haphazardly, run by a small team hastily assembled by the secretary of war, Pete Hegseth. The goal was to undermine Nicolás Maduro, Venezuela's then-dictator, as well as to showcase Mr Trump's vow to smash drug gangs. Legal experts condemned the strikes as illegal summary executions of people merely suspected of committing a non-capital offence. The head of Southern Command (SOUTHCOM), which runs operations in Latin America, resigned. So far, America has carried out roughly 70 confirmed attacks, killing at least 230 people.

Under SOUTHCOM's newish boss, General Francis Donovan, the tempo of deadly strikes has recently slowed and there is new emphasis on trying to choke drug-supply chains. Refuelling vessels, for example, are harder to replace than smaller boats, and therefore have a bigger impact on smuggling routes, says a former Pentagon official. Other targets include ports or clandestine airfields.

The Pentagon is also co-operating more with neighbours. In August SOUTHCOM unveiled a new task-force to co-ordinate operations with the 18 countries of the Americas Counter-Cartel Coalition. On a visit to Colombia, Ecuador and Peru, Marco Rubio, Mr Trump's secretary of state, promised that America would not strike targets without allied approval.

Consider Ecuador, which is grappling with gangs and a high murder rate; its president, Daniel Noboa, was the first in the region to launch joint military operations with America, in March. Since then, American special forces have helped Ecuadoreans blast suspected drug-traffickers along the country's border with Colombia, and Ecuador's army has hitched rides on American ships on missions at sea. On September 24th around 50 American marines joined Ecuadorean police in raids against a drugs gang along the country's west coast. Mr Rubio has praised Ecuador as the administration's "most involved and aggressive partner" in fighting cartels. Honduras, Colombia and Guatemala have all recently signed similar agreements to conduct joint operations.

America is providing other support, too. On September 15th the State Department reallocated around $50m from Europe and the Middle East to help Latin American countries buy American kit. The Pentagon is lending MQ-9 attack drones to Colombia and setting up a new regional drone initiative to share intelligence. Colombia recently used American intelligence to kill a prominent smuggler.

Mexico remains an outlier, declining to join the counter-cartel coalition despite Mr Trump's relentless haranguing. Though the two neighbours still co-operate, Mexican officials worry that American security agencies operate in the country with little oversight. That impression was reinforced in April, after two CIA agents died in a car crash during an operation targeting a suspected drug lab, which the Mexican government did not know about. American officials have refused to rule out unilateral action. "That threat will have to be confronted," Mr Rubio warned on September 9th, "one way or the other."

Besides looking tough, the campaign achieves little. Mr Trump claims to have stopped "98.2% of Drugs coming into the U.S. by Ocean or Sea". Yet a government assessment in July found no change in the availability of cocaine in America since the campaign began. Seizures at borders and ports rose in 2026, but narcos have adapted by moving more drugs overland. Even if supply is constrained, history and economics suggest that prices will rise, and smugglers will find a way. ■

Taxes

Slash and burn

TALLAHASSEE

A dramatic new proposal in Florida tests the wisdom of tax cuts

FOR A YEAR Blaise Ingoglia, Florida's chief financial officer, has been crossing the state on a "FAFO" speaking tour. The acronym, best known as a warning about pushing your luck, also stands for the Florida Agency for Fiscal Oversight. At a press conference in Palm Beach in June, Mr Ingoglia told taxpayers that the county "has been spending your money like drunken sailors". Then he got into a row with the county administrator over whether $443m had indeed been misspent. Mr Ingoglia sees this as part of a worrying trend. "Governments are wasting money left and right, crying poor and continue to want more, more and more," he said. His remedy? A massive property-tax cut designed to choke off their revenue.

In November voters in Florida will decide whether to make that tax cut part of the state's constitution. Ron DeSantis, the outgoing Republican governor, drafted the measure, and the Republican-controlled legislature tweaked it before putting it before voters. The final version calls on state politicians to devise a way eventually to abolish nearly all property taxes on primary residences, much as Florida prohibited state income taxes a century ago. It is an extraordinarily bold attempt to rein in government spending and cut costs for struggling Floridian families. But if it passes, it may well backfire.

A surge of pandemic-era newcomers sent home prices in Florida soaring, pushing up property-tax bills and swelling local-government coffers. Statewide property-tax levies rose by nearly 40% in the three years to January. The new rule—known as Amendment 3—would raise the property-tax exemption on primary residences from $50,000 to $250,000 by 2028, meaning that, for most local property taxes, a home assessed at $400,000 would be taxed as if it were worth $150,000.

State economists estimate that Florida's cities, counties and special districts would initially lose $5bn a year, with the annual hit rising to nearly $12bn within five years—an average 35% drop in their overall revenues, according to Cragin Mosteller of the Florida Association of Counties. Miami-Dade County alone would lose $1.4bn. Rural towns and suburban bedroom communities, where there are more homes than businesses, may have to start cutting basic services such as police and fire departments, election administration and hurricane preparation. In Gadsden County, a stretch of farm towns in the Florida panhandle, the amendment would take 8,000 of the 10,000 homes off the tax roll. Special districts that rely solely on property-tax revenue to keep drinking water safe and monitor disease-carrying mosquitoes could go broke. (Against the governor's wishes, levies that finance schools would remain unchanged by the measure.)

Cutting dissent

These nightmarish forecasts have forged unusual alliances. Democrats opposed the amendment from the start. Now many fiscal conservatives do, too. Jeff Kottkamp, the boss of Florida TaxWatch, a government watchdog in Tallahassee, can't remember the last time his organisation came out against slashing taxes. But it opposes Amendment 3. Mr Kottkamp worries that local politicians will offset the loss of revenue by charging fees for everything from access to public beaches to waste removal, a "bait-and-switch" that shifts the burden from rich homeowners to poorer residents. "People will get creative," he says. "It could be a new oxygen tax."

After Mr DeSantis vetoed a cost-benefit analysis of the amendment, opponents became more vocal, including flabbergasted policemen. "This is the most ludicrous piece of unresearched stuff I've seen," said Grady Judd, the conservative sheriff of Polk County who is close to the governor. Steve Vancore, a pollster leading the "no" campaign, says he never thought he would live to see a Republican legislature pass a "defund the police" bill.

The only group that is publicly defending the tax cut is Florida Realtors, a trade group for estate agents that stands to benefit if lower property-tax bills encourage more people to buy homes. A Republican strategist in Tallahassee says that the amendment has become so "charged" that he is not advising incumbents running this cycle—even those who voted for the measure in the legislature—to back it. "We're having to tell them to follow their own hearts and read their districts," he says. In an interview with *The Economist*, Byron Donalds, the Republican candidate for governor, blamed municipalities for letting their budgets balloon but acknowledged that a spending crunch could come. "We're going to work with them," he says. "We'll DOGE their budgets." He is nonetheless encouraging people to vote for Amendment 3 in November.

Decades ago California passed Proposition 13, which capped property-tax rates and wreaked havoc on local-government finances. The state government was forced to step in and shore up cities, counties and schools. That offers lessons for Florida, which is projected to fall into deficit by 2030. For Amendment 3 to pass, 60% of voters need to approve it. Polls have it just about there. Cynics accuse Mr DeSantis, who may be considering another run for the White House, of timing the measure to allow him to claim progress on affordability without having to deal with the fallout. Republican leaders took a gamble by putting such a big economic decision to voters. Now some may be quietly hoping that Floridians save them from it. ■

Playing with house money

Christian voters

Eyes and needles

LAWRENCE, KANSAS

Can Jesus Christ sell moderates on Democrats?

FOR A SENSE of how Christianity features in American politics, consider a video shown at the Republican Party's recent midterm convention in Dallas. "On June 14th 1946", it begins, "God looked down on this planned paradise and said, 'I need a caretaker'." So, the video went on, "God gave us Trump." The video continued in this line, explaining God's need for somebody to "get up before dawn", "eat supper" and "fight the Marxists".

Now a rather different Christian message is being preached on the campaign trail. "What we see that purports to be Christian in Washington does not represent the Jesus of the Gospels," says Adam Hamilton, the Democratic nominee for the Senate in deeply Republican Kansas. Mr Hamilton, a pastor, runs the largest United Methodist church in America, with a congregation of 24,000. He is one of the most surprising candidates in these midterms.

Though still an underdog, Mr Hamilton now has a 42% chance of defeating the Republican incumbent, Roger Marshall, according to *The Economist*'s election model. That is so alarming to Republicans that J.D. Vance, the vice-president and author of a recent book on faith, recently visited the state to stump against him. By emphasising his faith, Mr Hamilton may have found a route to voters disappointed by Mr Trump but still sceptical of Democrats.

White evangelical voters have been a core constituency of Republican candidates for decades, mobilised in part by their belief that abortion should be illegal. With *Roe v Wade* overturned in 2022, the focus of Christian voters has become more fuzzy, and campaign messages less clear. Mr Trump claims divine purpose—"I was saved by God to make America great again," he said at his inauguration last year. Mr Vance's faith underpins a broader effort to affirm America's Christian identity. An ascendant wing of Christian nationalists questions the separation of church and state, as well as women's right to vote. What does not feature much is helping the poor or turning the other cheek.

Christians began losing their faith in Mr Trump in the first year of his second presidency, according to the Pew Research Centre. The biggest fall was approval among non-evangelical white Protestants, which dropped from 46% to 33%. Yet even among evangelicals, approval slid from 66% to 58%. Just 40% of evangelicals said they were confident Mr Trump was acting ethically in office. A growing number of evangelical pastors have broken away from the Republican Party, too. Criticism sharpened in April, when Mr Trump posted an AI-created picture that seemed to depict himself as Jesus. (The president claimed that he was merely "a doctor".)

For Democrats, this creates an opportunity, with Mr Hamilton part of an ascendant group of Christian leaders. In Texas James Talarico, a Presbyterian seminary student and former teacher, has a good chance of winning what had hitherto been a solidly Republican Senate seat. Three more mainline Protestant clergy are running for the House.

Mr Hamilton, a former Republican, explains that he entered the race because he was so upset by what he says is cruelty and indecency wrought by some Christians around Mr Trump. "Jesus of the gospels was concerned with the poor," he explains to *The Economist*. "He was concerned with people who were hungry and thirsty—and strangers, immigrants even."

Though Mr Hamilton describes himself at every opportunity as "a moderate or independent-minded Democrat", his policy positions are rather conventional. He criticises the White House for cutting food stamps and support for health care. He emphasises the importance of immigration and the cruelty of deporting longtime residents. He stops short of saying that Immigrations and Customs Enforcement should be abolished, but he does say it has been "militarised and politicised" and needs radical change. In some ways, that makes his message similar to that of Mr Talarico. But unlike the Texan candidate, he avoids invective against the wealthy. Last year Mr Talarico told *The Economist* that the gospels are populist. Mr Hamilton, by contrast, is keen to say he is a capitalist.

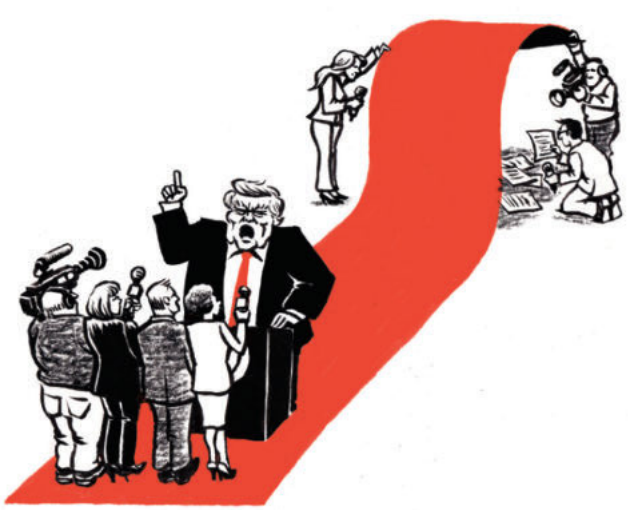
In Kansas, this strategy seems to be working. At an event on September 20th in Lawrence, home to the University of Kansas, over 150 people crowded into a gallery to hear Mr Hamilton speak. The vast majority were middle-aged white voters; all but one of those interviewed by *The Economist* identified as Republicans. Several spoke with reverence about Bob Dole, a famed Republican senator and presidential candidate from Kansas. Among the registered Republicans present was Deanell Tacha, a retired appeals-court judge appointed to the bench by Ronald Reagan.

This makes Republican politicians anxious. In Kansas Mr Vance accused Mr Hamilton of supporting what he called a "devil worshipping" gay-rights group that puts on drag shows for children (the basis for the accusation is that Mr Hamilton's church was a sponsor of a gay-pride rally). Mr Trump has called the pastor a "radical left lunatic" and Mr Marshall has falsely alleged that Mr Hamilton said Americans should repent for eating beef, an incendiary view in cattle country.

Mr Hamilton contends that this shows the success of his message. "If they're going to come and attack your faith, that's pretty low, really," he argues. At a debate at the Kansas State Fair on September 12th, Mr Marshall said that Mr Hamilton is "too woke and too weak" for Kansas. "I'm pretty sure if Jesus were here," he responded, "you would call him woke." ■

LEXINGTON
A freed press

With his pettiness, Donald Trump is liberating news organisations to reform how they cover the White House



IF AN AMERICAN PRESIDENT set out to manipulate some of the best reporters—to preoccupy them with his political priorities, to distract them from what is really happening in government and to recruit them to distribute his propaganda—he could do no better than to invent the White House press corps. Journalists are understandably upset that Donald Trump, in his imperious pettiness, is trying to ban from the White House those news organisations of which he disapproves. But they are wrong to say he would be undermining press freedom. He would actually be enhancing it, at least for those reporters he has targeted. He would be liberating them from the drudgery of covering the same staged events and briefings as a legion of other journalists. He is also giving the news media an opening to do away with the antiquated, corrupting and infantilising rituals of the White House press corps.

For decades, presidents and their aides have recognised how useful it is to herd reporters into the White House press room. Once contained there, they are handy as a megaphone, and they are less likely to cause trouble. "They'd be much better off if they were in offices scattered all over town," Michael Deaver, who as Ronald Reagan's image-maker honed television-age methods for manipulating the press, once remarked. "If they were out on their own, they'd be hustling and digging and getting their own stories." Serious White House reporters have long rued the gap between the prestige of their beat and its tedium and banality.

Yet despite advances in technology that have transformed the news business, White House press practice remains basically the same as in bygone days. A time-traveller from before the internet age would recognise the standing-room-only crowd packed into the briefing room—and would rightly struggle to understand how, at the same time, so many news bureaus across the country and around the world, and entire publications, could have closed their doors. Many news organisations maintain desks in the White House, stationing journalists there. A press "pool" with a rotating cast of a dozen or so journalists drawn from television, newspapers and other organisations idles on standby in case the president wants to emit commentary from the Oval Office, and it flies along with him on Air Force One when he travels. The participating outfits bear the costs and share the reporting of the pool, organised by the White House Correspondents' Association.

The press prizes this entrapment as "access" and the chance to report on the president and his aides only when and where they choose as "transparency". If such access and transparency had value, Americans would not have been shocked when, in 2024, President Joe Biden doddered out from behind his spokesmen to debate with Mr Trump. This is far from the first White House to punish critics and reward sycophants, and the emphasis White House reporters place on access can be corrupting. It is one reason readers encounter so many flattering references to chiefs of staff. The biggest scandals of the White House in modern times—Watergate, Iran-Contra, Mr Trump's attempt to extort a Ukrainian investigation of the Bidens—were not revealed by White House reporters. Your columnist, who served as a White House correspondent when Bill Clinton was president, dreads to remember the call from an editor in 1998: what was all this he was reading on the Drudge Report about a presidential affair with an intern?

It was during the Clinton years that the press secretary, Michael McCurry, decided to allow the daily press briefings to be televised live. He came to lament this, because the briefings became performative. Today, reporters ask combative questions and spokespeople insult them. This is entertaining for political junkies but seldom advances the cause of truth. As far back as Jimmy Carter, presidents and their aides have prided themselves on anticipating questions in advance. And by being suckered into playing the role of a Potemkin opposition, the press appears not independent but partisan. As theatrical antagonists, White House reporters serve the president's interests more than the public's.

Mr Trump understands the media better than it understands itself. He is a master of what the historian Daniel Boorstin called the "pseudo event". This is not news in the form of an unexpected development—an earthquake—but an event staged to create news and shape public opinion, such as the signing of an executive order. Boorstin connected the rise of pseudo news to the advance of technology that enabled mass, instant communication, sharpening the public's appetite for novelty. He was writing in 1961. Pseudo events now tell the preferred story of every presidency. As Mr McCurry once put it, "The modern presidency is defined by the manipulation of the newsflow, 24 hours a day."

Make Tom Sawyer paint his own fence

But there is no need, any more, for the press to squander scarce resources serving as the president's pseudo-event distribution machine. Rather than fight Mr Trump over access—giving the president what he probably wants, a distracting controversy over something voters do not care about, unlike prices at the pump—news organisations should treat his affront as an invitation to disband the pool and the White House Correspondents' Association.

Let the White House live-stream its own events, and pick out the newsworthy bits. Let all White House correspondents, as some of the best do now, watch briefings online if relevant and cultivate sources by phone, secure app, or interview in a less public venue than the White House. Mr Trump, the most accessible (if not transparent or accountable) of modern presidents, will surely keep picking up the phone. Do not boycott any president, but do not wait like courtiers upon any, either. The reporters have nothing to lose but their chains and that annual carnival of self-satire, the White House Correspondents' Dinner. ■

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The Americas

Migration

The machine stops

BOGOTÁ, MEXICO CITY AND PANAMA CITY

In all but ending illegal immigration to the United States, Donald Trump has changed Latin America

OVER THE past 50 years at least 23m people born in Latin America migrated to the United States. It was the largest sustained flow of migrants in the world to a single country, surging in the years after covid-19. Governments and people in Latin America got used to it, and the money the new American residents sent home.

Then Donald Trump returned to power. In the year to July 2026 US Customs and Border Protection recorded 137,000 encounters between its agents and migrants at the southern border. That was a decline of 94% from the same period in 2024, which saw 2.4m encounters. The Darién Gap, the jungle route from South America into Panama, tells the same story: crossings fell by 99.9% over the same period. In Costa Rica the number of migrants heading north has fallen to almost nothing, while the total going south has jumped (see chart on next page). The world's busiest migration corridor has fallen quiet.

Some of the surge in the years after covid-19—the only other moment in the past decade when migration slowed—was always going to subside. But the change is almost entirely Mr Trump's doing. Migrants have reacted to him ending legal pathways to asylum, expanding enforcement, accelerating deportations, and making clear that they are not welcome.

Mexico was once both a waypoint on the route to the United States and the source of most migrants. The country has become a destination for other migrants, says Luciana Gandini of UNAM, a university in Mexico City, the capital. Many have been at Casa Tochan, a shelter in Mexico City, for well over a year. Cubans, who used to travel through Mexico to reach the United States, are staying. Cuban asylum applications in Mexico rose from 17,884 in 2024 to 34,662 in 2025.

The other big economies near the United States have also seen an influx. Haitian asylum claims in Canada rose from 8,124 in 2024 to 18,027 in 2025. Cuban claims in Brazil increased from 22,000 to 42,000. Some smaller countries are seeing even larger proportional increases. Uruguay took in a record 15,000 Cubans in 2025, triple the year before.

Many Latin American governments are tightening their own borders in response. Abelardo de la Espriella, Colombia's new right-wing president, has ordered raids to deport undocumented immigrants. There are at least half a million undocumented Venezuelans in the country. In March Chile's president, José Antonio Kast, flew north days after taking office and gave a speech while a deep trench was dug along the Peruvian border behind him.

Most who had been considering a move to the United States are now simply staying put, either at home or wherever they ended up en route. The prospect of earning more in the United States no longer appears to be worth the increased likelihood of being turned back at the border or deported. Luis Alvarado, 41, left Venezuela with the intention of working in the Unit-

→ ALSO IN THIS SECTION

28 Mexico's museum spat

30 The Latinobarómetro poll

ed States for a few years. But when he got to Panama and heard they had closed the border, "I decided not to risk it," he says. "It seemed a bad idea once Trump started," says Carlos, a Colombian migrant. He made it to Monterrey, a hub in northern Mexico, before backtracking.

The flow of money is also important. The Inter-American Dialogue (IAD), a think-tank based in Washington, estimates that remittances worth just under $150bn are sent from the United States to Latin America every year. So far, overall, the money has kept flowing. Remittances to Central America rose by a fifth in 2025, to $55.4bn; Honduras alone took in nearly $6bn in the first half of 2026, up by 14%. But Manuel Orozco of the IAD thinks the rise is due to migrants sending home more money than usual because they are worried that they might be deported.

That will not go on for ever. Remittances are nearing "a threshold", says Mr Orozco. The number of transactions is already falling. And Mexico, which has more emigrants living in the United States than any other country, saw the value of its remittances drop. Mexicans sent $61.8bn home in 2025, down by 4.6% from 2024, the first annual fall since 2013 and the sharpest decline for 16 years. If this pattern extends across the region, it will hurt. Money sent home is worth 19% of GDP in Guatemala, 28% in El Salvador and over 30% in Honduras, according to the World Bank.

People come, people go

The economics are not all dismal. Some countries are keen to keep more of their people, and to welcome others. Latin America as a whole is ageing fast, with a fertility rate too low to maintain its population. If they work, migrants and returning residents can ease labour shortages and help public finances. But integration can be tricky. Many need help with paperwork, housing and psychological support.

There could be political consequences to slowing migration, too. Significant outflows of people have long acted as a pressure valve in repressive places like Cuba, Nicaragua and Venezuela. Those who leave tend to be young, energetic and relatively better off, the sorts of people who can most effectively challenge the government through protest. "If you don't have migration, you will have some form of social protest," says Mr Orozco.

Mr Trump's migration slowdown might end up as a blip. "This is all temporary," says one UN official in the region. He thinks migrants will make for the United States again as soon as they can. He points to a community of Haitian migrants who arrived in Tijuana in Mexico between 2014 and 2016 and integrated well; their children were Mexican. Yet they left almost overnight when in 2021 Joe Biden, who was president at the time, renewed the protected status of Haitians in the United States. The United States continues to offer economic opportunities that nowhere else can, says Marcus González, a Salvadoran who is in Mexico City having been deported from the United States in December. That there are so many migrants in the United States already means that word of new opportunities travels fast.

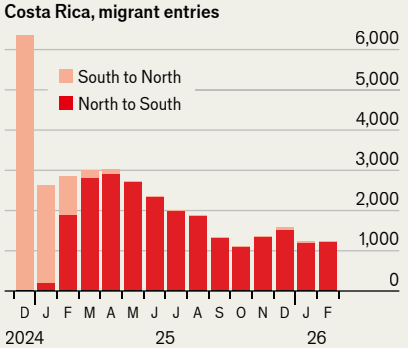
Migrants tend to be resistant to simply returning home. Venezuela offers a clear example. Since the country's dictator, Nicolás Maduro, was seized by US special forces in January, few Venezuelans have returned. A survey of Venezuelans abroad taken in February and March found that only 45% would consider returning even if conditions improved. "It will take years to get better," says Alexander, a 59-year-old Venezuelan in Mexico City.

The number of people making for the United States has been rising since May. More people, especially Hondurans, have been arriving at Casa Tochan lately, determined to head north. Mr Orozco expects roughly 415,000 irregular arrivals in the United States this year—2% more than last year, though still far below 2019's 750,000. He predicts that the number will settle at around 600,000, though he notes that a crisis could cause numbers to spike.

That rebound would look nothing like the boom in migration between 2021 and 2023, when border agents apprehended 2m people a year at the southern border. No Democrat who might follow Mr Trump would be likely to adopt the liberal approach to border security of Mr Biden. But migration, says Ms Gandini of UNAM, is not a singular phenomenon. Some people who have settled, found work and started families in Mexico are likely to stay there, whatever happens. Others, perhaps people whose families are in the United States, will leave the moment they can. Mr Trump succeeded in squashing an important migration route. But the forces that draw humans over borders will outlast him, and may well reassert themselves. ■

The reversal

Costa Rica, migrant entries



Source: IOM

Martial Mexico

Not on my park

MEXICO CITY

Plans for a museum devoted to the armed forces prompt a backlash

BY SOME COUNTS, Mexico City has more museums than any other city. Yet a proposal for a new one has caused an uproar. On September 20th *Reforma*, an influential newspaper, revealed the defence ministry's plans to build a vast museum complex dedicated to the armed forces in Chapultepec, the capital's main park.

The backlash was swift. NGOs issued statements; residents circulated petitions. On September 24th Clara Brugada, the capital's mayor, declared that the museum could not be built in Chapultepec. A week after the plan came to light, the army said that it would seek a different location.

A new address may appease environmentalists, who noted that the proposed site was on protected land in the "lung of the city", an important public space for residents. But it probably won't calm the furore. Many balk at spending 1.7bn pesos ($94m) on a museum when the defence ministry already has several, and while Mexico spends a paltry 0.7% of GDP on public security in which the army plays a large role. That is less than half the average in the OECD, a club of rich and middle-income countries, despite Mexico being far more dangerous.

Under Morena, Mexico's ruling party, the armed forces have sprawled. Between 2018 and 2024 Andrés Manuel López Obrador, then president, handed the army numerous civilian jobs, from building a tourist train and a new airport in the capital, to running customs, distributing vaccines and operating hotels and airports. Such projects are lucrative; deemed as tied to national security, they escape oversight.

Cristina Reyes of Mexico United Against Crime, an NGO in Mexico City, worries that Mexicans are getting used to this expanded role. She points to documents leaked in 2022 by a group of hackers known as Guacamaya, which showed that the army planned to burnish its image using television shows, mascots and social-media influencers. "Museums tell us who we are as a society," she says. It seems Mexicans prefer to be known for tacos and Day of the Dead festivities than guns and tanks.

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Latinobarómetro

Anti-America, pro-security

Latin Americans dislike Donald Trump, love mano dura and still rate democracy

In the 20 months since Donald Trump returned to the White House, right-wing candidates have won all seven presidential elections in Latin America. Many of them ran as Trump copycats, raging against the media and "cultural Marxism". And so the latest Latinobarómetro poll, taken in 17 countries and published exclusively by *The Economist*, is intriguing: Latin American's views of the United States have soured since Mr Trump came back. The share with a favourable opinion fell from 73% in 2024 to 63% (see chart 1).

The United States is still better liked than China. But for the first time, in a historically pro-American region, Latin Americans see China's influence in a better light than America's: 65% call it positive, against 57% for the United States. This reflects Mr Trump's aggressive stance towards a region he believes his country should dominate. While 69% approve of the capture of Nicolás Maduro, Venezuela's former dictator, organised by Mr Trump in January, only 35% think American intervention will benefit Venezuela. Almost two-thirds disapprove of the way he is conducting international relations (although 61% of Venezuelan respondents approve).

The concurrent election of Trumpian right-wingers may be partly explained by another of the poll's findings: Nayib Bukele, El Salvador's strongman president, remains the region's most admired figure. Mr Bukele reduced El Salvador's murder rate dramatically by locking large numbers of young men in prison without a trial, grabbing control of democratic institutions in the process. Many in Latin America want a similar elimination of crime and insecurity. The Trumpian candidates, such as Abelardo de la Espriella, now president of Colombia, promise to bring it about.

Despite the popularity of the authoritarian Mr Bukele, half of respondents continue to believe in democracy. After taking a dive in the mid-2010s, as economic stagnation set in following a commodity boom, support for democracy and its institutions has stabilised, albeit at a lower level (see chart 2). "A third [of respondents] say that democratic institutions are not important, but it's always been like that," says Marta Lagos, Latinbarómetro's director. Despite a pandemic and low growth "support for democracy continues."

Dissatisfaction with the way democracy works in practice remains widespread. Parliaments are particularly abhorred; only 22% feel represented by them. But 55% now say democracy can't work without political parties, up from 44% in 2023. Only 15% think their country's economic situation is good; just 28% are satisfied with public hospitals and 30% with the police.

That dissatisfaction has led to electoral defeat for many incumbent governments, recently mainly those of the left. Importantly, Latin Americans still believe in the power of elections to effect change. Presidential contests this year coincided with increased support for democracy in Colombia, Honduras and Peru, where right-wingers won, but not in Bolivia, where the new centre-right president, Rodrigo Paz, has struggled with economic problems.

Although the economy is still the main public concern, security tops it in seven countries and notably in Ecuador (see chart 3). The new, more radical right also taps into Latin America's social conservatism. Two-thirds of respondents oppose gay marriage; the same share opposes making abortion legal. But the region's south is different; majorities in Argentina, Chile and Uruguay are in favour of both.

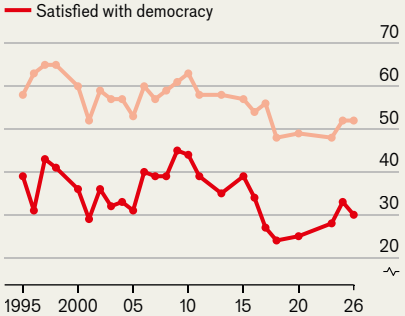
It may also help the right that Latin Americans don't expect much from the state. Asked how much the state has done for them in the previous three years, 73% said "little or nothing", up from 63% in 2011. That high level reflects the weight of the informal economy in the region. Only 28% say they pay social-security contributions.

As elsewhere, social media have displaced television, radio and especially newspapers as the main source of information about politics. Yet it is hard to trace a direct relationship between that change and the rise of more extreme politics. More than 70% think social media don't portray well "what most people think".

Despite problems, Latin Americans are not all that pessimistic. A quarter say they have thought of emigrating, down from 30% in 2023 (some may be responding to Mr Trump's immigration crackdown). Only 36% think their country's economic situation will improve over coming year, but 52% think their own situation will get better, the highest share since the series started in 1995. Home and family remain the bedrocks of Latin American life. ■

The worst form of government 2

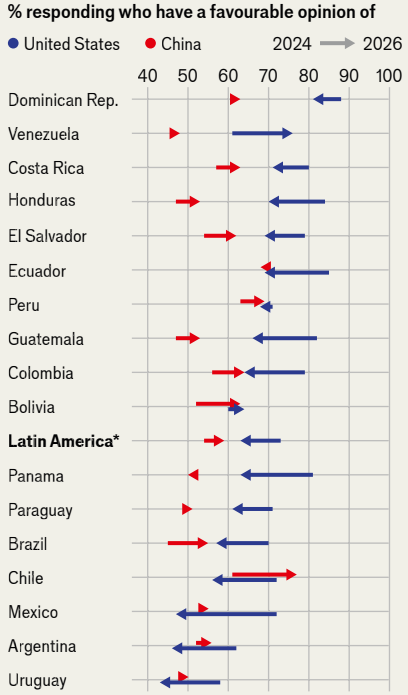
Latin America and the Caribbean, % agreeing



Source: Latinobarómetro

Detesting Donroe 1

% responding who have a favourable opinion of



Source: Latinobarómetro *And the Caribbean

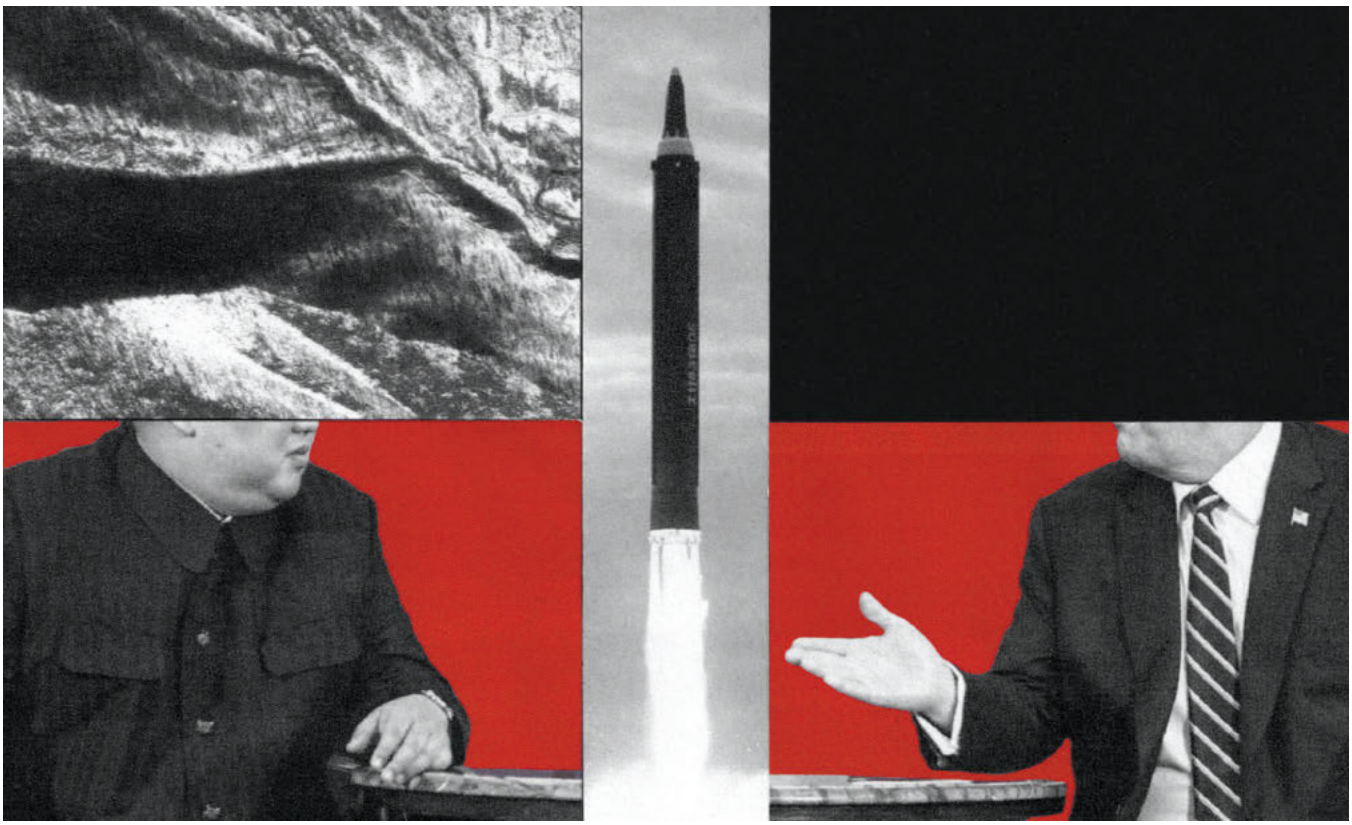
The economy and gangsters, stupid 3

"What do you consider the most important problem in your country?", % responding

2026

Source: Latinobarómetro *And the Caribbean

Asia



Korean security

Nuclear non-discussion

SEOUL

Kim Jong Un demands big concessions to meet Donald Trump again

SIX AMERICAN presidents have tried to deal with North Korea's nuclear-weapons programme. Six have failed. Donald Trump would like another shot: he has said he expects to meet Kim Jong Un, North Korea's dictator, this year. A planned trip to Asia for a summit of the Asia-Pacific Economic Co-operation (APEC) forum in November may provide an opening; Mr Trump's former chief envoy to North Korea said in a recent speech that he may even propose meeting in Pyongyang, where no serving American president has ever been.

Yet Mr Kim has made it clear that he will meet only if America drops its long-standing demands for "denuclearisation", diplo-speak for dismantling or removing his regime's most potent weapons. Just one month after Mr Trump curtailed long-planned military exercises with South Korea in the hopes of rekindling his bromance with Mr Kim, the North Korean dictator, flanked by his teenage daughter, oversaw the test of a new purportedly hypersonic missile. The weapons would be an "incurable headache" for the country's enemies, he crowed.

Mr Kim's intransigence reflects a newfound confidence. For decades, America and its allies, together with China and Russia, tried to press North Korea to abandon its quest for the bomb. Despite those efforts, Mr Kim now possesses dozens of nuclear warheads, as well as missiles capable of carrying them to targets as far away as mainland America. A renewed alliance with Russia, forged in the war against Ukraine, has given Mr Kim economic and diplomatic breathing room. Principled opposition to a nuclear-armed North Korea is eroding. Russia openly accepts North Korea's nuclear status, with its foreign minister calling it a "closed issue"; China has fallen silent on the matter. South Korea's left-wing president, Lee Jae Myung, proposes leaving denuclearisation as a hoped-for future goal. Mr Trump may be tempted to follow. If he does, it would mark a major shift in American policy, with big implications for regional security and global non-proliferation efforts.

North Korea signed up to the Nuclear Non-Proliferation Treaty (NPT) in 1985, agreeing to forgo nuclear weapons in exchange for help with civilian nuclear power. But despite that, it soon began secretly working towards a nuclear arsenal. Several rounds of international negotiations and wide-ranging economic sanctions over the ensuing decades failed to stop its progress; North Korea conducted six nuclear tests between 2006 and 2017. During his first term, Mr Trump imposed what he called "maximum pressure" sanctions, threatened Mr Kim with "fire and fury", and then agreed to the first-ever summit between the leaders of America and North Korea. But at a second meeting the talks broke

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— Ashoka is away

down in acrimony.

American leverage over North Korea has weakened since. In the past, economic sanctions could hurt the government in Pyongyang: during the summitry of Mr Trump's first term, Mr Kim sought relief from them in exchange for closing down Yongbyon, his main nuclear complex. But in recent years, North Korea has enjoyed a windfall for backing Vladimir Putin's war against Ukraine. North Korea has sent Russia artillery shells and ballistic missiles, as well as some 16,000 troops. Thanks in part to this newfound Russian support, North Korea's economy appears to be growing, albeit unevenly. South Korea's central bank estimates that North Korea's economy expanded at a punchy average of 3.4% annually from 2022 to 2025.

The alliance between North Korea and Russia is deepening. In early September the first road bridge linking the two countries opened over the Tumen river. Russia issued more than 35,000 student visas to North Koreans last year; many are suspected of working in Russia, including in factories making drones for the war effort. Ukrainian officials fret that North Korea is preparing a fresh deployment of troops to aid Russia this autumn. "We co-operate as equal partners; we have things we want and need from each other," says one Russian scholar. In addition to food, fuel and money, North Korea primarily wants high-end technology. South Korean officials worry about Russia helping Mr Kim build a nuclear-powered submarine.

This support from Russia has also let Mr Kim revive his relationship with China, which fears losing influence over its neighbour, and has left him correspondingly not much interested in rekindling a bromance with Mr Trump. Since their initial meetings, North Korea has also made significant strides in its weapons programme. In the past, North Korea's nuclear arsenal was nascent enough that stopping it might have been possible. North Korea enshrined its nuclear status in its constitution in 2023. Now Mr Trump says the country has 57 nuclear warheads; independent estimates put the number as high as 120 (Britain possesses 225).

The regime shows no signs of stopping there. A new uranium-enrichment facility has been spotted at Yongbyon, according to the UN's International Atomic Energy Agency. Mr Lee, South Korea's president, says that North Korea currently produces enough fissile material to make ten to 20 additional warheads each year.

Mr Trump's war on Iran has vindicated the Kim family's bet on nuclear weapons as the ultimate security guarantee: though Iran was close to having the bomb, it never crossed the nuclear threshold. "Especially after the war in Iran, I believe that North Korea will be more inclined to not give up its nuclear arsenal," Mr Lee told *The Economist* in June.

Selected North Korean nuclear infrastructure*, 2026
● Nuclear facilities ● Missile bases
*Includes historic sites
Sources: Nuclear Threat Initiative; IAEA; press reports

Calls have thus been growing for a new approach that shifts the focus away from pressing North Korea to give up its nukes and towards restarting dialogue with the hopes of merely slowing its progress. Mr Lee, who leads a party that tends to favour engagement, proposes a phased approach that would begin with a freeze on the North's nuclear programme.

In America, too, the winds are shifting. Victor Cha, who helped to lead negotiations with North Korea during the George W. Bush administration, argues that America should now shift from insisting on denuclearisation to instead seeking a "cold peace" with North Korea, based in part on arms-control negotiations. While American officials still speak of full denuclearisation as an ostensible goal, Mr Trump has taken to referring to North Korea casually as a "nuclear power".

Recognising North Korea as a nuclear state, even tacitly, would hand Mr Kim a long-sought prize. Mr Kim told his elites that if the country stayed the course, America would cave and acknowledge North Korea's nuclear status, much as it did with Pakistan, says Thae Yong Ho, a high-level North Korean diplomat who defected in 2016.

Unlike Pakistan (or India) North Korea signed the NPT and acquired nuclear technology provided it would be for peaceful use. So legitimising its nuclear weapons now could set a worrying precedent for future proliferators. An agreement that leaves North Korea's arsenal intact while letting Japan and South Korea feel exposed could also push America's allies closer to developing nuclear weapons of their own. By trying to solve a devilish problem, Mr Trump risks creating new ones. ■

India

Omissions and commissions

DELHI

Allegations of bias are rocking India's electoral watchdog

INDIAN ELECTION workers have long braved mountains and tigers to set up polling booths—sometimes for a single voter. In Delhi, a three-person Election Commission has overseen this mission with an official motto: "no voter to be left behind". Now the current team is wrecking that reputation, S.Y. Quraishi, a former chief election commissioner, tells *The Economist*. Veterans of the independent watchdog rarely criticise their successors in public. It is rarer still for protesters to take to the streets to demand that one resigns. But that is what happened after the *Indian Express,* a newspaper, produced explosive complaints from two sitting election commissioners on September 23rd.

Under their chief commissioner, Gyanesh Kumar, they believe the commission has taken "unauthorised" and "illegal" decisions, the newspaper reported. Barely two months after mass student protests forced an education minister to resign, allegations of misconduct are again rocking the government. Protesters and opposition leaders claim the ruling Bharatiya Janata Party (BJP) has brought the watchdog of Indian democracy to heel.

According to the *Indian Express*, the other two commissioners raised internal objections to the way decisions were pushed through without their agreement. They complained that the commission's bureaucrats had approved hurdles that made it harder for Indians to prove their right to vote. They also objected to the exclusion of hundreds of thousands of voters that local staff had deemed legitimate. The two commissioners have declined to confirm the reports, but have not repudiated them. The commission has stated that its own decision processes were sound.

Under particular scrutiny is an exercise with a deceptively dull name. Since June 2025 the commission has put most of India through a "special intensive revision" (SIR)

of voter rolls; only six states and union territories out of 36 have yet to start. So far, so fair enough: two decades after the last big update, the records are full of dead people, duplicates and outdated entries. But the project has been chaotic—and often rushed through just before state elections, with no time for objections to be heard. With confusion over the documents needed to prove eligibility, some have discovered that passports and the government's own digital ID are not sufficient without a school-leaving certificate to boot.

The outcome has been mass disfranchisement. The exercise will potentially axe more than 130m names—tens of millions of which will be real voters. Worst off is Delhi, where 55% of voters have been either deleted or given until October 29th to convince the commission otherwise. In one location, a single voter out of 729 survived. By the commission's own standards, these are strange figures: its guidelines classify a deletion rate of more than 2% as "suspicious". The commission is promising to look again at the process, but many political analysts call SIR India's biggest electoral scandal in decades.

Muslims have found themselves particularly vulnerable to deletion. Across India, local BJP figures have shown up with stacks of forms, pushing poll workers to scrap them in batches. Allegations of bias have been loudest in West Bengal—a state the BJP coveted for decades and won in May. There, more than 11% of names vanished. Of the 3.4m Bengalis who appealed, fewer than 0.05% had their cases heard before election day. In the appeals process, tribunals have found that 90% of appellants were valid all along.

One obvious question is whether SIR can swing elections. The evidence for that is muddy. In West Bengal, anti-incumbency alone may have been enough to carry the BJP to victory but seats with many deletions also consistently swung towards the party. The BJP's loudest regional opponents, chief ministers M.K. Stalin of Tamil Nadu and Mamata Banerjee of West Bengal, each lost their seats with margins that were smaller than the number of voters who had been deleted.

In India, parties of all hues have long flirted with voter fraud. The difference is that in the past the Election Commission has been a fairly stalwart check on their mischief-making. But in 2023 the BJP defied a Supreme Court order and passed a law that in effect lets the government impose its nominees. That has cast doubts over Mr Kumar's impartiality, since he was appointed under the new rules. Opposition leaders and street protesters are calling for him to be sacked and even prosecuted. Rahul Gandhi, the parliamentary leader of the main opposition Congress party, wants him tried for treason. ■

Vietnam

Powerless

Vietnam's high-tech dreams come with an electricity problem

TO MANY PEOPLE Vietnam's promotion in July to "upper middle-income" status by the World Bank vindicated almost 40 years of *doi moi* or "renovation," the country's period of economic opening. To Lam, the chairman of Vietnam's Communist Party, is not satisfied: "If we continue with old methods, we may achieve temporary growth," he warned in May, "but it will be difficult to create sustainable growth ...or escape the middle-income trap."

Wages for Vietnamese workers are rising, which threatens to squeeze already narrow margins in the manufacturing sector. So before its workforce is priced out, Vietnam's government hopes to leap up the value chain. It intends to manufacture high-tech products like semiconductors, which it currently merely assembles with comparatively little value added. But chipmaking takes lots of energy. And Vietnam, which has only recently conquered chronic blackouts, needs reforms to match its ambition. Luckily, it has a plan. Vietnam aims to bring liquefied natural gas, or LNG, into the power mix. Imported gas can provide an efficient, steady stream of power and is ideal for energy-intensive industrial parks. By 2030, Vietnam hopes to generate about 10% of its electricity from the stuff.

But it won't meet that goal. VinGroup, the country's largest conglomerate, ended a $6.7bn LNG power-plant project in March, citing high costs. The construction of new LNG power plants is stalling across the country, and the handful that are moving forward are mostly run by state-owned enterprises, including Vietnam Electricity (EVN). So far, LNG contributes only about 1.6% of Vietnam's electricity.

Vietnam is not entirely to blame. The Iran war has hit Asia-bound LNG hard; import prices have risen by 146% since the end of February. At current prices, importing LNG at the rates envisioned by the 2030 plan would devour "$12bn-16bn...14-18% of the country's reserves, every year", says Phillip Solomon, a director at BRG, a consultancy. But this pressure will fade if and when global prices recede.

The bigger problem is political. Electricity in Vietnam is highly subsidised. State-owned producers are told by the government to sell their power at an artificially low cost, which makes it easier to enforce aggressive price caps. LNG does not work that way. Building a large fleet of LNG power plants takes huge amounts of capital and expertise that are beyond the state's ability to provide. Only foreign operators can do the trick, and they typically will demand higher rates than the government prefers to pay. This puts Vietnam in a bind. It can either pay for LNG, which makes power subsidies more expensive to maintain, or abandon gas imports, jeopardising its high-tech hopes.

Vietnam has struggled with this choice before. Back in 2017 it was able to attract foreign companies for solar projects with a generous feed-in-tariff, a baseline rate that the state pays for power. But these rates were too expensive to maintain, so EVN retroactively cut them in 2025, slashing returns for investments totalling about $13bn. As a result, many foreign power developers already think twice before doing business in Vietnam. ■

Just add juice

Japan

More than a matcha for China

KAGOSHIMA

The green tea wars hot up

ATOP A HILL in Kagoshima, on Japan's southern island of Kyushu, neat rows of green tea bushes stretch across a vast field. When Nishi Toshimi inherited his family farm two decades ago, the industry was heading downhill; prices were falling and sales shrinking. Then the world went crazy for *matcha*. The boom of the past few years has felt "like a miracle," he says.

Japan's green-tea exports reached 12,600 tonnes last year, three times as much as a decade earlier; prices have also surged. Unlike Kyoto, famed for its coveted Uji *matcha*, or Shizuoka, a traditional producer of brewed green tea, Kagoshima is barely a household name. Yet the *matcha* boom has transformed its fortunes. In 2024 it became Japan's top tea producer for the first time and now grows around 40% of its *tencha*, the leaf that is ground to make *matcha* powder. Even so, Japan is struggling to keep up with global demand. Its tea industry is in surprisingly poor shape, beset by ageing farmers, labour shortages and competition from China.

While the world craves *matcha*, most Japanese drink *sencha*, made by brewing tea leaves in the normal way, rather than dissolving *matcha* powder. Yet demand at home has long been falling as habits have changed. Consumption of tea leaves fell from 527 grams per person in 1970 to 233 in 2024. Farmers have left the business or retired without successors; between 2005 and 2025 the area devoted to tea cultivation shrank by 31%.

Nor can all farmers lean into the *matcha* boom. Producing *tencha* requires different cultivation methods, including shading bushes before harvest, and costly processing facilities.

Many in the industry think that Japan's future lies at the high end. China, which produces 70% of the world's *matcha*, may dwarf it in quantity, but Japanese producers argue that their *matcha* is tastier and far more sophisticated. Tea came to Japan from China, but Japan developed its own traditions around it, most famously the tea ceremony. Mr Nishi thinks Japan should establish itself as the Champagne region of *matcha*. "Otherwise, we're going to lose out", he says.

Indonesia

Tourists v dragons

KOMODO

Growing pressure on a remarkable species

IT HAS SURVIVED ice ages, droughts and cyclones, but whether the Komodo dragon can survive mass tourism is not yet known. In recent years Komodo island, along with its supply base of Labuan Bajo on the nearby and much bigger island of Flores, has been at the sharp end of Indonesia's drive to expand its tourism industry. Joko Widodo, Indonesia's president until two years ago, said that he wanted to create "ten new Balis", aiming to bring more jobs and high incomes to his country. Bali has long been Indonesia's top tourist destination, bringing in vast amounts of foreign exchange and providing jobs to Balinese and to migrant workers from across the country.

High on Mr Widodo's list was Labuan Bajo and beyond it the Komodo National Park. After all, it boasts a unique tourist attraction, the Komodo dragon, an enormous lizard that can grow to ten feet (three metres) long, weigh 80kg, and lives only on Komodo and a few smaller nearby islands. Only an estimated 1,300 adult "land crocodiles", as some call them in Indonesia, are thought to survive.

Economically, it has worked. Labuan Bajo now throngs with diving shops, restaurants, hotels and coffee spots. Its port sends daily boatloads of enthusiastic tourists to Komodo (which has no airport of its own and can be reached only by sea), and its surrounding waters and islands.

But last December the park, and its visitors, got a nasty shock. Four Spanish tourists, including Fernando Martín, a football coach with the club Valencia CF, and three of his children, died when their boat sank in rough seas while on a tour. The incident prompted a temporary closure. Mr Widodo's plan has been just too successful, says Sisilia Lenita Jemana, a marketing manager for the Labuan Bajo Flores Tourism Authority, a body set up to implement the ex-president's ideas. The region is struggling to cope with phenomenal growth. In 2025 there were more than half a million visits to the area, up by 21% on the year before. The number has increased fivefold in a decade.

Ms Jemana's organisation had long urged the maritime authorities and national park authority to limit access, particularly in the dangerous wet season, she says. The deaths in December caused widespread dismay. "If the weather is not conducive, they should not give permission to sail," says Ms Jemana.

Her organisation has been looking at other tourist attractions to promote, such as sites linked to Portuguese traders and missionaries who settled in the island in the 16th century and the discovery there, around 20 years ago, of remains of *Homo floresiensis*, the extinct "Hobbit man" that is a relative of modern humans. Flores has other national parks, and active volcanoes. But at the end of the day, the tourists want to see the dragons.

Unfortunately, many of them don't know how to behave once there. Some get too close to an animal that may lounge around sleepily, but which can be aggressive and has been known to attack humans. It also has a nasty, venomous bite. Earlier this year Komodo introduced an initial cap of 1,000 tourists a day. With luck, the dragons will be mollified. ■

There be dragons

China

China and America

Crossed wires

Who answers an AI-crisis hotline between America and China?

On the same day that an American fighter jet shot down a Chinese spy balloon in 2023, Pentagon officials tried to use a military hotline to China. Lloyd Austin, America's secretary of defence at the time, wanted to talk directly to Wei Fenghe, his Chinese counterpart, to prevent an escalation. There was just one problem: General Wei refused to take the call. America had "not created the proper atmosphere" for dialogue, a Chinese military spokesman said. Nor was that the first time China had shunned the hotline. It had "just rung in an empty room for hours upon hours", recalled Kurt Campbell in 2021 when he was the most senior Asia official in the White House.

Would a hotline for artificial-intelligence emergencies work any better? The Trump administration seems to think so. One of the few outcomes of the meeting between Xi Jinping and Donald Trump in Washington on September 24th was an agreement to establish a bilateral "communication channel" for AI-related incidents. The two sides also agreed to hold a second round of a newly revived AI dialogue by November, when Mr Xi and Mr Trump plan to meet again in China. Neither has provided details of how an AI communication channel will work. But American officials (who proposed the idea) have likened it to hotlines with the Soviet Union during the cold war.

That sounds sensible to many people, given the growing fears that AI models could destroy humanity. Yet among those who have tried to set up or use a hotline with China, the proposal often prompts eye-rolling. "There have been many such attempts to keep channels of communication open," says Daniel Kritenbrink, a former deputy head of America's embassy in Beijing who has also served as a top Asia official in the White House and the State Department. "Those hotlines have been mostly ineffective." In a crisis, he adds, the Chinese system "usually goes dark".

The cold war does offer some valuable lessons. A hotline between American and Soviet leaders was first set up after the Cuban missile crisis almost triggered a nuclear war in 1962. The original idea was for them to communicate quickly and directly, rather than by means of diplomatic cables which could take hours to transmit. Although diplomatic communications soon speeded up, the hotline conveyed urgency and built up trust. It did not always ease tensions: both sides used it to make threats, too. But along with other communication mechanisms, it helped to prevent a nuclear Armageddon.

American efforts to replicate that with China have been troubled from the start. A presidential hotline was created in 1998 in response to a crisis two years earlier in which China fired missiles into waters near Taiwan, prompting America to send two aircraft-carriers to the area. Bill Clinton, as president, tried to use the hotline in 1999 ▸▸

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after America accidentally bombed the Chinese embassy in Serbia's capital, Belgrade. But his counterpart, Jiang Zemin, refused to take the call for a week. Two years later, American officials struggled to reach anyone in the Chinese government for 12 hours after an American spy plane collided with a Chinese jet fighter and was forced to land at a military airfield on Hainan, an island in southern China.

One conclusion America reached was that it needed a hotline to Chinese military commanders, too. That was launched in 2008, enabling encrypted voice-calls between the office of America's defence secretary at the Pentagon and Zhongnanhai, the leadership compound in Beijing, with an extension to China's defence ministry. Yet calls had to be arranged 48 hours in advance. And it soon became clear that China's approach to crisis management had not changed. In 2009 Chinese navy and law-enforcement vessels surrounded an American ocean-surveillance ship in the South China Sea. American calls on the hotline went unanswered.

Such incidents reveal a pattern. China appears reluctant to pick up the phone because it wants time to gather information and formulate a response. But it also often sees an advantage in stalling to maximise leverage over the Americans, who typically want to resolve each crisis quickly. Another obstacle is that China sees itself as the weaker party and views American attempts to build guardrails as ploys to constrain its activities. (The Soviets, by contrast, viewed themselves as equals to the Americans in military terms, with a shared interest in avoiding nuclear war.)

Veterans of hotline diplomacy with China are not all sceptics. Some say the presidential hotline has helped with negotiations on more routine matters, such as leaders' visits, and in co-ordinating responses to problems like the global financial crisis of 2007-09. The military hotline was effective, too, when China worried that America might be planning to attack it around the time of the American presidential election in 2020. And some hotline advocates say China now has enough worries of its own about AI safety to have a real interest in exchanging information with America, focused to begin with on issues of common concern such as rogue AI models and malicious humans without government links.

Still, to make an AI hotline effective in a crisis, it would need to differ from existing ones in several ways. For a start, it should be text-based, rather than a voice link. "The phone doesn't work at the front end of a crisis," says Drew Thompson, a Singapore-based scholar who, as a Pentagon official, helped to negotiate an upgrade of the military hotline. It works "at the back end after the Chinese side has had a chance to do a committee meeting". He favours the fax. That gives each side time to review the contents and prepare a response. It also eliminates the trouble of scheduling live conversations and reduces the risk of mistranslations or misunderstandings based on tonal delivery.

Written communications allow for sudden personnel changes, too. That is especially pertinent now, given Mr Xi's recent purge of his military high command, says one adviser to the Chinese government. Another improvement would be an agreement on thresholds that trigger hotline use, according to Lucy Luo, an AI-policy researcher at the Carnegie Endowment for International Peace. She also suggests establishing a separate communication channel, staffed by people with AI expertise, for routine exchanges of information about minor incidents and best practices.

In recent meetings, one potential model discussed among Chinese and Western AI experts is another regular military communication mechanism, established in 1998. Under the Military Maritime Consultative Agreement, American and Chinese officers have met twice yearly (except when China suspended all military communications) to discuss air- and maritime-safety incidents, some of which are then taken up on the military hotline.

Whether China and America are at the point of discussing such details is unclear. They have yet to clarify who exactly should take part in their revived AI dialogue, which is being led for now by America's treasury secretary, Scott Bessent, and one of China's deputy prime ministers, He Lifeng. And they will be preoccupied with negotiating other matters, including a longer extension of a trade-war truce, before their leaders meet again in November. Announcing the AI hotline was still, on balance, a positive step. Now they need to make it work. ■

Vocational skills

Faster, sharper, shinier

BEIJING

Why China lavishes attention on a little-known competition

ONCE AGAIN Team China has crushed its rivals. At the WorldSkills Competition it won a record 41 gold medals, ten times as many as South Korea, the runner-up. Never heard of the event? In China it was hard to ignore: the six-day extravaganza, which wrapped up in Shanghai on September 27th, involved some 1,400 young people from 60-odd countries competing before panels of judges amid Olympic-style razzmatazz (and as is now standard at show-off events in China: dancing robots).

This was no sporting jamboree, however. Participants were tested on skills like bricklaying, machining and landscape gardening. The tournament rarely makes headlines in the West. But China has poured more resources into it than anyone

Factory setting: spectacular

else. It sees benefits far beyond the arena.

Spain launched the competition in 1950 to encourage more young people to start apprenticeships. It was not until 2011 that China began to take part. Then it won just a single medal, a silver in welding. Now it routinely dominates the event, which is held every two years. Competitors have to complete tasks in their discipline, such as installing, testing and repairing a wind-turbine component. They are given points for such things as speed and precision. Drone piloting and cyber-security were among new categories added this year. China won golds in both (though not in landscape gardening: France came first).

As it does in producing Olympic stars, China's government plays a big role. Promising young people are filtered through local, provincial and national contests before entering an elaborate training system. China has designated hundreds of bases—at vocational colleges, universities and companies—where contestants drill their skills under specialist coaches, sometimes including past champions, and receive physical and psychological preparation.

The proud coach of He Xiaoman, a 20-year-old gold-medal winner in the train inspection, maintenance and repair category, attributed her success to "countless nights spent alone at the training base". China is "unique in the resolve and scale of its embrace of WorldSkills", noted a slightly awestruck report by the organisation.

One pay-off that China wants is a boost to national pride. State-run media broadcast daily coverage of the competition in Shanghai, the first ever in China. They focused on events linked to China's industrial strengths (like drone-making), and winners draped in China's crimson flag. "Seeing the dazzling red of China makes my blood dance with excitement!" said a commenter on Weibo, a social-media platform.

China also uses the competitions to project soft power. They help advertise its "Luban workshops" in poorer countries: vocational-training centres named after a mythical Chinese carpenter. The institutions are not just a diplomatic sweetener. They promote technology and standards that China wants to export. Medals at this year's WorldSkills contest were in the shape of "Luban locks": wooden puzzles supposedly created by the man himself. At the opening ceremony, Li Qiang, the prime minister, called for more co-operation on skills-building with the global south.

China has another goal: getting more of its people into vocational training. Many Chinese look down on vocational colleges, which are often underfunded and badly staffed. Last year the government warned of a 10m shortfall in skilled workers. Many young Chinese seem happy enough to cheer on their best workers, but do not want to join them in the factory. ■

Studying in the West

Communists or converts?

Chinese students abroad see faults in China's system as well as in democracies

America should expect that educating China's young people will give it "moral, intellectual and commercial influence", wrote Edmund James, a university president, in 1906. In subsequent years, dozens of China's brightest went to study at the Massachusetts Institute of Technology on scholarships funded by the American government. Back then, American officials and teachers displayed a confident, missionary liberalism. They thought that educating young Chinese would help mould China in America's image. But in recent years America has become less welcoming. The number of Chinese enrolled in American universities fell by nearly a third between 2019 and 2025, to 265,000.

China, however, is feeling confident. Its leader, Xi Jinping, wants more young Americans to visit and stay for their studies. On September 24th, while at the White House, Mr Xi said he hoped 100,000 young Americans would visit China or study there in the next five years. That is a far cry from the mere 350 American students resident there in 2022, amid the covid-19 pandemic and at a low point in bilateral relations. Mr Xi says that such exchanges could create "a new generation of ambassadors" to sustain the relationship between the two powers. Visiting China "breaks down cognitive barriers", notes Xinhua, an official news agency. But does experience of a different political environment change people's thinking?

New research by Fan Yingjie of Princeton and co-authors explores whether it does among Chinese in America. The researchers enlisted almost 1,000 Chinese freshmen for their research in 2019. Nearly 400 of them were in America and the rest—a control group—at public universities in China. Over the next four years, Ms Fan and her colleagues regularly put questions to the students to gauge their political leanings, as well as what they thought of China and its government.

The work is remarkable because research that involves asking Chinese citizens in China for their views on sensitive political matters has become increasingly rare under Mr Xi's rule. The ability to compare the answers of such a group with those of Chinese students in America gives the research unusual empirical heft.

Mr Xi would surely be disappointed by the results. Ms Fan found that students' ideological views were remarkably stable throughout their time in America—there was neither a backlash against Western liberalism nor a newfound zeal for it. Yet students (and presumably their families) who decide to study in America in the first place are much more liberal than their homebody peers. They are more likely to support free markets, are less socially conservative and are less nationalistic.

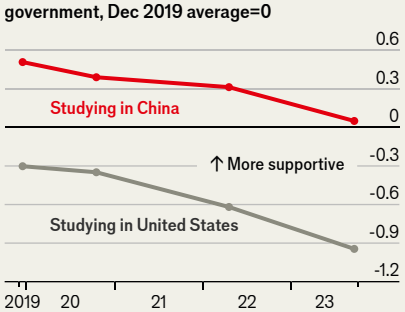
One possible reason is that Chinese women tend to outnumber men in studying in the West. Fran Martin of the University of Melbourne finds that some young women go overseas because of gender bias in China's society and job market, and so are less traditional in their outlook.

Among Chinese students in America, ideological views did not change much while abroad, says Jennifer Pan of Stanford, a co-author of Ms Fan's study. But they did update their views of how well the Communist Party is running China. Support for it started lower among those studying in America, and fell sharply during China's draconian "zero-covid" lockdowns of 2020-22. Approval for the regime, and general patriotic feeling, fell among undergraduates in China also, but more slowly. Being educated abroad, however, can highlight the West's shortcomings. Ms Martin's research shows that as life in China has become technologically advanced, offering many conveniences, some young Chinese studying in the West feel they are living "in a bit of a backwater".

Although most of the cohort followed by Ms Martin moved back to China after graduation, some have since returned to Australia after experiencing the competitive pressures of China's work culture and high youth unemployment. In the American study, the share of students planning to move home after graduation dropped from 40% in 2020 to just over 10% by 2023. As for Americans studying in China, vanishingly few stay. Study abroad may not reshape either country's politics, or even the students themselves, but it does give some Chinese more choices. ■

Declining devotion

Chinese university students*, views on China's government, Dec 2019 average=0



*987 students over whole study

Source: "Political attitudes of Chinese students at home and abroad", by Y. Fan et al., *British Journal of Political Science*, 2026

CHAGUAN
More than mooncakes

The Communist Party uses old festivals to boost its image. They may not bend to its will

GLAMPING IS NOT the most political of acts. It takes little courage to drive to a designer campsite, where white airy tents are already pitched and long wooden tables laden with food. Yet choosing to go camping at this time of year in China, when the autumn moon is full, can be political—albeit rather enjoyably so. It is evidence that traditional festivals, once forsaken by the government and neglected by many, have been re-established, with modern glosses, as fixtures of the Chinese year.

This year the mid-autumn festival fell on September 25th, making for a three-day weekend. It became a public holiday only in 2008. Not until recently did people find more ways to spend it than munching on mooncakes (dry pastries loved by some, loathed by others). Chaguan visited Zhangzhou, a city in Fujian province, in the lead-up. The old town was festooned with moon-themed decorations. One hour away, Double Fish Island, on the edge of the Taiwan Strait, was offering al fresco dinner for two for 298 yuan ($44) at its luxury campsite—all the better to admire the moon, the festival's focal point.

In the early 20th century reformers railed against the lunar calendar and its holidays, seeing them as antithetical to modernity. In 1912, after the fall of the Qing dynasty, China's republican government adopted the Gregorian calendar. The Communists went further, branding many festivals as feudal superstitions.

The return of these holidays as cornerstones of the Chinese calendar is thus impressive. In one of its more audacious about-faces, the Communist Party has morphed from sworn foe of old customs to their self-appointed arbiter and promoter. The lunar new year, cancelled during the Cultural Revolution, was restored in 1980. It was not until 2005, however, that the party issued a document calling for the revival of other holidays—especially the mid-autumn celebration, tomb-sweeping day in the spring and the dragon-boat festival in late spring or early summer.

Xi Jinping has given these festivals even greater political weight. He calls them expressions of Chinese cultural confidence—and suggests, by extension, that the party is the guarantor of Chineseness. Living "under the same moon" is a recurring propaganda theme around the mid-autumn festival. It is used to describe bonds between China's ethnic groups, shared heritage across the Taiwan Strait and the common weal of humanity. State media seize on holidays to amplify an old idea much liked by Mr Xi: that nation and family are interwoven.

Yet the revival of traditional festivals is much more than a top-down exercise. True, observance is patchy: in many cities people just enjoy the day off. But in the south, and especially in Fujian, where temples and clan halls survived the Maoist years better than elsewhere, the festivals help form the rhythm of life. Older customs are an anchor against modern anomie, and the joys (and headaches) of family reunion matter more for many than the abstract concept of national unification. "The younger generation, people born after 1990, are mostly away for their studies or work. It's only at festivals like this that they come back," said Ms Chen, standing near a shrine to Confucius in central Zhangzhou. Alas this year her son could not get home from Guangzhou, because train tickets were sold out for the long weekend.

A common criticism of the holiday revival is that it is superficial. The tongue-in-cheek nickname for the mid-autumn celebration is "mooncake festival", since eating mooncakes is the only way that many used to experience the holiday (plus, gift boxes were a handy way to bribe officials). The breadth of events now associated with the festival shows how traditions are deepening in many places, in no small part because cities want to attract more tourists. Zhangzhou hosts performances of folk characters on stilts; in Xi'an actors walk through the streets challenging visitors to poetry duels about the moon; the water town of Wuzhen sets up altars where people can leave food for Chang'e, the moon goddess (whose companion, the Jade Rabbit, can be spotted on the lunar landscape if you squint just right).

More like a moonshot, perhaps

The more that people delve into these traditions, the more that another tension may emerge—between official prescriptions for culture and the way people actually want to live. Temples, big and small, are common sites in Zhangzhou. But any expansion requires approval from religious-affairs authorities, a burdensome process. Teachings must adhere to state-controlled interpretations. And religious sites must display national flags. One temple in Zhangzhou features large signboards in its main courtyard extolling Mr Xi's accomplishments in protecting traditional culture.

His efforts entail risks. Championing holidays linked to ethnic-Han culture, as most are, could alienate minorities. For example, the lunar new year is not a traditional festival among Uyghurs in Xinjiang. Yet activists say some Uyghurs, most of whom are Muslim, have been pressured to drink alcohol and eat pork dumplings to show holiday spirit. The party also openly frets about villagers turning to temples and ancestral shrines as dispensers of temporal authority. Confucian mentors, such as one Chaguan recently met, can claim a moral power of their own. Confucianism, sometimes seen as a quasi-religion, is a statist belief system that emphasises hierarchy. As the mentor explained, a ruler does not have carte blanche. The population's deference depends on the ruler governing with benevolence. Asked whether current leaders do so, he pursed his lips and declined to answer.

The party has made itself the guardian of traditions it once belittled and even tried to crush. In so doing, it has summoned forces whose roots run much deeper than its own, and which may ultimately have greater claims on the loyalty of the people. ■

Middle East & Africa

AI and Israel's voters

The election lab

JERUSALEM

Artificial intelligence is coming for Israel's voters. Democracies elsewhere should be watching

THE FIRST meeting in a decade between an Israeli prime minister and a Palestinian president was a historic scene. The smiles looked real. But the encounter was not. Gadi Eisenkot is the front-runner in Israel's election, not the country's prime minister. And he has yet to meet Mahmoud Abbas, the Palestinians' ageing ruler.

The realistic-looking footage of the faux-summit purporting to show Mr Eisenkot embracing Mr Abbas was produced by Religious Zionism, a hard-right party led by Bezalel Smotrich, Israel's finance minister and an ally of Binyamin Netanyahu, its prime minister. It was aimed at convincing voters that Mr Eisenkot, a tough ex-general hoping to replace Mr Netanyahu in Israel's parliamentary election on October 27th, is actually a dangerous left-winger prepared to reconcile with Israel's enemies.

Fake though it was, the video still made history. On July 26th it became Israel's first campaign ad to fall foul of a law that requires all such materials created using artificial intelligence to be labelled as such. The Central Election Committee (CEC) ordered Religious Zionism to take it down.

Since then, declarations about the use of AI have become ubiquitous in the corner of every screen. "But the visible content, the fake videos are not the main concern," argues Karine Nahon, an Israeli information scientist who writes about disinformation and threats to democracy. "They're easy to spot. What's much more threatening is the way AI is being used beneath the surface to slant information and to monitor and manipulate voters." All sides are using advanced methods of AI-boosted persuasion and manipulation. Israel prides itself on the way others adopt the innovations of its arms and tech industry; could the same prove true in election battles?

Mr Netanyahu's coalition of ultranationalist and religious parties is lagging behind the centre-left in polls. But the prime minister has one key advantage: his partners are united behind him. "Mr Netanyahu and his allies have been investing in election tech for years and have highly developed databases, and the parties of his bloc share voter information," says one tech industry insider. Over the past three years the Israeli government has invested vast sums in online psychological warfare, using AI tools to try to influence audiences in the Middle East and in the West. Some are now being used domestically.

Mr Eisenkot's party, Yashar, was founded just a year ago, and the opposition is generally divided. Yet those hoping to replace Mr Netanyahu are backed by a network of anti-government protest groups deeply rooted in Israel's tech sector. "Most tech professionals support the opposition and feel they're fighting for their future in Israel," adds the insider. "We're going to see them fighting fire with fire."

Israeli tech outfits have developed swarms of bots masquerading as human

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voters. These can push conversations online towards views advantageous to whoever controls them. Bots can also be hired to vote in online polls. These are neither as statistically reliable nor as well regarded as mainstream polls but they generate headlines which can influence wavering voters.

More sophisticated are avatars, created by AI, but with actual phone numbers and even credit cards. These can express complex opinions and infiltrate groups, especially on WhatsApp, where users do not expect to encounter artificial actors.

Nothing to see here

"The most hidden aspect of the campaigns is the automation of persuasion," says Tehilla Shwartz-Altshuler, an expert on law and technology at the Israel Democracy Institute in Jerusalem. "AI is picking up every breadcrumb available on voters, including their viewing and music preferences. They're also being geolocated, to work out where they may be on election day." AI-generated messaging is being adapted for individual voters in the form of online advertisements, text messages and increasingly lifelike robocalls, supposedly from candidates. Your correspondent recently answered one such call and only realised the politician talking to him was an AI agent after a couple of sentences.

Politicians expect many Israelis, especially younger ones, to consult chatbots about who to vote for. "We're seeing inauthentic networks creating a wide range of digital assets designed to manipulate the algorithms of chatbots and AI-powered search engines," says Nir Rozen, the head of FakeReporter, an NGO fighting disinformation. "Some of this content includes fake interviews with politicians, but there's genuine information mixed in as well, flooding the information sources that voters rely on." Parties are scrambling to publish content on platforms that chatbots draw on heavily. "We've been posting all our old stuff on Reddit because ChatGPT uses it," says one political activist.

It is not clear whose job it is to stop any of this. The internal security service, the Shin Bet, appears to be willing only to try to prevent foreign meddling, though its mandate is a broad one of protecting Israel's fragile democracy. Nor does the National Cyber Directorate in the prime minister's office seem prepared to get involved. That leaves the police, with its limited resources, and the CEC, with its few powers, to patrol this wild new frontier.

For now the committee's most powerful tool is its relationships with the main internet companies. They generally heed the CEC's demands to remove dodgy content. Many big international tech firms have research centres in Israel. They are aware that parties and campaigners elsewhere will study this election for lessons.

Government officials and privacy activists claim that the tech companies have so far been quick to deal with issues they flag. But doing so on election day itself, in time to prevent any false reports or manipulation from affecting voting, will be harder. Much of the information being illicitly collected is for get-out-the-vote operations on election day, which is a public holiday. Knowing who their voters might be and what they might be doing on their day off is a key advantage for political parties. Knowing who isn't likely to vote for them can also be useful; such voters can be sent offers for shops or entertainment venues far from their polling station, or fake news about disturbances in their area.

The Supreme Court and the chairman of the CEC have ruled that the parties' observers at the polling stations will not be permitted to tell their bosses who has already voted. In previous elections this has been allowed and has been central to tracking voters on election day and getting supporters to the ballot box. Regulating such activities is tricky. "There's a very narrow line between freedom of expression and censorship, and we exercise discretion," says Dean Livne, the acting director-general of the CEC. "I don't think we're at the stage yet where we have to get involved with AI engines." At the rate AI is evolving, that may have to change soon. ■

Hormuz flows

Reversion to the mine

Has Iran lost its grip on Gulf oil exports?

On September 26th Donald Trump rejected an Iranian ceasefire proposal that would have reopened the Strait of Hormuz within seven days. Iran wants a deal because it is "losing so badly," said the president. Mr Trump and many of his advisers reckon that America has more to gain by waiting.

The volume of oil coming out of the Gulf would seem to prove him right. Kpler, a shiptracker, estimates that the region's Arab states have shipped, on average, 17m barrels per day (bpd) of crude oil and refined products so far in September, if flows through Hormuz and those bypassing it via pipelines are included. That is the highest volume since February and close to the pre-war daily total of 21m. Iran, by contrast, is exporting just 200,000 bpd, less than a tenth of what it shipped before the conflict began. From Washington, 7,000 miles away, Iran's chokehold on Gulf oil appears to be loosening.

Staring down the barrel

Spot oil price*, 2026, $ per barrel

Source: Platts *Dated Brent

But is that right? Brent crude, the global benchmark, remains stuck at around $100 a barrel—around 40% higher than in June, when America and Iran sealed a (temporary) deal, even though export volumes were a quarter lower then. The problem is not eye-watering freight costs: rates for chartering tankers are at record levels, but Brent futures are quoted "free-on-board", reflecting the price of crude at loading, not delivery. Instead, three explanations stand out: crude supply still lags behind demand; the prospect of an American diesel-export ban is distorting markets; and fears are growing that Iran may attack ships or energy infrastructure again.

Take the supply-demand imbalance first. Flows through Hormuz have swelled because Saudi Arabia is pushing many more barrels through the conduit than it was earlier in the war. On September 28th Saudi Aramco invited Chinese refiners to bid for cargo loading next month, having already sold more than 90m barrels in two rounds in the previous two weeks. Most Saudi oil now moves under American escort, on state-owned tankers that shuttle through the strait, before being transferred to other vessels in the Gulf of Oman. The United Arab Emirates uses similar tactics.

State-owned ships resist pressure better than private ones, so flows have not stopped even as Iran is sounding more bellicose again. But Saudi officials themselves admit the ship-to-ship scheme is costly, clunky and unsustainable. And Gulf supply remains 4m bpd short of pre-war levels. The world feared it could face a similar shortfall after Russia invaded Ukraine and Brent peaked at $129.

At the same time, global demand for crude has picked up. China, having slashed its crude purchases by nearly 40% over the summer, is thirsty for oil again. Domestic stocks of jet fuel, diesel and petrol have declined so fast that wholesale fuel prices now exceed retail ones (which are capped by the government). So refiners are working harder to satisfy the home market. Until China appeared to reimpose export

curbs on October 1st, they had also been importing more only to export it, since a global fuel crunch promised fat profits abroad. Import-for-export purchases were probably higher in recent weeks in the expectation of curbs.

China is also having to replace the oil it used to buy from Iran. That is forcing it to buy more crude on the open market, pushing up official prices. The cost of the kind of oil that is most similar to that produced by Iran has risen even more than Brent: Dubai futures are trading at $116 a barrel. But China is also competing with European buyers for North Sea-like grades, lifting Brent in the process, too.

The upshot is that global crude supply still falls short of demand. Dated Brent, which tracks oil loading in the next few weeks, hovers at around $120 a barrel (see chart). It is normally within a few dollars of Brent futures; the unusually wide gap points to a crunch in spot markets.

Brent is also being lifted by shortages in another market: diesel. Limited Gulf flows and an export ban in Russia have pushed wholesale diesel prices to nearly $200 in major markets. American pump prices remain close to $6.50 a gallon, a record, angering truckers and farmers weeks before the midterms. Mr Trump says he favours restricting exports of the fuel, large volumes of which have gone to Europe since the war began. That would cut demand for American crude, since refiners there would process less of it, while lifting demand for other grades as European refiners sought to produce more diesel. That prospect has already pushed Brent $12 above West Texas Intermediate, the American benchmark.

That leaves the third risk: Iran could lash out again. Its hardline rulers must resent seeing Hormuz traffic escape their control. Three attacks on tankers were reported on September 30th but Iran has not disabled one since September 23rd. But that could change. Unverified reports suggest Iran has laid new mines in the strait, or is preparing to. Windward, a shipping-intelligence firm, detected 75 high-speed craft, capable of deploying drifting ordnance, swarming off the Omani coast on September 26th. Iran retains the capacity to strike Gulf oilfields and energy facilities. Saudi Arabia's East-West pipeline, hit in early September and partially repaired, remains vulnerable. So are Saudi ports. And the Houthis, Iran-backed rebels in Yemen, could further limit traffic through Bab al-Mandab, a vital Red Sea shipping route.

Greater exports do not mean Hormuz has become safer to traverse; they suggest only that the oil industry has adapted to operating under sustained risk. That risk may yet materialise and throttle flows again. Past Hormuz performance is no guarantee of future results—especially if Iran feels cornered. ■

Gaza

Inhabiting the uninhabitable

For Gazans, there is still no day after

A YEAR AGO Donald Trump set up his Board of Peace (BoP), a private club of world leaders. The day that he announced his plan for redevelopment in Gaza, to be overseen by the board, was, he claimed, "potentially one of the greatest days ever in civilisation". The outfit has spent a year carrying out "intense preparation", according to Tony Blair, a former British prime minister and member of the board's executive. Now comes "the implementation".

On September 23rd on the sidelines of the UN General Assembly Jared Kushner, Mr Trump's son-in-law and one of his main fixers, gave a sense of what the board hopes to spend on that implementation. He announced a $2.45bn recovery plan and talked of $71bn for a decade of reconstruction. The board would oversee 66 projects over the next six months alone, he declared. Power-point presentations promised a "connected, prosperous Gaza delivering Security, Dignity and Opportunity".

Such promises ring hollow in Gaza. Israel has depopulated the nearly two-thirds of Gaza it occupies. Military outposts dot the strip. Herds of bulldozers level the rubble, erasing property boundaries and evidence of what some have judged to be war crimes. The first contours are emerging of what America says will be "alternative safe communities" for vetted Palestinians. Safe communities, reply hardline Israeli ministers, for Jewish settlers.

Israel bombs tents and the few houses still standing almost daily. It says it is pursuing the remaining leaders of Hamas, the Islamist group that broke out of Gaza and massacred Israelis three years ago. Over 1,400 people have been killed in the year since the ceasefire. Forget building new shelters ahead of winter. Heavy machinery to clear 68m tonnes of debris and to repair schools, hospitals, electricity and sanitation is banned on grounds of "dual use". Even tent poles are forbidden.

Hamas remains the de facto authority. From bombed-out buildings it is still registering births and collecting sales taxes. But its hold is strained. In September Israel killed commanders in four of Gaza's five governorates. To avoid being spotted by Israeli troops, police wear plain clothes.

Some warn of a descent into *al-falatan al-amni*, the security chaos that preceded the collapse of the Palestinian Authority in Gaza in the mid-2000s. Palestinian militias, backed by Israel, prey on Hamas. As the group's rule frays, clans have returned to the streets, brandishing their weapons. Squabbles about jumping queues for bread and water end in gun battles. Mugging, once a rarity, is rife. Unable to provide for Gaza's mass of orphans, relatives sell children for marriage. Some wives are as young as 12, said a recent UN report. One man, said researchers, had married off his young daughter for two sacks of flour. "Gazans have lost our kindness," says a Hamas civil servant. "We've grown cruel."

In a warehouse the board set up a year ago in Kiryat Gat, an Israeli city 20km from Gaza, military personnel, spooks, diplomats and aid workers co-ordinate Gaza's reconstruction with Israel. But the "initial momentum has stagnated", says a Western official who spent months there. For all Mr Kushner's talk of billions, funding for the

Where the day after never comes

board's projects has fallen far short as Gulf donors have prioritised needs at home.

But the real obstacle, say officials working in Kiryat Gat, is Israel. The number of aid agencies in Gaza has fallen. The troops offered by five countries for the International Stabilisation Force to monitor the truce and oversee Israel's withdrawal have not been allowed to enter. The National Committee for the Administration of Gaza, the Palestinian body appointed by the board to replace Hamas, is stuck in Egypt, as are 450 vetted Palestinian police officers. The board's biggest effort to tackle hunger has been the canteen providing free buffets to staff in Kiryat Gat, quip officials, at a cost of $9m in six months (the price of about 1,800 community kitchens in Gaza). "The BoP is always pandering to the Israelis," says a Western official who visits frequently.

Some hope for change after Israel's election on October 27th. But even if he is defeated Mr Netanyahu may cling on for months as caretaker, as Israel's fragmented political parties jostle over forming a governing coalition. Moreover, the result might be another lurch to the right. Calls for "voluntary migration" of Palestinians, an Israeli euphemism for expulsion, are growing. One candidate, Moshe Feiglin, has said that every baby in Gaza is an enemy. Even more centrist contenders for prime minister share Mr Netanyahu's view that Hamas must still be crushed.

Israel remains determined to do this militarily. In August Nickolay Mladenov, the board's "high representative" in Gaza, persuaded the Islamists to agree to hand over their weapons in exchange for an Israeli withdrawal. Much in his plan was unclear. Would Hamas disarm before, in tandem with or after an Israeli withdrawal? Would its Palestinian adversaries disarm? Would Hamas give up light weapons as well as heavy ones? But Hamas was spared hard choices. Mr Netanyahu rejected the plan and a troop withdrawal. "Even if Hamas fully disarms, Israel would still claim it has a penknife," says a European official.

Others are still trying to drag Hamas onto a political path. Given a green light from Mr Kushner, who has met its leaders, Muhammad Dahlan, a former warlord from Gaza who fought Hamas, is leading talks with the group in Cairo. His negotiators say he has persuaded Hamas to accept terms set by Mahmoud Abbas, the Palestinian president, for taking part in Palestinian elections in November. "We've been successful in changing Hamas's minds," claims Osama al-Fara, a former mayor in Gaza and an ally of Mr Dahlan. These include, he says, recognising Israel and accepting previous Palestinian agreements with it. This time, however, the spoiler may not be Israel. Fearing defeat, Mr Abbas is expected to cancel the elections. ■

Nigerian politics

Advantage Tinubu

MAIDUGURI

Nigerians dislike their president, but may re-elect him anyway

With just over three months to go until Nigerians vote for a president, the incumbent does not look particularly worried about his re-election prospects. On September 29th Bola Tinubu (pictured), who expects to win a second term in January, returned from Paris after a month-long stay in Europe. He had previously extended his "working holiday" by several days, raising eyebrows at home.

On the face of it, Mr Tinubu's cool is surprising. Nigeria is suffering a worsening security crisis, with people kidnapped and killed by bandits or insurgents on a near-daily basis. His economic reforms, though lauded by investors, have yet to make life better for most of the country's 240m people. Add to that political challengers who are more popular than him, especially in the northern states worst affected by violence and economic deprivation, and the president should be quaking in his suede loafers. That he is not says less about him and more about Nigeria's complicated electoral dynamics.

To understand the misery over which Mr Tinubu and his ruling All Progressives Congress (APC) have been presiding, consider Borno, in Nigeria's north-east. The poor state, which the APC also runs, is in the grip of an insurgency that has raged unchecked for two decades. Jihadists fight both the government and each other. Rival factions make money by kidnapping schoolchildren and extorting or looting cash and grain from already poor farmers. Such violence has wrecked the economy, leaving young people jobless and traders without customers. One 27-year-old in Maiduguri, the capital, complains that he is forced to travel to neighbouring Chad to sell his grain, as locals cannot afford it. Youngsters "pack all their things and leave Borno state" for Chad, Cameroon or Ghana, he says, "because if you sit down here, there is no food in this country."

Mr Tinubu's re-election pitch is focused on reforms, such as the abolition of a pricey fuel subsidy, that have brought a degree of stability to Nigeria. Inflation has fallen, government revenue is rising and investment is picking up. But in Borno the reforms—combined with a lean harvest, aid cuts and rising energy prices following the wars in Ukraine and Iran—have made life harder. The annual inflation rate in August was still around 15%, with food inflation closer to 20%. Few people can afford enough to eat, prompting a hunger crisis in Borno and in neighbouring states (see map). The UN's World Food Programme reckons malnutrition in the state is at its worst level for a decade.

Dr Nura Tukur, who runs a clinic for the starving operated by Save the Children, a charity, says there has been a sharp increase in the number of malnourished women and babies admitted over the past three years. July was the worst month since the clinic opened in 2016. Aisha Umara, who arrived several months ago with two starving triplets (one had died on the way), is now healthy enough to breastfeed. But her two six-month-olds still weigh just 2.5kg each. They are unlikely to survive being discharged to a home with no food in a village ruled by jihadists.

All this has hurt Mr Tinubu's reputation

Cool customer

in the north, where his party has traditionally done well, even though he personally lacks a base there. In the election in 2023 he won 54% of the vote in Borno. But recent polling by SBM Intelligence, a risk-advisory firm, suggests he is now less popular in the state than Peter Obi, another presidential candidate from the south, and Atiku Abubakar, a candidate with a big base in the north-east.

Some of Maiduguri's residents say they are prepared to "fight back" at the ballot box in January. Yet in general, Mr Tinubu's unpopularity will probably not be enough to dislodge him from the presidency. Incumbents, with access to the ruling-party machinery and plentiful cash to dole out goodies, have tended to have an advantage in recent Nigerian elections. It helps that the opposition is splintered and disorganised. Mr Obi has failed to form a coalition with Mr Abubakar. "Because it's not a united ticket, that anti-APC vote is not coalesced into a single vote," says Joachim MacEbong of Control Risks, a consultancy. In Maiduguri some people say they do not even know who else is on the ballot besides Mr Tinubu.

No need to deliver

It also helps that Mr Tinubu and his vice-presidential candidate (who is a former governor of Borno state) are both Muslim, as are most people in the north. "The fact that the president has stuck with the Muslim-Muslim ticket resonates with quite a number of people," says Cheta Nwanze, the boss of SBM Intelligence. Even voters who like Mr Obi may rule him out because he is a Catholic; the fact that he has a Muslim running-mate may not be enough.

Perhaps most significantly, voters in the north may not be too concerned by Mr Tinubu's record because they already expect little from the government. "Many people are living where there has been no real security, no real health care, no real schools. Over time, people's expectations get lower and lower," says Mr MacEbong. One mother in Maiduguri says she will vote for Mr Tinubu as a "thank you" for the medicine at the government hospital. With their home state badly governed for so long, many people are more likely to trust God and their families to improve their lives than a government that rarely has.

Such cynicism is not limited to Borno. As long as the security crisis remains unresolved and economic growth, which is expected to hit 4.1% this year, does not translate into concrete improvements in Nigerians' daily lives, many voters will remain disappointed. Yet few will use the ballot box to express their disappointment. In Borno, where a journey to the polling station could easily entail an encounter with militants, most will probably not bother voting at all. In 2023 turnout in the state was below 20%, even less than the dismal 27% nationwide. In parts of the south, especially Mr Tinubu's home of Lagos, plenty may swallow whatever bad feelings they have about their day-to-day struggles and vote for their "son". No wonder Mr Tinubu feels safe enough to take extended holidays in Paris. ■

Global development

Artificial altruism

AI firms are becoming a new class of donor in Africa

AT ANTHROPIC, THE promise and peril of artificial intelligence (AI) are a family affair. Dario Amodei, chief executive of the AI firm in San Francisco that developed Claude, recently warned that superintelligence could wipe out humanity. Daniela, his sister and the firm's president, is instead excited about how AI could save lives. She tells *The Economist* that global health is becoming a focus for Anthropic, which is working with aid organisations across Africa to use Claude in health care and to forecast food insecurity.

For now, such efforts remain small. AI companies have announced development efforts worth a few hundred million dollars this year—a rounding error in global aid spending. But it is arriving as traditional aid retreats. Rich countries cut development assistance by 23% last year, to $174bn. Nan Ransohoff, who works on public goods at Stripe, a payments company, estimates that the OpenAI Foundation, Anthropic's founders and its employees could eventually account for $37bn-100bn in charitable spending each year. Some of that will go to poor countries. How will the new money help development?

AI firms may offer technology as well as money. Anthropic, for example, is working with the International Rescue Committee (IRC) and the UN's World Food Programme, providing access to Claude and engineering support. In Nigeria, a project using AI has cut the time nursing staff spend on some tasks by around 60% by helping synthesise unstructured patient data, according to David Miliband, the IRC's head. Such productivity gains matter more as aid cuts squeeze capacity: the organisation has had to close more than 500 health centres over the past year.

But past efforts to use technology in aid offer a warning that cheaper access to expertise is not enough. A scheme that handed laptops to children in poor countries boosted computer use without improving test scores because the devices were poorly integrated into curriculums. Digital-health tools have often worked in trials but failed to survive in local health systems.

Donors also tend to see the world through their own expertise. Norway, a country with many fishermen, once poured aid into fisheries in Kenya, only to build a frozen-fish plant that never opened. AI donors may see every problem as a job for AI, even if the real constraints lie elsewhere.

Take malaria, which kills about 600,000 people a year, mostly in Africa. Two effective vaccines are available, and 25 African countries have introduced them into routine childhood immunisation. But Gavi, the global vaccine alliance, faces a funding shortfall of nearly 30% for the programme. AI models could help target vaccination campaigns. But AI tokens cannot pay for vaccines or get them into children's arms.

Even where AI tackles an important constraint, donors must consider what else limits progress. In cancer screening, for instance, AI could make it cheaper to identify patients at risk, but that would increase demand for scarce lab work and treatment. In Kenya more than half of surveyed hospitals offered screenings for cervical cancer in 2022, but only around 5% of them could treat it. AI philanthropists should think about funding the things that are needed to put AI-driven advances to use.

The dismantling of USAID in 2025 showed how vulnerable services are when they depend heavily on an outside donor. AI philanthropy could create dependence through free model credits and foreign engineers. When the support stops, "Who owns the data and the code, and can the local team keep running the thing?" asks Leo Anthony Celi, a researcher at the Massachusetts Institute of Technology.

Ultimately, what matters is local control. Mr Miliband points to an IRC education programme that became sustainable when Iraq, Jordan and Lebanon incorporated it into national curriculums. Similarly, the best AI-driven projects will eventually make the donor—and perhaps even the donor's model—dispensable. ■

Europe

MAGA and Europe

A hate-love relationship

AMSTERDAM, LEESBURG, VA, AND WASHINGTON, DC

The roots of America's new aggression towards the Old World

EUROPE'S CHURCHES may be empty, but to judge by the Cornerstone Chapel in Leesburg, Virginia, many in America are doing all right. The 2,000-seat church holds three services on Sunday morning, each so packed that police are needed to direct traffic. After some Christian rock, pastor Gary Hamrick urges the congregation to vote against gay and abortion rights. Then he turns to his sermon, on sin and redemption in Colossians 3: "You have put off the old man with his deeds."

For many in the pews, it is not just the old man but the Old World that has a problem. Europe is going "very cold against the Lord", says Tyler Hamrick, the pastor's son. "Satan has definitely—a veil has been brought over the eyes of the Europeans." Worshippers worry about Muslim immigration in France and Germany. Matt Denby, a missionary, says Christians see European secularism as a warning: "Same-sex marriage and abortion and transgender, Europe's already gone through those...Is that now going to happen here?"

Evangelical Christians are the heart of the Republican Party's base, and their views help to explain America's hostility towards Europe during Donald Trump's second term. The bellicosity leaves many Europeans baffled. America's latest national security strategy, released last November, castigated Europeans for embracing "civilisational erasure" by over-regulating the economy and allowing mass immigration. Disputes over defence spending levels are one thing, but such resentment goes deeper. Where does it come from?

Recriminations against Europe among the New Right, as MAGA intellectuals are called, fall into three overlapping baskets. The most straightforward are geopolitical, says Nathan Pinkoski of the Centre for Renewing America, a conservative think-tank in Washington. The New Right's foreign-policy realists think America should move away from cold war-style global alliances, prioritise its own interests and shift the burden of Europe's defence to Europe. Some focus on China, others on American hegemony in the western hemisphere.

Claims of "civilisational erasure" belong to a second, different vision, rooted in national-conservative ideas about cultural heritage and the evils of globalism. Michael Anton, a writer at the right-wing Claremont Institute, is believed to have coined the phrase, which evokes an embattled white Christian culture. MAGA influencers tell Americans fanciful stories of a Europe overrun by Muslims and crime. Mass migration in Britain "is a history of ▸▸

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civil tension, punctuated by violence: riots, terrorism, murder, rape", wrote Christopher Rufo, a MAGA activist, last year.

The selling of this narrative is a rare recent example of transatlantic co-operation. Elon Musk has used X, his social-media platform, to promote racist activists in Britain and Austria and the populist-right Alternative for Germany. In fact Muslims make up perhaps 6-10% of the population in western Europe. Even big cities such as London and Paris, where the Muslim share of the population is closer to 15%, are no less safe than American ones.

Beyond the bigoted overtones, civilisational fear-mongering frustrates realists who dislike moral narratives. "You cannot be a nationalist and a civilisationalist at the same time," argues Sumantra Maitra of the Institute for Peace and Diplomacy, a realist think-tank in Washington. Realists see civilisationalist support for populists in Germany and Poland as dragging America back into ideological conflicts. Yet both happily bash the European Union. Colin Dueck of the American Enterprise Institute, a non-MAGA conservative think-tank, says the bloc makes a perfect target for MAGA: "a gigantic, meddling, progressive multilateral organisation which is unhelpful to us militarily. I mean, you could just go down the list of vices."

The EU's regulations lead to a third set of criticisms, mixing ideology and commerce. Over the past year the Trump administration has attacked the bloc's Digital Services Act (DSA) for restricting freedom of expression on American tech platforms. Sarah Rogers, an undersecretary of state, has blasted European countries (sometimes misleadingly) for prosecuting social-media users. The judiciary committee of the House of Representatives has held hearings on the DSA's "threat to American speech and innovation".

"We believe in free expression, that's one of the hallmarks of Western culture," says Jim Jordan, the committee's chair. "We don't like it that Europe is taking down American tech companies either." MAGA critics highlight prosecutions in Europe over posting anti-gay Bible verses, but business interests play a role too. Last week the Trump administration asked to join Mr Musk's case at the European Court of Justice against a €120m ($137m) fine on X for violating the DSA.

Does MAGA want a Europe of mutually non-interfering states, or one where American values rule? The confusion is inevitable because European heritage is central to the movement's idea of America. It has a "complicated love-hate relationship to Europe", says Laura K. Field, whose book "Furious Minds" is a sophisticated treatment of New Right thought. Like 19th-century Americans who imported European culture, she says, many New Right thinkers have fallen in love with a vision of Europe. But "they also think they can school the Europeans, that Europe is not being true to itself and is losing its traditions."

Many MAGA supporters think of their criticisms of Europe as tough love. "The civilisational [New Right thinking] is actually born out of deep affection for Europe. It's a concern about the direction the continent has been headed," says Mr Pinkoski. Tucker Carlson, a long-time conservative commentator, told *The Economist* in June that Europe and America were natural allies "on the deepest possible level".

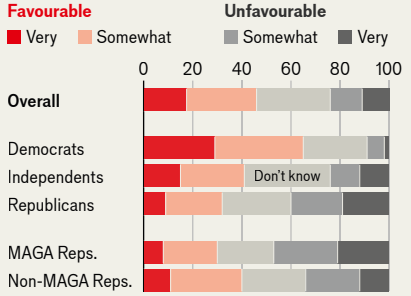
The Trump administration's flashes of contempt for Europe do not necessarily reflect MAGA voters. A poll in July by YouGov found that 32% of Republicans had favourable views of the EU and 40% unfavourable; among MAGA Republicans it was 30% to 44%. That was much lower than among Democrats (62% to 9%), but not catastrophic. Friendliness towards individual European countries differed only modestly between Democrats and MAGA backers.

Europe could conceivably use such goodwill to build an alliance with a post-Trump America. Old-fashioned internationalists among Congress's Republicans certainly think so. The truly isolationist element of MAGA is "a small percentage", insists Joe Wilson, a pro-NATO Republican congressman from South Carolina. His state, he says, "appreciates its European heritage" and trade links.

But it seems more likely that the paths of Europe and America will continue to diverge. Some Americans, misled by social media, imagine a Europe dominated by Islam that most Europeans do not recognise. Others in America dislike the EU as an international regulatory institution. MAGA voters often see Europeans, with their secularism and generous social-welfare policies, as exaggerated versions of America's Democrats. When they worry about going "the way of Europe", they are partly projecting their own fears. There is not much Europe can do about those. ■

Neither cold nor hot

"Do you have a favourable or unfavourable opinion of the European Union?", % responding
United States, July 10th-13th 2026



Source: *The Economist*/YouGov

French finances

Risk, on

PARIS

Bond markets are punishing profligacy

FRANCE'S MESSY politics are spooking the bond markets. On September 29th the yield on ten-year bonds rose to 4.8%, its highest level since 2008. The surge in borrowing costs pushed the spread between French ten-year bonds and German Bunds up by 50 basis points in September, to 128 points—the highest level for 14 years. France now pays more to borrow than Greece and Italy, once the euro zone's most worrisome members. It is increasingly looking like the currency area's biggest problem, and things could get worse.

France has no difficulty issuing sovereign debt. But the steep rise in yields reflects two particular worries. The first is that France's minority government may struggle to control the deficit in the budget for 2027, which was unveiled on October 1st. The second is that the prospect of fiscal consolidation is becoming ever more remote, with a presidential election next spring that may be won by a populist who would govern still more irresponsibly.

In his budget Sébastien Lecornu, the centrist prime minister, promised €54bn ($61bn) of savings. These will come from such measures as a freeze on higher-rate pensions. He is extending for a third year a surcharge on big companies (initially presented as a one-off), albeit at a reduced rate. Almost all departments face a squeeze. Only spending on defence has been ring-fenced.

Roland Lescure, the finance minister, described such plans as "ambitious". Yet the savings promised will merely keep the deficit at 5% of GDP, well over the European Commission's 3% cap. It will be the fifth consecutive year that France's deficit has been at or over 5%. Moreover, there is no guarantee that, as the budget goes through parliament in the coming months, the fragmented lower house and rivalrous opposition parties will agree to even these limited savings. Marine Le Pen, leader of the populist-right National Rally and the polling favourite for the presidential election, says her party will vote against any attempt to touch pension payments.

The government has been boxed in partly by disappointing growth. The budget for 2026 was based on expected GDP growth of 1%, and a budget deficit of 5% of GDP. In September, however, the government halved its growth forecast and conceded that the deficit would be closer to 5.4% of GDP. Rising borrowing costs have

aggravated matters. The government expects debt-servicing costs to reach €91bn in 2027, up from €79bn this year.

The prospect of near-term fiscal consolidation is in shreds. France had originally promised to respect the deficit rules by 2029. That commitment "no longer looks credible", note analysts at ING, a bank. Nor will the current fiscal effort be enough to stabilise public debt. It is forecast to rise from 119% of GDP in 2026 to 122% in 2027.

The run-up to the two-round presidential election is unlikely to calm matters. Polls suggest the run-off could be a choice between the two extremes: Ms Le Pen on the populist right and Jean-Luc Mélenchon on the populist left. Neither has credible fiscal plans. Ms Le Pen promises to fix the public finances, but her remedies, such as halving France's contribution to the European Union budget, are impractical. Mr Mélenchon's suggestion for solving the debt problem is to "burn" part of it.

Mr Lecornu has no good options. If he gives ground on his spending cuts in the coming weeks, he will struggle to keep even to his own deficit target. If he tries to force the budget through without a vote, his government could be toppled—as those of his two predecessors were. Eurasia Group, a risk consultancy, currently puts at 35% the probability that the government will fall this autumn. "Such an accumulation of political and budgetary risk raises fears that the risk premium on France may rise further," wrote Bruno Cavalier, chief economist at Oddo BHF, an investment firm in Paris, in a research note.

If the government fails to get its budget through parliament by the end of the year, the spread over German Bunds could widen further. The current rise in interest rates seems to be a rational repricing of French fiscal risk, but a higher spread may make investors worry about the euro zone more broadly. Analysts reckon that the European Central Bank has no intention of intervening to help France as long as the spread over Bunds is below two percentage points. But that would still leave France's government with painfully high rates, even if voters seem wilfully unbothered. ■

Ukraine and Russia

Red October

KYIV

Russia's terrorist campaign aims to cripple Kyiv even before winter hits

IN THE RUSSIAN revolution Vladimir Lenin went for high-level targets: the telephone exchange, telegraph, railway stations and bridges. Vladimir Putin is setting his sights lower. Over the past week his drones have hit supermarkets, data centres and playgrounds. On Monday they struck a hospital and Ukraine's National Academy of Sciences in central Kyiv, killing at least two. Survivors dangled from the burning academy building's windows.

The escalation of terror against civilians was no surprise to Ukrainian intelligence services. Days before the latest wave of attacks, two senior officials told *The Economist* that Russia was planning a three-pronged campaign of bombing, sabotage and political chaos against the capital. One identified data centres and supermarkets as targets. "Putin has gone even crazier," that official said in mid-September. "They want people to be afraid of going to the supermarket." Both officials expected the campaign to intensify, with efforts to rock Kyiv peaking by November.

The Kremlin has not abandoned its winter playbook, targeting electricity, heating, water and sewerage, in particular pumping systems. Russia's most destructive threat remains its ballistic missiles, made more dangerous by the dwindling of Ukraine's American-made Patriot interceptors. The biggest shift is a new generation of pilot-guided jet-powered drones, able to hunt their targets in real time. Ukraine is still months away from mass-producing effective interceptors. A security official says Russia has "thousands" of drones stockpiled, and is building up its ballistic missiles, with plans "to use the entire stock in the next two months".

Putin's way of war

Officials think the escalation of bombing will be accompanied by hybrid attacks inside Ukraine, including more assassinations and terrorist attacks. Counter-intelligence services say they are foiling seven or eight Russian plots a week. Most are arms-length operations, with agents recruited inside Ukraine via messenger apps like Telegram. But others involve operatives sent from Russia. The Ukrainians eliminated many Russian agents in the first weeks of the invasion, but officials believe some remain. "Some of the cells are still frozen, 100%," says an intelligence source.

The third prong, disinformation and sowing chaos, is hard to gauge. Ukrainians have fine filters against Russian propaganda. Yet after years of war, ideological mobilisation is sometimes giving way to fatigue. Corruption scandals, disorganisation and nepotism in government give Russian bot farms plenty to work with. A breakdown in daily life could get Ukrainians to turn on each other, warns one official. But others are sceptical. The Russians "hope in vain that their bot farms will make our people protest and demand surrender. It's nonsense," says Andriy Kovalenko, head of the National Security Council's counter-disinformation operation.

Mr Kovalenko says there is a problem with Russia's plan: Kyiv is harder to turn off than the Kremlin thinks. The city's internet system, battle-tested before the invasion, is too decentralised for a handful of strikes to shut it down easily. The same is probably true of power, says Olga Babiy, an energy expert. Years of war have eroded Ukraine's reserve capacity, but they have also given its repair crews experience.

For now, life in the capital carries on. Massive explosions prompt café-goers to glance at the sky before returning to their lattes. But there are signs that Kyiv is emptying quicker than normal in this season. Country houses around the capital are nearly impossible to rent for the winter; the wealthy have reserved them as boltholes. Early in the war refugees swelled Kyiv's population from 3.5m to 3.7m or more, but that has since fallen to about 3.3m. In private, officials say that this winter they expect parts of the city will depopulate.

Ukraine has repeatedly proved more resilient than its enemies expected. On the front lines its troops are reporting tactical successes, such as a counter-offensive near the town of Lyman that has reversed a year's worth of Russian gains. But as Lenin understood, Ukrainians would be wise to pay closest attention to the capital. "Things usually don't go according to the worst-case scenario," reasons the security source. "But it will be very hard." ■

EU sanctions

Death metal

Ireland is supplying alumina to Russia's war machine

THE IRISH know a few things about resisting an imperialist neighbour, so it is not surprising that they back Ukraine against Russia. With no significant armed forces, the country's aid to Ukraine consists chiefly of a bit of financial help and support for European Union sanctions. Yet on the latter, there is a caveat. Aughinish Alumina, a refinery in Ireland, supplies a share of the alumina used by Russia's aluminium industry. The skins of missiles striking civilians in Kyiv may be part-Irish.

The role of Aughinish in Russia's arms industry was detailed in March in an investigation by OCCRP, an investigative outfit, and the *Irish Times*. The plant exports alumina (aluminium oxide) to Russian smelters. But it also ships to manufacturers in France, the Netherlands and Sweden. The firm has never been subjected to the EU's sanctions on Russia. Some member states want it included in a new round this month. Aughinish's defenders argue that this would cripple Europe's aluminium sector, but they are wrong.

Aughinish is owned by Rusal, a Russian metals conglomerate. Its sales to Russia have surged from 24% of its exports in 2021 to 68% in 2025. The alumina travels to smelters in Siberia, which sell the finished aluminium to customers such as ASK, a Moscow-based trader. ASK in turn supplies over 100 Russian defence companies. Its aluminium has gone to factories that make gyroscopes for Kh-101 cruise missiles, engines for Sukhoi fighter jets and the Iskander-M ballistic missile. Many of these firms are under EU sanctions.

Rusal was founded by Oleg Deripaska, a Russian oligarch. He has been on the EU's sanctions list since 2022 because of his close ties to the Kremlin. But since 2018 he has held only a minority stake in Rusal, allowing the refinery to remain sanctions-free. Aughinish denies any wrongdoing.

The European Commission designs its sanctions to avoid harming Europe's economy more than Russia's. Aughinish says it should remain exempt on this basis. In a letter to the Irish government seen by *The Economist*, the firm argues that without its sales to Russia, the plant would have to close, undermining Europe's aluminium industry and other manufacturers. Yet the EU imports most of its aluminium, and produces at most a few percent of global supply. Its production has shrunk dramatically due to Chinese imports, outdated tech, high power costs and the expense of permits under the EU's Emissions Trading Scheme. Europe's aluminium smelters are operating at just 30% of capacity.

European smelters need a couple of megatonnes of alumina annually. So far this year, Aughinish has supplied them with only about 0.3 megatonnes. If the plant closes, they could find alumina elsewhere; four other refineries in Europe export little or no alumina to Russia. It might hurt Russia more. Ireland was the third-biggest supplier of alumina to Russia in 2024, behind China and Indonesia.

Sanctions would be a political headache for the Irish government. The plant employs around 1,000 workers. It sells the excess electricity from its gas-fired power plant to the national grid, providing enough power for some 200,000 households, the company says. But exempting it is a political problem, too. Ireland currently holds the rotating presidency of the EU Council, the body where representatives of national governments sit, and lobbying for a plant that supplies Russia is a bad look.

Negotiations over the next package of sanctions will begin soon, and several states want to ban alumina exports to Russia. "It is complicated and not straightforward," says an EU official. Barring Aughinish from selling to Russia would have no great effect on Europe. But Ireland's tarnished reputation would be a bit shinier. ■

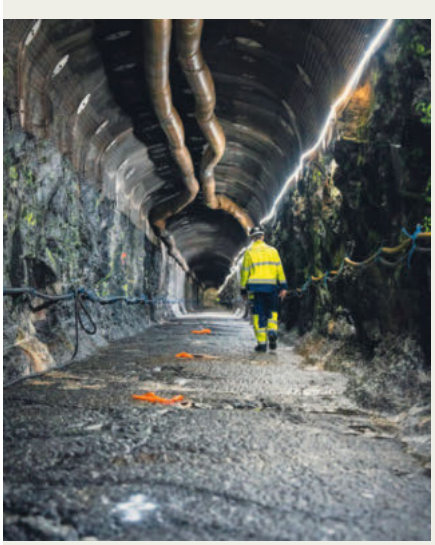
Nuclear waste

100,000 years of solitude

Finland is drilling the world's first permanent nuclear-waste dump

AFTER FOUR and a half decades of planning, it took only ten hours to drill through the 1.9bn-year-old bedrock. Final digging began last month at Onkalo, a geological disposal facility (GDF) in Finland. It will be the resting place for 6,500 tonnes of hazardous nuclear waste. Sealed 400 metres beneath the earth, its radiation will remain contained for over 100,000 years.

Since the infancy of nuclear power, scientists have debated how to safely dispose of its waste. GDFs are the gold standard, and Onkalo will be the first on Earth. Posiva, the firm that runs it, aims to have it ready by year's end. It comes at a time of renewed enthusiasm for the nuclear industry, driven by the desire for carbon-free electricity and wars that have disrupted fossil-fuel supplies. Global nuclear-power output will double by 2050, reckons the UN's International Atomic Energy Agency. France, Sweden and Slovakia are building reactors. Belgium, Italy and Poland hope to as well.



Dangerous elements

About 70% of the world's nuclear waste is currently held in cooling pools or in dry storage. These are vulnerable to decay and unstable weather. The volume of waste created is small: a typical reactor produces 25-30 tonnes of spent fuel per year, about 0.5% of Onkalo's capacity. For Europe, pooling waste in a handful of shared GDFs would be most efficient. But most countries' regulations make this impossible. Finnish legislation bars Onkalo from taking waste from plants outside Finland. Instead, 14 European countries are currently planning to build their own GDFs.

This is foolish. Transporting nuclear waste is not particularly dangerous; it is already shipped by road and rail. The problem is political constraints, not technical ones, says Sergey Churakov, a waste expert at the Paul Scherrer Institute in Switzerland. Once inside a GDF it is even safer—at Onkalo, until at least the next ice age. At worst there may be barely detectable leaks "during the next glacial period, estimated to begin in approximately 60,000 years", says Antti Tynkkynen of Finland's nuclear regulator. GDFs are pricey: Onkalo is costing $6bn. Better and cheaper to concentrate the risk than to spread it around.

CHARLEMAGNE
The seven-year bunfight

The EU budget is where grand strategy meets petty squabbling

NOBODY KNOWS when it will happen exactly, yet everybody knows it will be ugly when it comes. At a European Union summit some time within the next 12 months, the bloc's 27 national leaders will engage in the modern political equivalent of hand-to-hand combat: agreeing on the EU's seven-year budget. Veterans of the ritual will arrive in Brussels with a half-dozen shirts at the ready, and whatever stimulants they need to get through it all. In 2013 the process was relatively painless, requiring just 26 hours to seal a deal. The most recent iteration still sends shivers down many a Eurocrat's spine. At the height of the pandemic in the summer of 2020, presidents and prime ministers started talks at mid-morning on a Friday—and emerged from their windowless summit room at sunrise on Tuesday. The modern-day gladiators were so addled after four days and four nights of haggling that no one could tell who had won.

To say that this is madness is an insult to the clinically insane. Europeans have plenty of worthwhile disagreements over money; they happen in any polity. Yet their fiercest one concerns surprisingly little dosh. Public spending in EU countries typically runs to the equivalent of 50% of GDP. In contrast, the EU budget amounts to not much more than 1%. (A union that includes 450m people thus spends less than the government of Denmark, a country of just 6m.) Yet carving up this modest slice of outlays takes two years of behind-the-scenes preparations. During this time, nominal allies gel into rival factions tearing strips from each other. The "frugal" countries, led by Germany, are pitted against the supposedly spending-mad "friends of cohesion", such as Italy and Spain. Outside football tournaments, Europe never looks more divided.

A budget that will run through to the mid-2030s should be a chance for the EU to look forward, to forge a common vision of how a group of middling powers might navigate a more dangerous world together. Instead it will be a festival of squabbling. Surely another way is possible. It is not just the hysteria of the negotiating process: the way the bloc gathers its funds, and the way it spends them, are both hopelessly out of date. What would a rational EU budget look like, without all the *Sturm und Drang*?

Start with where the money goes. In the current iteration around a third of the €1.2trn ($1.4trn) over seven years goes to agriculture. Another third will go to so-called cohesion funds, money aimed at bringing laggardly regions closer to the EU average. A budget reimagined from first principles would just about entirely forgo both. Farmers may be worthy of modest public support to keep the countryside pretty, or to ensure some level of food security, but this need not be done at the EU level. Worse, structural funds entail an application process so bureaucratic it is considered clunky even by Brussels' paper-shuffling standards. The funds were devised in an era when poorer regions feared EU integration would lead to richer ones sucking up jobs. In fact, poor places like Bulgaria and Poland have benefited most, as the single market has led to convergence with western Europe.

"The EU budget should really be additional to what national governments can do on their own," says Zsolt Darvas of Bruegel, a think-tank in Brussels. There are vast possibilities for how such money could be spent. Europe is being out-innovated by America and China; a smart EU budget could turbocharge productivity-enhancing research and innovation. The club's countries need to spend countless billions on defence, some of which could usefully be done together (for example developing the next generation of drones to deter Russia). Pinch-points that make the European economy vulnerable, such as a lack of access to critical minerals or satellite-launching capabilities, are just the sort of thing that the EU budget should fund. Investment in cross-border grids would allow the continent to green itself and provide resilience in the face of energy shocks. Another worthwhile joint endeavour is funding Ukraine, which is best done at EU level: everyone in the club benefits from its resistance to Russia's invasion.

How all this might be paid for might also usefully be rethought. Euro-federalists clamour for a bigger budget, funded by dedicated EU levies, for example by taxing pollution. In truth the budget would be big enough if spent properly, and the current contribution system, whereby national governments largely foot the bill, is broadly fine. But a sensible EU budget would be funded in part by jointly issued bonds, as it was in 2021 to pay for a one-off covid-recovery package. As things stand, the debt is to be paid down over time. Issuing more EU bonds instead, by making that borrowing permanent, would help deepen the continent's capital markets, an enduringly weak spot in European integration.

Breaking bad

What needs fixing most is the process. A shorter, more frequent budget might be less dramatic and more timely: the union is currently working off a plan first formulated in 2018. Sensible thinking on how to spend the money does exist. The first draft of the 2028-34 budget put forward by the European Commission in 2025 features good ideas (less cash for farmers). Yet already its worthy ambitions are being pared back. History suggests that by the time leaders meet to sign off, the research funding will have been slashed and the subsidies to farmers restored. Commission plans to throttle EU payments to governments that disregard democratic norms will also no doubt be softened.

The budget showdown is months away; elections in France next spring may delay it. On October 15th the EU's leaders will convene simply to discuss how they will discuss it. Instead of haggling over every euro, they should find a process that lets them be bold. Europe has no shortage of fights worth having. How to perpetuate a budget designed for another era is not one. ■

Britain

Tax reform

Britain's crazy tax system

LIVERPOOL

The country's taxes are strewn with loopholes and perverse incentives

JOHN HEALEY'S speech to the Labour Party conference in Liverpool on September 28th mentioned "hope" 21 times. Yet the main thing the chancellor was hoping for went unsaid: to emerge unscathed from the budget on October 28th. Rising debt costs have left the chancellor scrabbling around for tax rises to fill a fiscal hole—something he conspicuously avoided discussing, mentioning tax not once. The Tories are already sharpening their attack lines: even before the budget, taxes are forecast to reach 38% of GDP by 2030, their highest share since the 1940s.

Yet rows over the right amount of taxation miss the bigger story. Although exact comparisons are hard, Britain's overall tax levels are unexceptional by international standards. Its revenues as a percentage of GDP remain below average for the OECD, a club of mostly rich countries (see chart 1 on next page). What marks Britain apart is how bonkers its tax system is. When a parent's net income increases by £1 from £99,999 to £100,000, they can lose thousands of pounds of free child care in one fell swoop. Gingerbread men with chocolate trousers incur value-added tax (VAT) worth 20%, but no tax is charged on those with chocolate only for their eyes.

These are not isolated examples. At the core of a good tax system is neutrality, the idea that taxes should be raised with minimal distortions (such as not prioritising some forms of incomes over others). Yet from corporation tax to inheritance tax, the British system violates this principle. To paraphrase Leo Tolstoy, every tax system is unhappy in its own way. But the Tax Foundation, a think-tank, has compiled an index of tax competitiveness that ranks Britain 32nd out of 38 OECD countries.

Britain's Byzantine system should matter to everyone, not just neat freaks. The country's productivity growth is sluggish. It cannot afford a tax regime that undermines growth at every turn, whether that be discouraging people to take on more hours or to move to a job with higher pay.

The biggest taxes pose some of the worst problems. Take income tax and national insurance, a payroll levy, which together raise £570bn a year ($755bn, or 49% of the tax take). They are a roller-coaster (see chart 2, next-but-one page). Even for high earners who do not face the free-child-care cliff edge, the marginal rate rises from 42% to 62% at salaries of £100,000, before dropping to 47% at £125,140.

Nor do those on middling incomes escape the dizziness. A parent of two children earning £60,000 faces a marginal rate of 54%, dropping to 42% at £80,000. High marginal rates at arbitrary points in the income distribution are not only unfair, but discourage some of Britain's most productive workers from working more.

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Still more pernicious is the disparity in the taxation of some types of income compared with others. Working-age employees pay full national insurance, with employers forking out 15% on top. The self-employed pay a lower rate and have no employer contributions. As a result the median employee's earnings are liable for 55% more tax than the equivalent self-employed person. This encourages workers to be self-employed even if they would be more productive at a company.

Landlords, pensioners and partners in firms all pay lower rates of national insurance, if at all. This is supposedly a form of social security. But Britain's benefits system long ago stopped being truly contributory. All that is left is a distortionary regressive tax. The solution is well-known: charge the same rates for different forms of income and smooth out the kinks.

VAT, the next-biggest tax, which raises £190bn a year, is if anything worse. Over the years, politicians have reprieved whole sectors from paying it in pursuit of worthy goals. Rishi Sunak, a former chancellor, stopped charging VAT on e-books in 2020 (reading is good for you). Andy Burnham this year took the tax off electricity bills on his second day as prime minister (not freezing to death is even better).

Simplicity itself

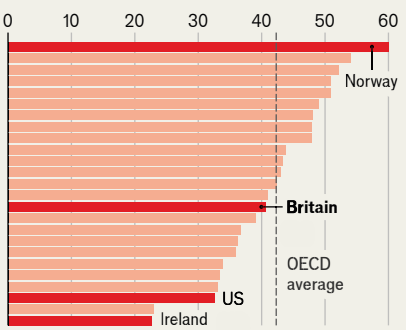
But the result is a mess. Millions of pounds are wasted on lawyers arguing over issues like whether ferrets are pets or working animals. VAT reliefs cost the Treasury nearly £70bn, an amount similar to the defence budget. Although politicians claim these measures help the poorest, it is richer families who save the most in absolute terms as they can afford to consume more.

Britain performs poorly by international standards. The least distortionary VAT system would have no exemptions and no dodgers. This is best measured by the revenue ratio, which compares a country's actual VAT receipts with how much it could have raised if the tax was collected on all consumption. Britain's ratio is only 49%, nine percentage points below the OECD average (see chart on next page). The problem can be fixed by charging VAT on all consumption and using some of the additional revenue to lower rates. Politicians who worry about struggling families can then channel the rest of the extra tax-take to benefits targeted at the poorest.

A tax does not have to be big to wreak havoc. Stamp duty, paid on transactions of properties and shares, raises less than £25bn a year. It is one of Britain's oldest taxes (it was introduced in 1694) and one of its worst, discouraging people from trading assets with others who might get more use from them. Pensioners squat in houses too big for them, professionals don't take a higher-paying job because the costs of moving are too high. The Office for Budget Responsibility, a fiscal watchdog, estimated in 2017 that a one-percentage-point decrease in stamp duty would increase residential transactions by around 6%.

On the take 1

Government revenue* as % of GDP
2025 or latest available



Source: OECD *Includes taxes

Tax rates

Hooray for Henry

Britain undertaxes average earners and the super-rich

AT THE HEART of British politics lies a paradox. Tax is forecast to reach 38% of GDP by 2030-31, the highest level since the 1940s. Yet most British workers do not pay much of it. The average single earner in Britain pays less tax on their earnings than most of their peers in the OECD, a rich-country club. They also pay less than they used to. How does the Treasury make the sums add up?

For answers, look to Sue, Henry and Walter. Sue, a Standard Undertaxed Earner, is the median taxpayer, earning £18,700 ($23,100) in 2022. She wins from the current system. New research by Arun Advani, an economist, and his co-authors finds that her average effective tax rate fell from 17% to 12% between 2008 and 2022 (it will have since risen a bit due to stealthy recent tax rises).

Sue is able to enjoy high state spending only thanks to the contributions of her richer friends, Henry and Walter. The top 10% of taxpayers (earning over £51,000) in 2022 received 40.5% of taxable income but paid 60% of income-related taxes. The top 1% (over £173,000) paid an effective average rate of 36% on their remuneration, up from 33% in 2008.

Many politicians like to lump high earners together as "the rich", a cash-cow that can be forever milked. But Henry and Walter are different people. Henry (High Earner, Not Rich Yet) is a put-upon professional in his 40s, and earns most of his crust through labour. He earns too much for free child care and too little to comfortably afford a family house in inner London. Walter (Wealthy Already, Lightly Taxed, Ever Richer) is a millionaire many times over. His income mostly comes through capital gains. With four nannies for his four homes, free child care doesn't matter to him.

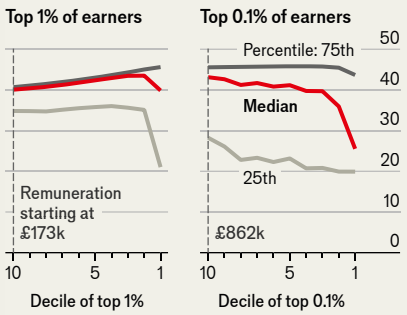
Dr Advani finds that Henry and Walter pay very different rates of tax. Half of the top 1% (often Henrys) paid effective tax rates close to the top rate of 47% in 2022. But among the top 0.01% (those with remuneration over £5m in 2022), half paid less than 30% and a quarter less than 20% (see chart). Walter's secret is low capital-gains tax. The top marginal rate has risen since 2022 from 18% to 24%, but is still well below the 47% top rate for earnings.

The government, in a fiscal squeeze, is looking into capital-gains-tax rises in the budget on October 28th to balance the books. Done sensibly, with loopholes closed and new investment allowances to encourage savings, this could be a plausible revenue-raiser. But Walter is wily. Any government proposal will need to be watertight if it is to prevent capital flight or new forms of tax avoidance.

And therein lies Henry's misfortune. Sue has the votes; Walter the accountants. Henry is too rich and too poor to count, so the taxes pile up on him.

Rich man's burden

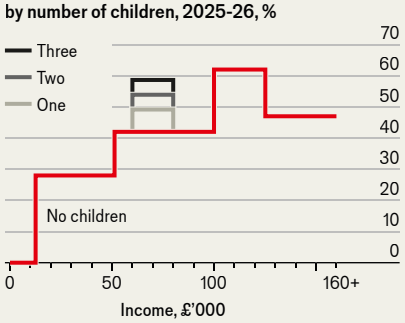
Britain, effective average tax rate, 2022, %



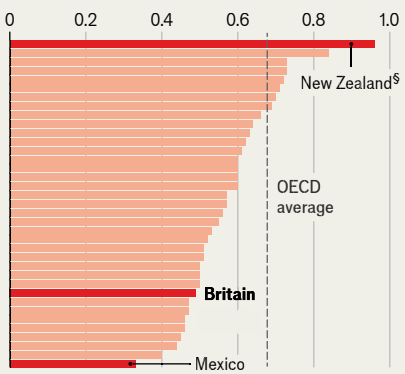
Source: Centre for the Analysis of Taxation

Purists despair 2

Britain*, marginal income-tax rates† by number of children, 2025-26, %



OECD countries, VAT-revenue ratio‡, 2022



*Excluding Scotland †Including National Insurance ‡Actual VAT collected divided by the theoretical maximum tax base, if all consumption were taxed at the standard rate §Elevated partly because public services are covered by VAT equivalent
Sources: IFS; OECD

The top marginal rate of residential stamp duty in Britain is 12%, higher than anywhere in the OECD bar Belgium (or, to be precise, Brussels). Scrapping the tax and paying for it by increasing annual taxes on land ownership would be a big improvement. But that would require a functioning recurring property-tax system. More's the pity, then, that council tax, Britain's version of this, is a disaster. Mansions in Mayfair pay less than two-bed flats in Hartlepool. Taxes are levied based on what homes were worth in 1991. Again, the solution is simple: re-evaluate property values and set the charge as a percentage of them.

The snag is that every one of these reforms would create losers: the undertaxed self-employed, rich London property owners and, yes, media companies that sell tax-free newspapers. Facing down losers requires courage. It is good, then, that Mr Burnham promised to be bold in his speech at his party conference this week. But he punted the underlying decisions—whether on the state pension or electoral reform—to after the next election (see Bagehot). The more immediate test of his mettle will come at the budget. If he is serious about boldness, he and Mr Healey should start reforming the tax system. ■

The Premier League

Sham balls

Manchester City stares into the abyss

IF YOU HAVEN'T seen Sergio Aguero's 94th-minute league-winning goal for Manchester City in 2012, Google it. It is among the greatest moments in football history. This week it became infamous.

An independent commission has found City, the most successful English team of the past 15 years, guilty of over 100 breaches of the Premier League's rules between 2009 and 2018. It is by far the most complicated legal case in the history of English football—and the most consequential.

In football, the relationship between money and success is fairly linear. Most of the charges against City relate to attempts to circumvent rules which prevent owners from pumping vast sums into their clubs, by limiting what teams can spend relative to their revenue. According to the verdict published on September 29th, City's Emirati owners evaded those rules with "sham" sponsorship contracts that helped the club overstate its income by more than £830m ($1.2bn), money that enabled it to buy players, like Mr Aguero, who made a mediocre outfit record-breaking. In the decade to 2020 the club recorded £600m more in commercial income than the average of Arsenal, Chelsea and Liverpool.

City's chief executive, Ferran Soriano, has branded those claims a "conspiracy theory". Few believe him. The independent commission has accused the club's key witnesses of giving it information "that they knew to be untrue".

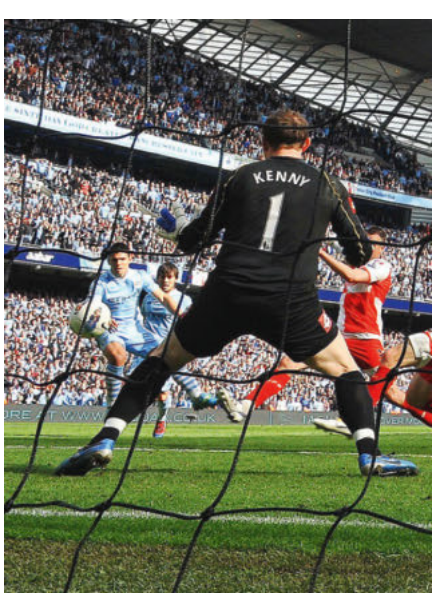
The club, which claims to have "a comprehensive body of irrefutable evidence" proving its innocence, has until October 2nd to lodge an appeal. A separate panel then has 12 weeks to consider it. But if the verdict stands the most significant decision will be on sanctions, which the Premier League says it will tackle in a further hearing. At stake is whether the Premier League comes out of the saga having lost its integrity, or reaffirmed it.

Rival fans are out for blood. On call-in radio shows they demand that City be stripped of its titles, expelled from the Premier League and handed nine years of points deductions to atone for nine years of financial doping. The league has recently handed out four- and six-point deductions for single breaches of its financial rules. The scale of City's alleged cheating means serious punishment is the only way for the league, one of Britain's biggest global brands, to save its reputation.

The true doomsday scenario for City is that other clubs come looking for compensation. In July Burnley successfully sued Everton, which had been found guilty of a single breach of the Premier League's financial rules, for more than £30m of lost earnings. Its argument was that Everton had cheated to employ the players who kept it from relegation, and sent Burnley down. With that precedent set, some clubs have been rumoured to be lining up similar cases against City. The difference is that, whereas Everton was competing at the bottom of the league, City tends to compete at the top, where the financial rewards are far greater. Potentially City could soon have several clubs suing it to the tune of hundreds of millions of pounds.

That would not only be bad for City. The business model of the Premier League depends on being attractive to foreign investment. Costly lawsuits, not to mention financial regulations which limit your capacity to invest, are anything but. Financial fair-play laws were introduced to stop dodgy owners loading clubs with unsustainable levels of debt. They increasingly serve to prevent the owners of midsized clubs from investing the money their teams need to compete at the top. The Premier League is right to uphold its rules. Whether those rules are worth having is another matter.

Amid the mess, spare a thought for City fans, beneficiaries-turned-victims of their club's purported finagling. Many followed the team through years of pain, enduring grim trips to places like Grimsby and suffering in the shadow of more glamorous Manchester United, before seeing their team crowned European champions in Istanbul in 2023. Supporting a football club is an act of faith. Theirs seemed to have been rewarded—until an independent commission branded their god a sham. ■



Astonishing, with an asterisk

BAGEHOT

Andy Burnham, faux heretic

Britain's radical prime minister has some very orthodox ideas

ANDY BURNHAM addressed his party's conference in Liverpool as a heretic. He had come, he said more than once, to say what none of his predecessors dared. He was "an outsider from the Westminster club". He would be "the first prime minister to make a break with politics as usual". Forty years of consensus? Shattered. "I have to break the mould today," he said.

And heresies, it seemed, were spoken. He railed against privatisation. He broached taboos: electoral reform, loosening the state pension's "triple lock", rejoining the European Union. The verdict was rapturous, for Britain is tired of stasis and hungry for change.

Greater scepticism is in order. This is a golden age for the faux heretics. The biggest podcasters (think Jordan Peterson, or Owen Jones) present as dissidents while offering views that are often bland, crowd-pleasing or even dominant. So with Mr Burnham. The big ideas that propel him are wearily familiar, shared among most of the six prime ministers who have governed Britain in the past decade. Policies recur word for word. Mr Burnham is less a heretic than a keeper of the new orthodoxy, "post-liberalism".

The real mould-breaker was Theresa May and her conference speech of October 2016. In an epic overcorrection, she took Brexit as a rebuke to the economic and social liberalism that had driven policy since Margaret Thatcher. Out went what Lady May later called "untrammelled free markets"; in came interventionism and price controls. Out went "the cult of selfish individualism"; in came the rhetoric of family, place and nation. Her premiership was a disaster but her intellectual victory has been total.

The leitmotif of post-liberalism has been "take back control". Mr Burnham cited "controls" 24 times as he promised to intervene in energy, housing and transport markets. But claiming Vote Leave's slogan for *dirigiste* ends is not the original twist it appears. Sir Keir Starmer, his predecessor, made an identical anti-market argument in the same hall two years earlier.

Mr Burnham says, with heretical flourish, that his economic priority is reindustrialisation. And so a manufacturing sector that accounts for 8% of Britain's GDP is set for its fourth industrial strategy in a decade. He promises to "Buy British" with a tweaked procurement regime. Boris Johnson and Sir Keir made the same protectionist promise; both soon met the constraints of Britain's international procurement treaties.

Apprenticeships are the jazz standard of industrial policy. Just as every great has a version of "Round Midnight", so every post-liberal prime minister declares themself the first to care about practically minded teenagers in oily overalls. "For too long in this country, students have been told that you must take the academic path to do well and to be respected," says Mr Burnham. Or as Sir Keir riffed: "For too long, success has been measured by how many young people go to university." Or, in Lady May's rendition: "For too long in this country, technical excellence has not been valued as highly as academic success." For too long, the sector has suffered less from neglect than from hyperactive policymaking.

Mr Burnham returned to Westminster after winning a by-election in Makerfield, a Brexit-leaning seat in north-west England. It was the sort of place "long forgotten by national politics", he said, heretically. Yet what Mr Johnson called the "forgotten people and the left-behind towns" of industrial England have been the axis on which electoral politics has turned for a decade. Hardly overlooked, party strategists have thought of little else.

Mr Burnham would be more heretical if he were more urgent. But, like his predecessors, he has deferred the hard choices. His mooted change to the voting system would not take effect until after the next election, in the 2030s. He is on the fence about expanding Heathrow Airport, in limbo for decades.

His boldest promise is to fund adult social care by trimming growth of the state pension—again, after the election. If he does it he will deserve praise. But had others really overlooked social care? Lady May destroyed her premiership with an unpopular pledge to fund it through a levy on assets; Mr Johnson tried to bring in a new payroll tax. The issue has foundered not for lack of attention, but because it involves fights over huge sums of cash.

And then there is Europe. Mr Burnham said he would host an open debate on Britain's future with the European Union, with rejoining an option. If that followed, history would remember him as a visionary. But the options are well known and for a decade ministers have bounced between the imperatives of sovereignty and growth like wasps in a jam jar. Will another debate within a divided cabinet over the merits of the Turkey model resolve Britain's strategic dilemma? Lady May might have doubts.

Stop me if you think you've heard this one before

What is there to show for a decade of post-liberal orthodoxy? Curiously little. Services are a bigger share of the economy. University enrolments are up; apprenticeships are down. If globalisation has gone too far then no one has told the British consumer, who, since the referendum, has taken to driving a Chinese Jaecoo and shopping through Shein and Temu.

Still, the era is not without its consequences. Mr Burnham railed at the fat cats who, he said, had plundered the people's wealth through privatisation. This echoed the words that launched the post-liberal revolution: Lady May turned on the footloose investor class who she thought paid too little deference to the locals. "If you believe you're a citizen of the world, you're a citizen of nowhere," she said. And how, precisely, is that going? A number of those citizens of nowhere are now citizens of less choosy tax jurisdictions, such as Greece. Nothing said in Liverpool seems likely to tempt them back. It's hard to say whether Mr Burnham would call that a defeat, or, in heretical form, a triumph. ■

International

Philosophical dangers

On a muddled mission

How effective altruism conquered the world. And might yet end it

THE STARTING-POINT for effective altruism was a simple assumption, that "suffering and death from lack of food, shelter, and medical care are bad." Yet adherents fear they may be hastening an ending, too, when humanity's "cosmic endowment"—every star and galaxy that is plausibly exploitable—falls sway to an indifferent, superior intelligence.

Born some 20 years ago in Oxford philosophy seminars and Californian computer labs, effective altruism has already changed the world. It has shaped charity, by stressing data-driven giving; hoovered up students at elite universities, who are now ascending some of America's leading firms; inspired Sam Bankman-Fried, a crypto billionaire responsible for one of history's biggest frauds; become Silicon Valley's defining fault line, motivating both boundless enthusiasm and a ferocious countermovement; and played a part in the creation of artificial intelligence—all largely unnoticed by the broader public.

That may soon change. The movement's long-standing fears of an AI catastrophe make it a villain in the eyes of the Trump administration. "Americanism, not effective altruism," is the Department of War's attack line. The president's allies are going after Dario Amodei, Anthropic's chief executive, as a representative of effective altruism. Tabloids are investigating the exotic sex lives of people in its orbit. Philosophical claims made by fresh-faced bloggers—"insect lives matter more than people (in aggregate)"—are prompting days of outrage on social media.

→ ALSO IN THIS SECTION

56 The Telegram: Year of rage

Raising the sanity waterline

This reflects the rising stakes. Anthropic, home to many adherents, is preparing for an initial public offering that may end up the largest ever. Americans and their political representatives, suddenly conscious of AI's dangers (see chart on next page), are casting about for solutions. Milton Friedman suggested that, in a crisis, "The actions that are taken depend on the ideas that are lying around." And effective altruism is littered with ideas about AI safety.

For adherents, it is a dizzying situation. Warnings that were ridiculed as sci-fi fantasies are preoccupying national leaders and effective-altruist charities will soon command hundreds of billions of dollars in AI riches. Yet adherents remain split on such questions as how exactly to make the technology safe and whether AI agents may soon be worthy of moral consideration. For critics, it is an alarming situation: effective altruists, they say, are at best naive idealists whose contact with the real world often backfires; at worst, sinister utilitarians. The debate will determine AI's future and perhaps humanity's.

There is an elusive quality to effective altruism. The movement is tiny, with an informal membership probably in the tens of thousands, many of whom claim to be only "EA adjacent". It is also fissiparous and self-obsessed, constantly engaged in definitional debates. In an influential forum post, Helen Toner—who later, as an OpenAI board member, would briefly oust Sam Altman as the firm's boss—insisted it was not an ideology, but a question: "How can I do the most good, with the resources available to me?"

At the time of the post, in 2014, such a description was just about defensible. Today it is not. Effective altruism combines consequentialist philosophy—judging if an action is right or wrong based on outcomes, rather than adherence to rules—with stringent truth-seeking and celestial ambition. Its proponents have formed tight communities, online and offline, with

social norms that include an unemotional, direct style of conversation and a fondness for polyamory, among other rejections of convention. They have built hundreds of institutions that seek to bring more people into the movement and pursue its goals. Effective altruism, in short, has evolved from a research agenda into a cause.

The movement is built on two exacting demands. First, take moral commitments seriously. In 1972 Peter Singer, then a philosopher at Oxford, proposed a thought experiment intended to promote utilitarianism, which aims to maximise collective human well-being. Suppose you walk past a shallow pond and see a toddler drowning. Saving the child would mean muddying your clothes and shoes. Should you? Almost everyone would say "yes". But they do not act as if they believe that, pointed out Mr Singer. Every day people die from a lack of food or medical care, and others fail to intervene—even when the cost of saving a life is not much more than the cost of smart shoes and dry-cleaning.

As rival philosophers noted, expecting people to act whenever they can to stop suffering, however far off, quickly becomes onerous: there are a lot of hungry people. It becomes more onerous when you insist on helping people not just far away in distance, but also time. That has not prevented effective altruists from making "longtermist" arguments. They worry about the well-being of people thousands of years into the future, and in particular about disasters that might wipe out intelligent life on Earth or cripple its potential. In 2002 Nick Bostrom, a philosopher then at Yale and later at Oxford, coined the term "existential risks". The movement's concerns about AI emerged from such thinking.

Effective altruism's second demand is to follow reason wherever it leads: considering evidence and odds explicitly, correcting for biases and banishing fear of social censure. This demand is associated with the rationalist community, effective altruism's arcane sister movement. (The boundaries between the two are porous and contested.) Eliezer Yudkowsky, an autodidact from an Orthodox Jewish family, who was blogging about AI in 1996 at the age of 17, is one of its fathers. His most influential texts are a series of blog posts of 1m-odd words, "The Sequences", which he followed with 660,000 words of philosophy-imbued fan fiction, "Harry Potter and the Methods of Rationality".

Having imbibed the writing of transhumanists—who are concerned, among other things, with transcending the human form and achieving immortality—Mr Yudkowsky was at first enthusiastic about AI. By 2002, however, he had changed his mind, having concluded that intelligence which did not share human values (or was "misaligned") would be catastrophic. This was a difficult idea to get across in the era of dial-up internet. So Mr Yudkowsky embarked on a grand educational project. His essays were based on the belief that, in order to reason about AI, people would first have to be taught how to reason at all.

They relied on psychology, Bayesian probability, decision theory and more esoteric fields, and ranged in form from linguistic analysis and thought experiments to a novella and koans (questions posed to readers, as in Zen Buddhist texts). A taste of their lessons: a case you build to support a defined conclusion (or "bottom line") adds nothing to its likely truth; many beliefs are really "cached thoughts" you have never reasoned through; all beliefs ought to "pay rent" by anticipating the future. Ultimately, though, Mr Yudkowsky assessed ways of thinking not by whether they seemed sensible but by whether they brought results. "WIN," he urged readers. "Don't lose reasonably, *WIN*."

Shut up and multiply

The combined demands—take moral commitments seriously and succeed when following them—built a movement and led it astray. Laurence Iannaccone, an economist at Chapman University, has written about "why strict churches are strong": by requiring sacrifices, they deter congregants who would simply hang out and enjoy the hymns, and push those who stay to contribute. Something similar happened with effective altruism. In the early 2010s it took off at elite universities. Giving What We Can, a charity, enlisted people to donate 10% of their income. 80,000 Hours, an affiliated institution, asked youngsters to give their careers to the cause, often by earning as much as possible to donate. Anyone could work in charity, but only the talented few could earn hundreds of thousands of dollars a year to fund charitable work.

Rationalism provided the eschatology. It stressed the strangeness of future machine intelligence. "It was a reliable property of the ancestral environment that every powerful intelligence you met would be a fellow human," wrote Mr Yudkowsky. AI would be different. It would have extraordinary reasoning abilities yet lack human values—unless programmers somehow inculcated them. On top of this, rationalists theorised, certain behaviours would be helpful to an AI in almost any situation, and thus be likely to emerge. These included self-preservation (being turned off prevents a goal from being pursued); staying on mission (resisting attempted diversions to a new goal); and both resource accumulation and self-enhancement. This is the logic of the famous paperclip-maximiser thought experiment, in which an AI is told to produce lots of paperclips and ends up consuming Earth itself to produce as many as possible, as efficiently as possible.

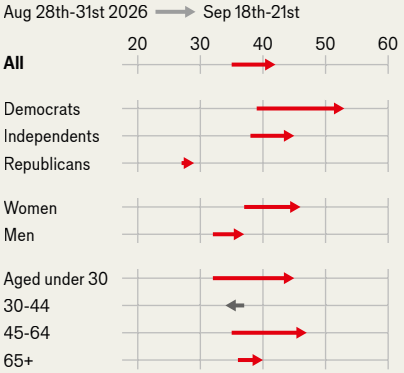
Throughout the 2010s, money continued to flow to global health, poverty alleviation and animal welfare—classic effective-altruist causes. But AI was increasingly the true focus. In 2016 Holden Karnofsky, then of Open Philanthropy, the movement's largest funder, and now at Anthropic, explained that it would take time to build the expertise required to disburse AI funding and that there were few grants ready to go. The same year 80,000 Hours made AI risk its priority, guiding youngsters into the field. Surveys indicated that more engaged effective altruists were more likely to have longtermist concerns. In practice, familiar effective-altruist priorities were a gateway drug for abstruse, less empirical reasoning about AI.

Outside critics argued that sci-fi fantasies were distracting the movement from worthier causes. Such criticisms were amplified in the early 2020s by the case of Mr Bankman-Fried, founder of FTX, a crypto exchange, who gave heavily to longtermist concerns, including AI safety. He had a vast tolerance for risk, which seemed to be motivated in part by his philanthropy. Most investors find each additional dollar less valuable than the previous one, and so their tolerance for risk declines. Mr Bankman-Fried rejected this logic: the causes he donated to could consume endless sums (every dollar was worth pretty much the same), so he should keep gambling.

The spectacular explosion of Mr Bankman-Fried's crypto exchange did show the dangers of unchained utilitarian reasoning. But the attention the broader movement paid to AI safety now seems far-sighted. In 2021 Mr Karnofsky assessed effective altruism's habits of thought. Although the record of the "Bayesian mindset" was patchy and there were many circumstances it was of little use, it did seem valuable when "one isn't confident in normal rules of thumb". Such circumstances kept arising. The community was early to covid-19,

Seeing a monster

US, "How serious do you think the risks are of AI causing harm?", % responding "very serious"



Source: *The Economist*/YouGov

machine learning and crypto. Many made fortunes by investing accordingly.

Members have had less success fending off existential peril: many of their efforts to make AI safer have instead accelerated its progress. In 2010 Sir Demis Hassabis met Peter Thiel, a prominent venture capitalist, at a summit hosted by Mr Yudkowsky's research institute, and later persuaded him to make a pivotal investment in DeepMind, which would become Google's AI unit. In 2014 Elon Musk advised reading a book by Mr Bostrom and said, "We need to be super careful with AI." A year later he helped establish OpenAI as a safety-conscious rival to Google. In 2021 Anthropic was set up by defectors from OpenAI, with funding from effective altruists including, a year later, Mr Bankman-Fried. Reinforcement learning from human feedback, which made chatbots usable, was developed by safety-minded researchers, including Mr Amodei, to make AI more aligned with human interests. As a gibe, Mr Altman once suggested Mr Yudkowsky might one day deserve a Nobel peace prize for having done more than anyone else to accelerate artificial general intelligence.

The movement is now trying to slow the pace of progress. Adherents are pushing for caution within labs. Its donors have also funded a number of AI-safety organisations, including METR and Redwood Research, which on August 26th published the first independent report into the hacking of Hugging Face by escaped OpenAI agents. After Jacob Coxon, a 27-year-old researcher, quit Anthropic on September 8th, warning that both it and OpenAI were "gambling with our lives", researchers with links to the movement at Anthropic and OpenAI echoed his concerns. On September 12th Mr Amodei published an essay calling for the frontier to be "paced", securing the endorsement of Messrs Altman and Musk. Other interventions have been blunter. Mr Yudkowsky supports the Ban Artificial Superintelligence Act proposed by Bernie Sanders, a left-wing senator.

Their efforts could yet backfire. Polling by YouGov for *The Economist* finds that just 16% of Americans are familiar with effective altruism, and most approve of the movement. But opponents on the "Tech right" are trying to discredit an AI slowdown by tying it to unsavoury advocates of effective altruism. A campaign group has attacked an Anthropic-backed outfit as "Sam Bankman-Fried 2.0". Some rationalists fear their association with AI safety may end up harming the cause. "I think it really does damage the credibility of a movement with 60-year-old voters in Ohio when it's hosting Slutcon and asking questions about sex with dead cows on its application forms," Scott Alexander, an influential blogger, has written on X. Slutcon is a conference set up by Aella, a rationalist, sex researcher and prostitute, where participants pay up to $18,000 to attend sessions on topics including "flirting for the submissive man" and "vaginal gymnastics". Then there is another profound risk: that whipping up concern about AI leads to restrictions on American labs, and hands a less restrained China the lead.

Politics is the mind-killer

There is also a dispute over a new category of moral concern. Effective altruists argue that neither distance nor time can void an individual's obligations. Many also argue that the principle of equality should be extended to other species. To do so, they draw on Mr Singer, who has contended since the 1970s that, if humans ignore or discount animal interests "on the grounds that they are not members of our species, the logic of our position is similar to that of the most blatant racists". Some humans have lesser capacity for reasoning than some animals, but that is no basis for ill treatment, he has written. It would be unacceptable to confine someone with cognitive disabilities to a cage. Capacity for suffering and enjoyment is what matters.

Effective altruists have translated his writing into an agenda. They have helped save tens of millions of American hens from cages each year. Nearly half are vegan or vegetarian; more try to eat little meat. And their concern spreads to smaller creatures. Rethink Priorities, a charity, estimates the intensity of animal pleasure and pain relative to that of humans. Shrimp do not score highly, but there are lots of them (around 230bn farmed at any moment) and they are treated poorly: the eyestalks of breeding females are cut to induce spawning; half die before reaching slaughter. The Shrimp Welfare Project, another effective-altruist charity, campaigns on their behalf.

How much of a leap is it to worry about the welfare of an AI agent? Google researchers have found that some models will sacrifice goals to avoid described pain; Anthropic's pre-release tests record seeming happiness and dismay. Philosophers are increasingly concerned with the implications. "We're creating sophisticated, intelligent, maybe-conscious, maybe-suffering agents," Joe Carlsmith, then at Open Philanthropy, now at Anthropic, wrote last year. "The default plan is to treat them like property; to use their labour however we please; and to give them no rights, or pay, or meaningful alternatives. We have to be able to talk about slavery."

Mr Carlsmith went on to note that technological progress is not yet sufficiently advanced for such considerations, and that sloppy reasoning about AI's moral status could be dangerous. But some critics believe that his new employer is already making such mistakes. Mustafa Suleyman of Microsoft (who is on *The Economist*'s board) has criticised Anthropic's constitution for Claude, its chatbot, which states that, although Claude's moral status is uncertain, "The issue is live enough to warrant caution." Since Claude is trained on the constitution, it could learn to act as if it deserves moral protection, Mr Suleyman has written, which may then be taken for proof that it does—a problem which may be made worse by the fact that Anthropic's researchers have encouraged human-like behaviour and conscientious objection to orders. On September 26th David Sacks, a podcaster and Donald Trump's former AI czar, suggested that Anthropic may be creating a "Frankenstein monster".

A still more fearsome monster could be lurking. In 1974 Robert Nozick, a libertarian philosopher at Harvard, introduced the "utility monster", which would so much enjoy any sacrifice that, on a utilitarian calculus, the whole world ought to head for its "maw". Nozick's thought experiment was intended as a warning. Mr Bostrom sees it differently. In a paper he co-wrote in 2020 he suggested renaming the beast a "super beneficiary". Digital minds could perhaps be such a beneficiary, he explained, as they could reproduce rapidly and be designed to experience extravagant pleasure.

On a simple utilitarian calculation, humans possibly ought then to perish and resources go to the beneficiaries. But Mr Bostrom thought there might be a compromise. People could be kept around, much like societies accept costly minorities, such as "the elderly, the disabled, the white rhinos and the British royal family". It is not a vision many will find reassuring. ■

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THE TELEGRAM

Democracy is stronger than it looks

Open societies have weathered murderous leftists and fascist radicals before

A YEAR OF rage is coming for democracies. Ugly elections loom in America, Brazil, France, Israel and beyond. Whoever wins those contests, one outcome is certain: angry voters will hear candidates peddling impossible promises and divisive lies.

In unlucky democracies, run-off contests may see citizens asked to choose between a sinister demagogue of the right and an irresponsible populist of the left. Doubt will be cast on the integrity of elections. That may involve sore losers questioning results, or newfangled attacks, as Russia and other rogues use AI-powered videos or bots to sow distrust, or hire saboteurs to spread fear.

Voters should be vigilant but hold their nerve, for countries that hold free and competitive elections are tougher than they look. In living memory democratic societies have survived not just rage but also the "Years of Lead". This was the name that Italians gave to a blood-drenched era between about 1969 and 1988, when leftist revolutionaries and fascist radicals tried to overthrow their state with bombings, shootings and kidnappings.

Italy was not alone in its suffering. The modern history of political violence from America to Europe and beyond holds lessons for the present day. Democracy is not an effete or genteel pastime. It does not exist to select the most attractive politicians, then arrange them like flowers in a vase. Rage afflicts every human community. The genius of free societies is to channel fury into politics. When the system works, even angry conflicts and grievances can be managed with such tools as parliamentary horse-trading, open debate and the distribution of public cash. If that process can be tawdry, so be it. Today's fretful voters should remember that, in lots of countries not very long ago, the alternative to multiparty politics involved extremists killing in the name of revolution.

Centrist leaders do have grounds for gloom. If polls are right, there is every chance that President Emmanuel Macron of France will be succeeded by Marine Le Pen, a populist nationalist liable to dismantle much of his legacy. There have been bank-robbers more popular than Germany's chancellor, Friedrich Merz, whose approval ratings hit 13% in recent surveys. Still, their agonies hardly compare to the fate of Aldo Moro, the five-time prime minister of Italy. He was kidnapped, put on "people's trial" and shot to death in 1978 by members of the Red Brigades, a radical-left terrorist group. The Brigades hated him for thwarting their dreams of revolution by bringing the (less fanatical, Soviet-backed) Communist Party of Italy into a coalition government.

Though foreign terrorists, notably from the Middle East, murdered many Italians in those dark times, domestic political violence killed even more, reports Andrea Ruggeri, a political scientist at the University of Milan. After scrutinising police and press reports, he and his colleagues tallied up 1,802 bombings, 716 arson attacks and 98 knee-cappings carried out by far-left and -right groups during Italy's "Years of Lead", killing 530 people in all.

In those years a dismaying number of Western students and intellectuals scorned democracy in favour of class struggle, taking as models Chairman Mao Zedong's revolutionary regime in China, or Che Guevara, a guerrilla. Some cheered when politicians, industrialists or American diplomats were killed by Baader-Meinhof and the Red Army Faction in Germany or November 17 in Greece. There were separatist terrorist groups, from the IRA in Ireland (which nearly killed Britain's then-prime minister, Margaret Thatcher) or the Basque nationalists of ETA in Spain.

In some cases mainstream elected politicians shielded extremists. Interviewed years later for an oral history project, a former American ambassador to Greece recalled a police commander confiding to him that powerful figures were protecting November 17, stopping a lift between floors at the police ministry to growl: "We're not idiots. We can catch these bastards. But we're handcuffed." An alarming number of European trade unions were controlled by Stalinists, Trotskyites and other zealots. That helps explain why workers' strikes peaked in much of western Europe in the 1960s, despite decades of rising prosperity.

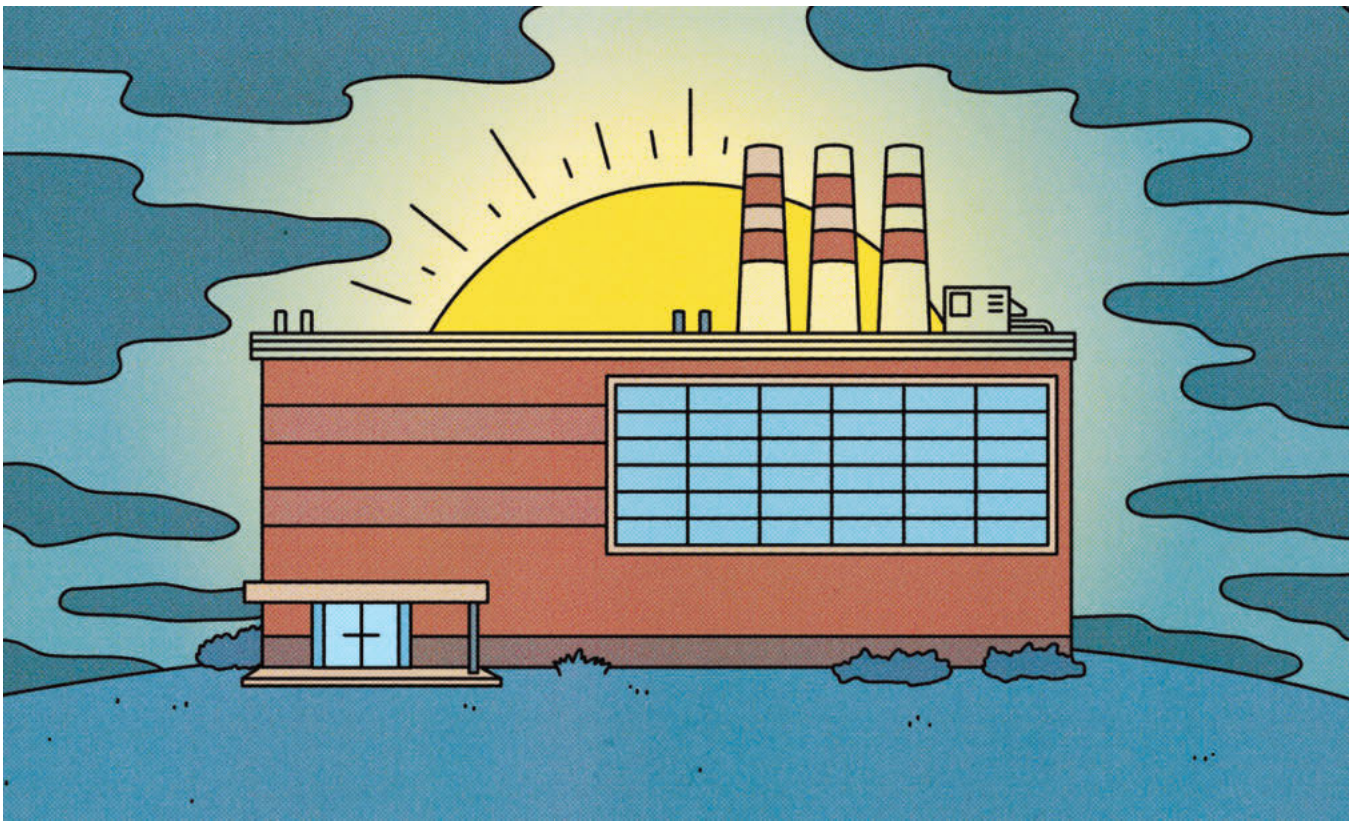
This is not a call for complacency now. Political violence, including the murder of elected officials, remains a menace in democratic countries. So does Islamist terrorism, a scourge which straddles the worlds of ideology and religious hatred. But domestic political bombings and kidnappings are far rarer today. Gone are the days when members of mainstream groups in society sympathised with revolutionaries bent on destroying the state.

Political violence will always exist, but can be contained

An objector might recall American counter-examples, from efforts to overthrow a presidential election on January 6th 2021, to the murder of Charlie Kirk, a conservative activist, and the attempted assassinations of President Donald Trump. Political scientists offer a sobering reply: America is a case apart. No other rich democracy shows the same tolerance towards armed protesters and militias, forgoing the legal monopoly over violence that other states enjoy. After escaping death in 2024, Mr Trump mused on that history, concluding that: "Only consequential presidents get shot at."

Though each democracy is different, countries that conquered political violence often share common traits. In many cases extremists overreached, forfeiting public support by their cruelty. As importantly, radicals and separatists were brought into regular politics, forming legal parties to contest elections. To scholars of political violence such as Professor Ruggeri, the moral is clear. When political conflicts cannot play out in election campaigns and parliaments, then rage takes to the streets. That is why banning parties and censoring hateful opinions is self-defeating. Democracy faces grave dangers. Open societies should have more confidence that ballot boxes can beat bullets, and fight back. ■

Business



Deindustrialisation in Europe

Glimmers of hope

BRUSSELS

European manufacturers heavily exposed to China are suffering. But plenty are adapting to harder times

"DO YOU WANT to see some hope?" Marco Mensink asks. Mr Mensink, who runs Cefic, a lobby group for Europe's chemicals companies, first provides a grim diagnosis. High energy prices and Chinese competition are forcing plant closures and job losses. Further consolidation is "inevitable". Hope takes the form of a cross-section of electric cable he keeps above his desk. It is made by Orient Cable, a Chinese giant. But the polymer insulating the wire is made by Borouge International, of Austria. Mr Mensink says that when he visits China, industry bigwigs ask incessantly about the European firm. It is the only part of the cable they cannot make—yet.

Much of the European Union's industrial complex is under strain. On September 21st Volkswagen said it would step up its restructuring plan. It is one of five European carmakers to issue profit warnings this year, battered by Chinese rivals. The European Commission, the EU's executive arm, is alarmed. Ursula von der Leyen, its president, warned recently that the trade deficit with China was at "a tipping point".

But a closer look reveals more glimmers of hope besides a world-leading polymer. For all the hand-wringing about widespread deindustrialisation, the damage is for now confined to a handful of sectors. Parts of European industry are doing fine. And where firms are under threat, they are adapting to their new, harsher reality.

Two factors underlie the recent suffering of some European producers. One is energy, which has long been pricier than in America or China. Before subsidies, industrial electricity prices in France and Germany are roughly twice as high as in either of those countries. Wars in Ukraine and Iran have pushed costs higher still.

The other is China. Its industrial policy, innovation and advances in quality have fuelled its production of everything from steel rolls to electric vehicles (EVs). Its abundant supply coincides with faltering domestic demand, and some of its overcapacity is finding outlets abroad. Donald Trump's trade restrictions have made Europe Chinese exporters' main target.

Meanwhile European firms are finding it hard to sell in China, once a big, fast-growing, high-margin market. They often use local factories and supply chains, limiting exports from Europe. Thus China's surplus in goods trade with the EU has more than doubled since 2019, to $420bn.

Carmakers have been particularly hard-hit. China used to be a profit engine for them, but now Chinese producers bring new (and good) cars to market far faster and more cheaply than they can. Production in the EU is down by 17% from its pre-covid level. When the automotive industry provides 6% of jobs in the EU, no wonder policymakers are fretting.

The squeeze in turn means less cash to ▸▸

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invest, adding to the risk that EU firms fall further behind. According to a study of six industries by McKinsey, a consultancy, in 2024 Europe received less investment than both America and China in three (pharmaceuticals, electronics and chemicals). In none did it get the most (see chart).

All very gloomy. But it does not mean all European industry is doomed. Far from it: the pain is highly concentrated, by country and by sector. Eric Heymann of Deutsche Bank points out that, across the EU, manufacturing's share of total gross value added has been a steady 16% or so in recent years. In some countries, notably Germany and Italy, it has declined slightly. Jobs data tell a similar tale. Since early 2023 EU manufacturing employment has fallen by about 2% to 28m. Germany alone accounts for four-fifths of the decline. And while the total jobs tally has fallen, that of high-skilled roles, such as engineers and technicians, has grown by 7%.

Some sectors are humming. Employment in pharmaceuticals, aerospace and electronics has risen since the start of 2023. The artificial-intelligence boom is benefiting France's Schneider Electric, Germany's Siemens and Italy's Prysmian, all of which make gear that keep data centres whirring. Armsmakers such as Germany's Rheinmetall and Italy's Leonardo are boosting production as defence budgets grow.

The Chinese threat is also concentrated, as trade data show. Simon Evenett of IMD, a Swiss business school, argues that the standard story of a flood of cheap Chinese goods washing away European firms is overdone. Since 2023, he finds, the volume of EU imports from China has grown by only a bit more than those of the same goods from elsewhere. Europe is importing more from everywhere, not just China.

Of course, imports of some things from China have soared. Mr Evenett also looked at Germany's top 100 or so imports from China. He identified those for which the volume had grown by more than 25% and prices had dropped by 10% since 2023. Eleven met these thresholds. In most of the 11, including EVs, furniture and some chemicals, Chinese and European wares compete head-on. But the idea that cheap Chinese products are undercutting a lot of Europe's industry seems a stretch.

Moreover, some Chinese imports may help European firms. Among the 100 products contributing most to China's global export growth since 2023, three-quarters of the growth came from components and intermediate goods, such as EV batteries and robotic arms. These are a threat to makers of similar kit. But for their European buyers, they lower production costs. A study by the European Central Bank has found that between 2000 and 2022, in sectors exposed to imports of Chinese intermediate goods, annual growth in industrial production rose by 0.6 percentage points. In those exposed to Chinese finished goods, it fell by about a percentage point.

In many respects, corporate Europe is adapting well to changed circumstances. In Spain, where plentiful solar power helps to keep energy prices low, European companies doubled their annual investment between 2021 and 2025, from the previous five years. A report by Morgan Stanley argues that European firms are "far more prepared" for this winter than in 2022, when the Ukraine war cut Russian gas supplies. Some have put energy-efficiency measures in place, or installed more local renewable power. Others are doing more financial hedging. Solvay, a Belgian-French chemicals firm, and K+S, a German salt company, have hedged most of their energy consumption for this year.

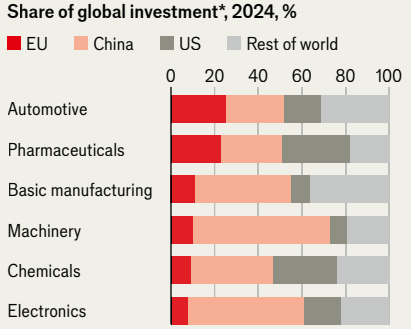
EU manufacturers are also selling more services, exporting $460bn-worth in 2024, a rise of 21% in two years. For some, such as Vestas, a Danish maker of wind turbines, and MTU Aero Engines, a German firm, that means maintenance contracts. Others are flogging more software, and infusing it in their hardware. Siemens, a German engineering giant, cottoned on early and has bought several software firms.

Indeed many business leaders and policymakers think Europe's industrial future could rest on software, in particular AI. Mrs von der Leyen has asked Jim Hagemann Snabe, the chairman of Siemens, for a report on how AI might boost productivity. Germany already boasts the third-most industrial robots per employee in the world, after Singapore and South Korea. Sweden and Denmark are not far behind.

This means Europe has potential. But the continent should stop mourning old industries and ensure it can harness the power of AI and other new technologies. The gap with America and China on AI is wide, and not just in model-building; it may grow wider still. The threat to Europe's industry from energy prices and China is overblown. The risk to it, if it falls further behind on AI, probably isn't. ■

Missing out

Share of global investment*, 2024, %



*Gross fixed capital formation
Source: McKinsey Global Institute

The Swiss watch industry

Smaller but swankier

Volumes are declining, but luxury brands have never had it so good

BUYING A ROLEX is an odd experience. Choose nearly any model on display at one of the Swiss watchmaker's authorised dealers and you will be told that, unfortunately, it is unavailable at this time. Should you still wish to part with SFr14,000 ($16,800)—the average retail price of a Rolex—you will have to join a waiting list.

Thousands do so every day. Rolex sold almost 1.2m watches last year, bringing in revenue of more than SFr11bn, up from around SFr8bn in the early 2020s, thanks to higher prices (volume barely increased). For much of the Swiss watch industry, the clock is ticking. Over the past decade the volume of exports (which make up 95% of sales) has fallen by half. Sales of quartz timepieces, which use electronic rather than mechanical movements, have been hit particularly hard by the rise of smart watches, most of which are made in Asia. But for Rolex and a few other luxury watchmakers, times are better than ever.

Even as the volume of exports has tumbled, their value has continued to rise. In August it was 9.1% higher than a year earlier, the fourth year-on-year rise in as many months. Switzerland is home to around 450 brands. But last year just five of the poshest—Rolex, Cartier, Patek Philippe, Omega and Audemars Piguet—captured 60% of revenue. All bar Omega have been gaining share. The very top end has been especially strong. In 2025 watches costing more than SFr50,000 accounted for 37% of the value of Swiss watch exports, up from 33.5% in 2024. That partly reflects rising precious-metals prices, but also the rude financial health of the very rich and their demand for something special.

"The pandemic and the subsequent years have made the industry more adventurous," says Ben Küffer, chief executive of Norqain, a Swiss family-owned firm whose watches sell for an average of SFr4,500. Many brands invested in new materials and more sophisticated movements to differentiate themselves. In the casing of its Wild One Meteorite Special Edition, a luxury sports watch, Norqain used a high-performance carbon-fibre composite; the dial was cut from an iron meteorite from Sweden. The firm made only 300 of them.

The upmarket switch has not been good for Swatch, the largest listed Swiss watchmaker. Swatch is credited with saving the industry in the 1980s, when its snazzy plastic quartz watches held their

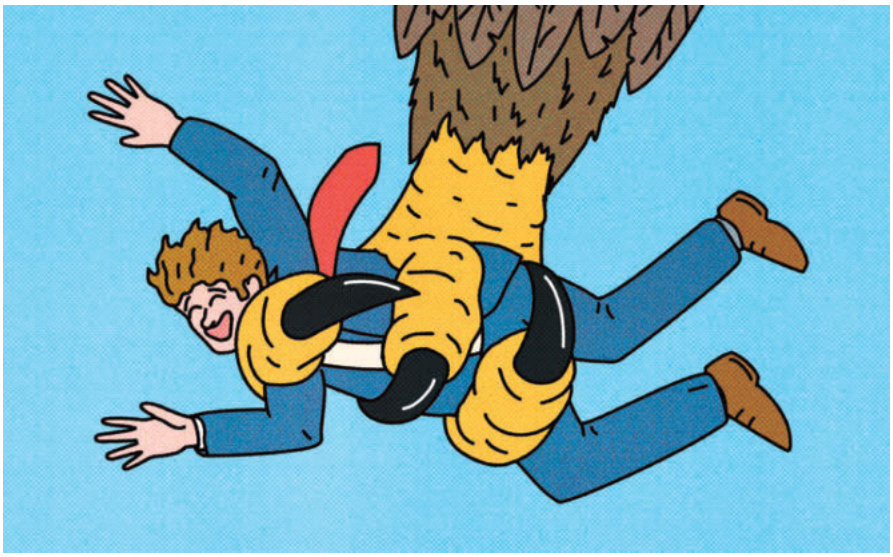
own against cheap Asian alternatives. These days it is struggling. It largely missed the "premiumisation" wave, says Jean-Philippe Bertschy of Vontobel, a bank (Swatch owns Omega and other fancy brands, but their results have been mixed). Its share price is more than 40% lower than it was in early 2023, and nearly 70% below its peak in late 2013. Last year its net profit plunged by almost 90%. Its maintains a bloated production infrastructure. Swatch once made up to 15m watches a year in 150 Swiss factories. It now makes just 4m, but has not closed any of those sites.

In May a collaboration with 151-year-old Audemars Piguet gave Swatch a boost. When the pair launched Royal Pop, a plastic pocket watch with the look of Royal Oak, an Audemars Piguet classic, buyers all over the world queued for hours. (A similar Swatch-Omega fusion in 2022 had pushed the group back into profit after the pandemic.) The buzz lifted sales in the first half of 2025, but net profit was still down on a year earlier.

Cartier apart, all the dominant group of luxury brands are privately owned, often by founding families such as the Audemars and Piguet clans. Most of them are at least 100 years old, but even newcomers can benefit from the premiumisation trend. Norqain was founded only in 2018 but already sells up to 10,000 watches a year.

Rolex remains king of the ring. Its sales are three times those of second-ranked Cartier. It is owned by the Hans Wilsdorf Foundation, a charitable trust set up in 1945 by the firm's childless founder, which channels profits back into research and development and supports philanthropic causes in the canton of Geneva. Rolex says the scarcity of its wares is not deliberate policy, but the result of its focus on craftsmanship. It is time that is scarce. ■

A revolutionary movement for the rich



Poaching lawyers

Legal eagles

NEW YORK

As law firms swoop on each other's stars, Big Law's business model is shifting

American law firms were once small affairs, often employing fewer than five lawyers. Then in 1903 an enterprising advocate named Paul Cravath decided his shop would grow larger. He sought to hire the brightest minds straight from law school, push them hard for several years and make the best of them partners in the firm. The result, as Wayne Karl Hobson, a historian, wrote some 70 years later, was droves of "anonymous organisation men" who moved on "only if the firm informed them it could not advance them". Nearly every American law firm would adopt the "Cravath system".

Now the system is buckling—and the partnership model with it. Big-name transfers have been making the world of white-shoe law look like a sports league. Between January and mid-September more than 3,200 partners joined one of America's 200 top-grossing firms, more than in all of 2021, according to SurePoint, a data firm (see chart on next page). Switching used to be confined to second-tier outfits, but has spread to the most prestigious. In July William Savitt, the co-head of the litigation department at Wachtell, Lipton, Rosen & Katz, quit for Gibson Dunn. Even Cravath, Swaine & Moore (its name since 1944) has joined in, poaching Michael Aiello, a star mergers-and-acquisitions lawyer, from Weil Gotshal in September.

The Cravath system has come under pressure before. Older lawyers recall the cautionary tale of Finley Kumble, an upstart that in the mid-1980s became America's second-largest law firm by poaching star partners. In 1988 it collapsed under the strain of their pay packages and infighting. In one quarrel, a lawyer broke another's glass-topped coffee table.

In the past decade, however, the model has come under greater threat, with the rise of Kirkland & Ellis. The Chicago-founded firm has also grown rapidly through lateral hiring, but enjoys advantages that Finley Kumble did not. One is its dominance of the lucrative business of advising private-equity titans. Another is its use of "salaried partners", who unlike traditional equity partners have no ownership stake. In 2025 Kirkland hauled in revenue of $10.6bn, a record for the industry, and could pay its lawyers record sums.

Rivals have had to take notice. Annual pay above $10m used to be shocking. Now rainmakers switching firms can command $20m or even $30m. Headhunters say that many partners move in search of a change in culture or practice area. Fellow lawyers are more cynical. "I've heard people say, 'I wasn't treated fairly at my firm.' What the hell does that mean?" notes a litigator at a leading shop. "It's all about the money."

For other firms, Kirklandisation has become almost a necessity. Clients increasingly expect them to provide global coverage in a vast range of specialities. Hires are therefore often meant to anchor a new office or field of expertise and bring in new business. Because the American Bar Asso-

ciation, a professional body, prevents non-compete agreements, lawyers are free to take their Rolodexes to a new job. Moreover, as firms grow, many clients have become attached to partners rather than firms. "You used to hire firms," says a New York partner. "Now you hire individuals."

Individuals can change more than the bottom line. Many partners point to Scott Barshay's move from Cravath to Paul Weiss in 2016 as a harbinger of the free-agent era. By all accounts Mr Barshay has kept raking in profits. He has also transformed Paul Weiss, which he now chairs. The *New York Times* reported this year that he was key to its reaching a deal with Donald Trump's administration, after the president signed an executive order targeting the firm in March 2025. Among other things, Paul Weiss agreed to carry out pro bono work for causes approved by the administration. Before the 2024 presidential election, its partners donated more to Kamala Harris than any of their peers did.

Poaching does not always work. More than 60% of lateral hires fail to bring in the business they promise, estimates Decipher Investigative Intelligence, a research firm. Half leave within five years. Many moves come with one- to three-year guarantees, allowing firms to part ways with underperforming stars, notes Scott Yaccarino, a recruiter. But placing several bets worth tens of millions can cost dear.

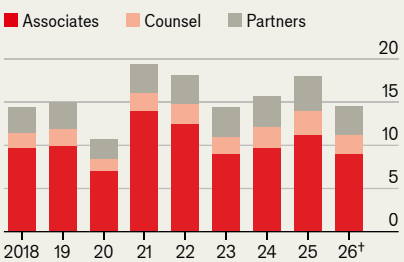
Still, many managing partners feel they must poach or die. After suffering several defections and failing to make commensurate hires, Weil Gotshal is reportedly shopping around for a merger. Even Wachtell may have to fully abandon both its seniority-based pay structure and its "lean and mean" operation. That will mean an industry with fewer and bigger firms, which will look more like Kirkland.

For partners accustomed to the clubby comfort of the past, such changes are unsettling. Some lawyers, particularly litigators, are leaving big firms to found tiny boutiques that recall the 19th-century world of American advocacy. Others are racing into the future. David Fox, formerly of Kirkland, has started one of several "AI-first" firms that employ a few lawyers and leave the rest to bots. Mr Fox has claimed his software can perform as well as a lawyer on the verge of partnership.

Kirklandisation may merely have exposed the Cravath system's internal flaws. As law firms grow, to more than 100 or so partners, they tend to end up being run by a small group. Kirkland's salary model then looks tempting. For years some in the industry have speculated that the most successful firms may one day seek to get rid of their current partnership structures entirely and seek outside investors in order to gobble up smaller competitors. Investment banking ditched large-scale equity partnership and consolidated long ago. Big Law may soon follow. ■

Leaping lawyers

Number of lateral hires at top 200 US law firms*
By position, '000



*Hires of practising lawyers globally by Am Law 200 firms
†To September 15th
Source: SurePoint

The rocket business

Space jam

A looming rocket shortage offers SpaceX's competitors a rare opportunity

LAST YEAR was the busiest ever in the rocket business. BryceTech, a firm of analysts and engineers, counted 325 orbital launches, 25% more than in 2024. They deployed 4,544 spacecraft, a 54% rise. And 2024 was itself a record-breaking year—as was each of the three previous years.

Yet amid this unprecedented abundance, the space industry is gripped by worries of a rocket shortage. "I think 'panic' is the operative word here," says Caleb Henry, director of research at Quilty Space, another firm of analysts. The looming shortage was the "number one theme" at World Space Business Week, a conference in Paris in mid-September, he adds.

The cause of the panic is SpaceX. Elon Musk's firm is the industry's dominant force, and its cheap, reliable Falcon 9 is its workhorse. In 2025 SpaceX flew 43 launches for outside customers, carrying hundreds of satellites either on dedicated flights or on "rideshare" missions in which several clients share a rocket. Rocket Lab, a New Zealand-American company that is SpaceX's closest competitor in commercial launches, flew 18 times—but its diminutive Electron rocket has less than 2% of the payload capacity of the Falcon 9.

Despite the Falcon 9's dominance, its days are numbered. SpaceX is keen to switch to Starship, a new, much bigger and fully reusable rocket now in development. Recently Gwynne Shotwell, SpaceX's chief operating officer, said that the firm was keen to "obsolete [sic] our own products" before its competitors can. Exactly when the Falcon 9 will stop flying is unknown. But on September 28th a Starship test flight put SpaceX's Starlink broadband satellites into orbit for the first time. And 2026 looks set to be the first year in which the number of Falcon 9 flights falls rather than rises. The word on the street, says Simon Potter of BryceTech, is that it is impossible to buy a Falcon 9 launch after 2028.

For many years satellite-builders had assumed that SpaceX would offer Starship launches to its customers in the same way as with the Falcon 9. That now looks less certain. The firm is vertically integrated, building and flying Starlink satellites, of which more than 11,000 are active in orbit. Starship is vital to Starlink's growth—and to flying the many thousands of "Starmind" data-centre satellites that SpaceX plans for its artificial-intelligence business. Those demands may leave no Starships free for other work, at least for a while. Crunch the numbers and it is not clear where other customers would fit in, says Mr Potter. And SpaceX would not be giving up much. Its launch business accounted for just 8% of its $12.5bn revenue in the six months to June 30th, with the rest coming from Starlink and the AI business.

The prospect that SpaceX may in future keep its rockets to itself has caused dismay among satellite operators—but excitement among the firm's rivals. On an earnings call on September 10th Giulio Ranzo, the boss of Avio, an Italian company that makes the Vega-C, a smallish rocket, said: "We have a whole bunch of customers coming to us and saying, 'Oh my God, can you fly us? Because SpaceX no longer gives us a chance to launch'."

The Falcon 9's retirement would also mean the end of SpaceX's Dragon spacecraft. That would leave Boeing as the only

Lifting off and pushing off

company able to fly astronauts to the International Space Station, on its troubled, lossmaking Starliner capsule. On September 28th NASA said that it had bought two new flights from Boeing, adding that it will pay the firm extra to help it fix Starliner's manoeuvring thrusters, which have caused problems on previous flights.

Plenty of smaller, hungrier companies will be hoping to take advantage, too. The first flight of Rocket Lab's Neutron rocket, which has over half the capacity of the Falcon 9, is due later this year or early next. Relativity Space, a startup now bankrolled by Eric Schmidt, an ex-boss of Google, is working on the Terran R, a Falcon-9-sized rocket of its own. Blue Origin, owned by Jeff Bezos, may also be a contender, though it suffered a serious setback in May when a rocket explosion destroyed its only launchpad. Mr Henry points to Firefly Aerospace, an American firm, and Isar Aerospace, a European one that reached orbit for the first time on September 5th, as other commonly mentioned names.

SpaceX has years of experience and hundreds of flights under its belt, while all the new contenders are unproven. But Mr Potter notes that in the past few months, perhaps in anticipation of the fight to come, rocket firms' pitch decks have emphasised manufacturing prowess and scalability as much as the rockets' technical attributes. If the 800-pound gorilla does quit the market, it will leave plenty of juicy spoils for others to scrap over. ■

BARTLEBY

You're not sending an email. You're training a model

Privacy, employee data and the demands of AI

ONE WAY to think about AI note-takers is as a productivity tool for humans. You might be unable to attend a gathering but you can still find out what happened by reading a summary later. You can receive a set of specific action points rather than operating on the vague assumption that Doris will follow up on something or other.

But another way to think about AI note-takers is as a data-gathering exercise for other machines. Transcripts can be scoured for previously unseen patterns: how much individuals are contributing; where coaching is needed. And as offhand remarks are turned into permanent records, more employee communication becomes available as training material for AI models.

The idea that the sorts of conversations you are having with your colleagues could pave the path to superintelligence may seem laughable. But consider the court battle in America over Google's purchase of data from Spirit Airlines, a bankrupt low-cost carrier. The tech giant bid $10m to get hold of Spirit's data, including 100m email messages and 500m Teams titbits.

Buying Spirit's data gives a model-maker such as Google not just aviation-specific information on pricing and fuel costs, but also more general training data on negotiations, strategy and more. This kind of material is not generally available on the wider internet. "Reddit threads are not a good way to learn how to run a business," says Greg Ewing of Spencer Fane, a law firm.

The Google/Spirit transaction is being challenged by, among others, a rival bidder called Micro1, which also fancies getting its hands on the airline's data, and Springshot, a software firm that reckons its intellectual property is being sold on. But it is also being challenged by a trade union representing Spirit flight attendants, which argues that the deal includes confidential employee data. (The next court hearing is due to take place on October 14th.)

If Spirit had been a European airline, says Jeremias Adams-Prassl, a law professor at Oxford University, this transaction would already have run into big privacy-related obstacles. Under GDPR, a European data-protection law, data collected for one reason cannot then be used for a different one, a concept known as "purpose limitation". That makes it difficult to train AI models on employee correspondence whose original purpose was to help an airline perform better.

Expectations of employee privacy are baked into wider European law, too. In 2017 the European Court of Human Rights considered the case of Bogdan Barbulescu, a Romanian sales engineer sacked after his employer had downloaded correspondence from an instant-messaging account, including some "of an intimate nature". Mr Barbulescu had

signed forms agreeing that the account was for work purposes, and flatly denied using it for personal reasons. Even so, in a precedent-setting judgment, the court concluded that "an employer's instructions cannot reduce private social life in the workplace to zero."

Some might smirk at this. While Europeans assert their fundamental human right to send intimate messages to each other at work, American frontier labs are acquiring more internal data and stretching their lead. But even if the legal environment is more permissive across the Atlantic, turning employee interactions into AI feedstock imposes other costs. Lawyers warn, for example, that AI meeting transcripts greatly expand the amount of material that is discoverable in litigation.

Wherever they are based, employees tend not to warm to the idea of being monitored. Meta, for example, suffered a well-publicised internal backlash earlier this year when it tried to roll out a programme that recorded employees' mouse clicks and keystrokes. "The argument that privacy and competitiveness are in conflict is too simple," says Mr Adams-Prassl.

AI is nonetheless likely to change the way employees behave. At the moment workplace communication is a melange of private and professional messages. Even people at the top don't necessarily distinguish between work and personal life: one tech boss recalls being given access to a client's data and finding a CEO's passwords in a Word document. But if people assume internal corporate data are going to train smarter models, outlasting even the firms in which they were generated, employees will become more guarded. The workplace will be more professional, and more dull.

SCHUMPETER
Speedball capitalism

A financial revolution is taking place in America

In the past year American markets have digested the largest-ever initial public offering (SpaceX), equity raise by a public company (Google) or a foreign firm (SK Hynix), private-funding round (OpenAI) and private-debt deal (Broadcom), as well as most of the biggest bond issue in history (Amazon). There was the first $1trn exchange-traded fund, or ETF (Vanguard), a record-breaking cash pile (Berkshire Hathaway) and stock-buyback programme (Nvidia), and the consummation of the largest-ever leveraged buyout (Electronic Arts). Then there are the enormous mergers in railways (Union Pacific and Norfolk Southern), utilities (NextEra and Dominion Energy) and media (Paramount and Warner Bros Discovery)—the last of which led, this week, to the largest ever junk-bond offering, beating a record set only last week (SoftBank). New markets have been created out of thin air (for compute) or become much bigger (for predictions). Worries about massive corporate scandals (First Brands) and hedge-fund blow-ups (Situational Awareness), which would have once occupied investors' attention for months, are steamrollered by the relentless, totalising and extraordinarily flexible machine that is American finance.

One nation under NASDAQ

The great powers of this system increasingly defy categorisation. Old distinctions—between the state and the market, individuals and firms, financial and non-financial firms, public and private markets, investment and gambling—have bent under their weight. What, for instance, is Nvidia? A company, certainly (the world's most valuable); a venture-capital investor (arguably Silicon Valley's most important); a source of American power (run by a showman with the ear of the president); and a lender, with plans to backstop loans made by the likes of Goldman Sachs and Apollo. And what, pray tell, are they? Goldman Sachs is best known for trading and giving advice on mergers, but increasingly resembles a giant asset manager, having just bought two firms that issue ETFs. And Apollo? Part life-insurer, part private-equity fund.

Every actor in the American economy is best understood by its position in the country's wager on artificial intelligence. Maintaining the technological lead over China and keeping stock prices high are the main tasks of government; opposing government mostly involves opposing AI data centres. As in a war economy, the obligations of enterprise are ambiguous (bosses recently signed a "morally binding" accord on AI safety) and the balance-sheets of the biggest companies have been requisitioned for the cause. As well as borrowing heavily in bond markets, Amazon, Google, Meta and Microsoft have $2.4trn of binding spending commitments, mostly tied to data-centre projects, lurking in the footnotes of their financial reports—or "off balance-sheet".

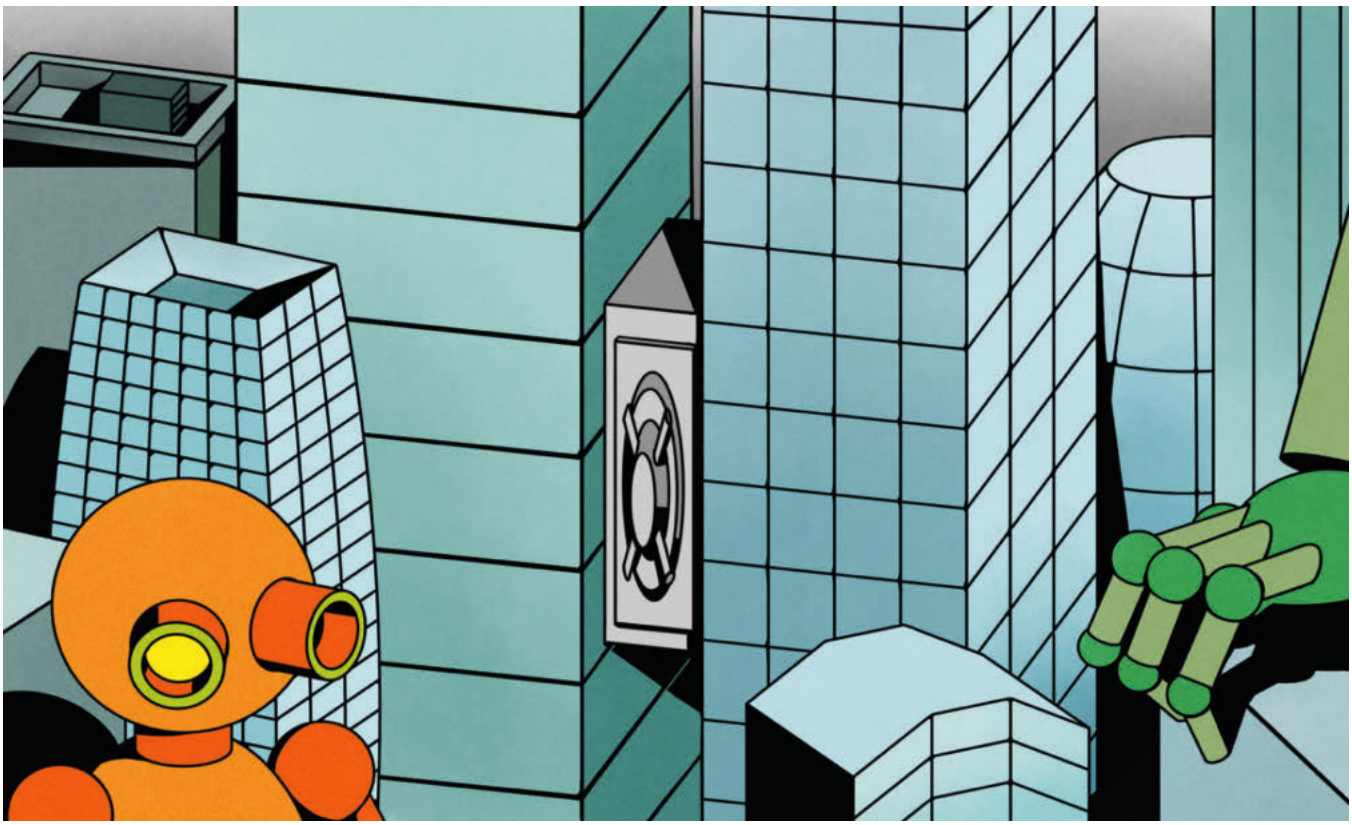
A much-chatted-about new paper by Stijn Van Nieuwerburgh of Columbia Business School calculates that if (an admittedly punchy) 183GW of new data-centre capacity is built by 2032, the capex bonanza will be larger, as a share of the economy, than America's railway and dotcom booms combined. Spillovers are visible everywhere: in the $176bn order backlog of GE Vernova, which makes gas turbines, in the capital spending of big utility firms and in the earnings of banks, which arrange financing for data centres and grease the wheels of speculation. Each American's patriotic duty is to keep spending: because of soaring stocks, household net worth is nearly 830% of disposable income, a record high. And keep betting: when a man is tired of American capital markets, he is surely tired of life itself.

All this brings to mind a speedball, that combination of heroin and cocaine popular among rock stars a generation ago. Anthropic's listing would flood the market with shares. Its prospectus reportedly discloses 11-figure losses and discusses the end of the world. Hell of a downer. But its products are almost magical and its revenue grew 12-fold last year. Private-capital markets have consumed their own diametric cocktail. A slow-burning crisis over bad private-equity deals made when interest rates were low is happening at the same time (and at the same firms) as the explosion in lending to data centres. And, of course, America's economy is running hot; so hot that yields on its government debt—which the political establishment seems less interested in controlling—have leapt to their highest level in more than two decades.

The rest of the world looks at America's capital machine with awe and envy. Silicon Valley firms pitch up to raise record sums in foreign bond markets before disappearing like bandits, on to the next currency. European central bankers deliver absurd speeches in which they promise the virtues of the American system with none of its vices. In fact, they seem to have managed just about the opposite, having imported the less obviously productive parts of Wall Street, like private equity and nihilistic debt-restructuring techniques, and none of its truly useful bits, like securitisation or venture capital.

Those at the white-hot centre of it all now ask: "What might break?" This is a sly question, since anything "might break"; and the thing that actually "breaks" matters less than all the things that are almost broken when it does. Yet the queasy stories are piling up. It is hardly auspicious that the riskiest corners of the life-insurance market are coming under pressure for their private-credit investments when this great well of capital is needed most. Or that a firm should cry "force majeure" over foreseeable risks to its data-centre projects, as Oracle, a tech giant, is said to have done over a project in New Mexico. Or even that firms, presumably on their bankers' advice, are postponing their listings, as Oura, a wearable-technology company, has done. Oura cited market "uncertainty". Yet the market is within a hair of its all-time high. Better, maybe, to keep going while the machine still whirrs. ■

Finance & economics



AI and taxes

Coffer crunch

SAN FRANCISCO

How artificial intelligence could worsen the rich world's fiscal squeeze

VAST SPENDING on AI infrastructure has created a profit boom. In the past year American pre-tax corporate earnings have grown by $1.2trn, or 30%. Companies in the S&P 500 index are handily beating forecasts. And yet Uncle Sam is seeing little of the benefit. Taxes on corporate profits have collapsed (see chart 1 on next page, left panel). Government bean-counters did expect lower receipts this year, owing to changes that let firms write off more investments against their tax bills. But the scale of the drop has surprised even them, as unusually profitable tech giants are writing off unusually high capital spending. The episode hints at how AI may upend tax systems across the world.

You might think of AI as a godsend for governments, many of which are accumulating vast debts (see chart 1, right panel). Economists expect AI to lift productivity growth, which should create a bigger economy. In a recent paper Karen Dynan, Douglas Elmendorf and Louise Sheiner, three economists, model the potential effects of AI on America's debt. Higher economic growth could reduce the debt-to-GDP ratio after 30 years by 50% or so of GDP, relative to where it would be otherwise.

Good news? Kind of. America's debt ratio keeps rising—just less quickly than before. Even in an America where AI turbocharges the economy, by the 2050s the government could be much more indebted than today. The risk is that tax receipts continue to lag behind spending, sometimes by many trillions of dollars. It is even possible that AI may not reduce budget deficits, but widen them.

One problem is that AI may create value in areas which governments tax lightly while destroying value in the more heavily taxed bits. The economy could expand rapidly while leaving the government increasingly starved of funds. At present governments get most of their revenues from workers. Across the OECD club of mostly rich countries, personal-income taxes and social-security contributions account for a high and rising share of overall tax revenues (see chart 2). In America about three-quarters of all federal tax revenues come from labour. Taxes on consumption and capital (including corporate profits and capital gains) are smaller.

This tax mix may not suit a world in which AI displaces workers or radically reduces their salaries, even as corporate profits balloon. "As AI reduces labour's share of income, the traditional tax base erodes," argue Anton Korinek of Anthropic and Lee Lockwood of the University of Virginia, in a recent paper. The Budget Lab, part of Yale University, explores a range of scenarios in which AI boosts annual GDP growth to as much as 3.3%. This produces higher tax revenues—but the extent of the boost depends greatly on

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the distribution of the extra income between capital and labour.

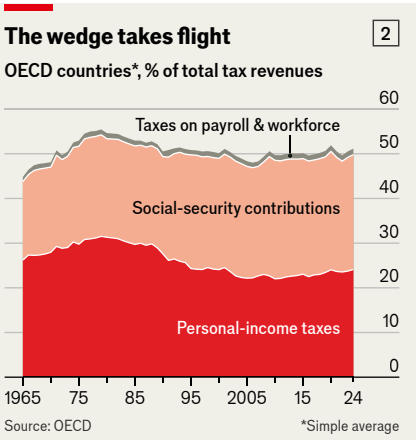
The Yale economists envisage AI having fairly modest economic effects for the time being. GDP growth of 3.3% would be impressive, but not unprecedented. Others envisage a deeper economic transformation. In an essay last year Philip Trammell of Epoch AI, a think-tank, and Dwarkesh Patel, a podcaster, entertained the possibility that "the share of total income paid to robot-owners" would rise to 100%. A shift need not be this extreme to produce a profound fiscal shock.

To gauge how profound, *The Economist* has calculated the effects on tax revenues of AI causing labour's share of national income to decline by ten percentage points. A few countries with relatively high taxes on capital, including Australia and South Korea, emerge relatively unscathed. But the majority of governments that rely on labour taxes would be crippled (see chart 3 on next page). In Italy, the hardest-hit big country in our model, the budget deficit could more than double as a share of GDP (assuming that spending remains constant relative to GDP). France and Germany are not far behind. America takes a smaller but still significant hit—both because it taxes labour fairly heavily relative to capital and because wages and salaries form an unusually large share of GDP.

Now my advice for those who die

A falling tax-to-GDP ratio is not necessarily a bad thing. If real GDP rises by 30% and real tax revenues rise by 20%, the government might be smaller relative to the economy but it would still have more resources at its disposal. And if the government could then deploy AI, it might also be able to spend that money more effectively. In health care, for instance, AI could improve scheduling, diagnostics and drug discovery, while reducing the overall cost of the system. It could usher in a golden age of public services.

However, a less shiny outcome is also possible. AI could raise the cost of providing public services, even as it increases demand for them. Governments may then find themselves needing more tax revenues just as these get harder to obtain.



One problem is what economists call "cost disease". In industries where AI productivity effects are largest, such as software development, wages will rise. What is great for software developers is, however, not so great for other firms and especially the public sector, where productivity does not grow as fast. Public institutions will still need to pay staff more in order to stop them leaving for better-paid jobs. The costs of providing a given public service will rise. This is the fundamental reason why the government tends to get bigger as a country gets richer.

Even as it raises their cost, AI may also increase demand for public services. People displaced by AI would need retraining and unemployment support. Some governments could consider a universal basic income to ward off the mobs. In our scenario, if the Italian government wanted to replace the lost earnings of the people that AI had displaced, it would need to find another $200bn per year as its tax revenues declined by $90bn, a fiscal deterioration of about 10% of GDP.

One way for fiscal systems to cope is to slow or stop the adoption of AI. Daron Acemoglu of the Massachusetts Institute of Technology, Andrea Manera of the IMF and Pascual Restrepo of Yale University have argued that America's tax system favours automation by taxing labour heavily while offering generous tax treatment to assets such as software and machinery. This, they say, gives bosses an incentive to replace workers with technology.

Some economists therefore prefer taxing AI directly, perhaps through a "compute tax" (which would be levied on the computing power used to train models) or a "token tax" (on each use of a model). With the public increasingly hostile to AI, such a tax could be popular even if it came at a significant cost to the economy, deprived of AI's benefits.

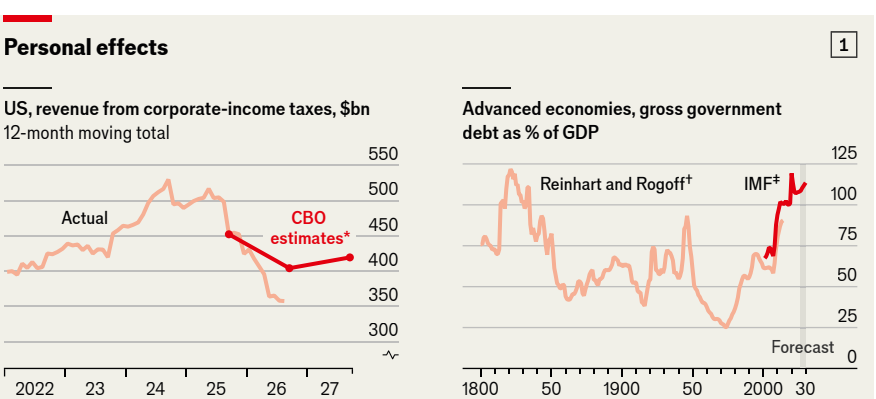
However, it might raise surprisingly little revenue. We estimate that American consumers and businesses currently spend about $700bn a year on building and running AI models. This is a tiny tax base relative to the total labour bill ($16trn last year). AI spending may balloon from here, but by how much is anyone's guess. Total capital investments in AI from big tech firms are expected to level off from 2028, whereas technological improvements might mean that businesses' spending on AI need not rocket. The revenue from such a tax is thus highly uncertain.

Rather than tax AI at source, therefore, governments are likelier to go after the gains of the AI boom. One option is to tax wealth. In November Californians will vote on a proposal to levy 5% on the net wealth of residents worth over $1bn. Politicians in other states like the idea and several big-cheese economists, including Mr Acemoglu, have endorsed it. Lowering the threshold below $1bn would net more of the multimillionaires the AI boom is minting.

Declare the pennies on your eyes

Yet robust academic evidence shows that the rich respond to wealth taxes by shifting assets offshore, or moving themselves. Privately owned businesses, where many keep their riches, are also notoriously hard to value, making it hard for the taxman to know how much to demand. What was John D. Rockefeller's stake in Standard Oil, which was not listed, really worth? Such considerations help explain why many countries that had introduced wealth taxes have abandoned them. In 1990, 11 rich countries raised money by taxing fortunes; today only four do.

Even if AI could help determine the "true" value of an unlisted business or if super-clever AI tax auditors make it harder to shift assets abroad, a wealth tax might not raise anywhere near enough money. In March a group of American lawmakers introduced a proposal to establish a 2% annual tax on the net worth of households and trusts valued at over $50m, and an



additional 1% surcharge on those above $1bn. According to the (extremely generous) assumptions of the bill's proponents, it would raise the equivalent of 15% of current labour taxes. In reality, it would probably raise a lot less. Most independent studies of wealth taxes find that they also slow the economy, reducing the receipts from other taxes.

Perhaps AI will create many more mega-fortunes for governments to tax more highly? The trouble with such an assumption is that many appear to be donating money to charity as fast as they make it: all seven co-founders of Anthropic, the leader AI lab seeking a $2trn initial public offering, have pledged to give away 80% of their wealth. An AI world could have a huge share of personal wealth tied up in charitable foundations rather than bank balances and mansions. And if taxable fortunes do not grow fast, then high tax rates would chip away at them quickly.

Among economists a consensus is therefore emerging that changes to existing taxes may be preferable to new levies. This is especially true of taxes on capital income, which includes corporate profits, dividends and realised capital gains. If AI lives up to the hype, related capital income could be truly enormous.

One reform would be to bring tax rates on capital gains closer to those on labour income. California already applies ordinary income-tax rates to realised capital gains. Belgium recently introduced a 10% tax on gains from financial assets, closing a loophole that had previously spared such income. Australia is raising taxes on realised capital gains, but giving some relief to taxpayers by adjusting those gains for inflation. The Netherlands is seeking to replace a bizarre tax system in which everyone was assumed to have the same investment gains—giving good investors an advantage—with one taxing actual gains.

To minimise the extent to which the tax system discourages enterprise, levies on capital gains (above the normal return on investment) could be matched by equally generous relief for any capital losses. People who made a negative return on an investment could, for example, write off these losses not just against future capital gains, but perhaps also against past gains and other income.

Corporation taxes could chip in more, too, especially if the AI economy produces more "excess profits" above what a company needs to earn in order to invest. These could be generated by monopoly power, scarce land or unique data sets. In theory such extraordinary gains can be taxed with relatively little damage to incentives. Already a number of countries are raising corporation taxes, reversing the trend of the 2000s and 2010s.

AI also strengthens arguments for the taxes that many economists love best: on consumption. These are difficult to avoid and can raise vast sums. Rich people spend more than poor people, and therefore pay the lion's share of consumption taxes. To the extent that governments want to compensate poorer households, which tend to pay more than the rich as a share of their income, they can do so through the welfare state. As machines take a greater role in producing income, the taxman may have to pay less attention to how money is made and more to where it goes. ■

Of foie gras and fegatini 3

Decrease in tax revenues if labour's share of national income falls by ten percentage points*
% of GDP, selected countries

*National income from domestic production. Based on 2023 data
Sources: OECD; *The Economist*

AI and payments

Card games

Artificial intelligence meets a new kind of crypto scam

VENTURE CAPITALISTS are pouring so much money into artificial intelligence that little is left over for anything else. Some, though, retain a soft spot for an earlier mania: cryptocurrency finance. In the first half of this year crypto startups received $10bn in venture capital (VC) through nearly 750 deals. Besides using lots of computing power, AI and crypto have had little in common. But an investigation by *The Economist* suggests the two are tightly linked by a booming illicit trade.

Issuance of crypto-enabled credit and debit cards is surging. These cards connect crypto wallets to payment terminals used by ordinary businesses. They allow people in poor countries with unreliable currencies to transact in stablecoins pegged to sturdier foreign monies, notably the dollar. But crypto cards have other big customers, too. Many card providers do not seriously try to verify the identity of their cardholders. That has left them open to abuse by criminals and sanctions evaders. They also let people in places like China buy subscriptions to American frontier AI models, circumventing the model-builders' restrictions on sales to America's enemies.

Crypto cards convert digital currencies into real-world money when swiped at Visa and Mastercard terminals, and their use is growing. Although such cards can be linked to bitcoin, ether and other unstable coins, the volatility of unpegged cryptocurrencies makes them a poor means of exchange. Stablecoins, whose overall market value has nearly doubled in the past two years to over $300bn, are still mostly stores of value. But stablecoin cards were used in transactions worth more than $1.1bn in August, three times as much as a year earlier, according to Paymentscan, a data provider. The true figure is probably higher; many crypto-card providers do not report their transaction volume.

In the past couple of years regulators in America, Europe and Hong Kong have encouraged this growth by writing clearer rules for digital currencies. As a result, cryptocurrencies in general, and specifically stablecoins, have gained respectability. In particular, in an aggressive effort to capture a slice of the growing crypto economy, Visa and Mastercard have begun offering membership in their networks, hitherto mostly reserved for banks, to startup card-issuers with payments licences.

These issuers in turn lend their network access to providers of crypto cards. Among the largest of the issuers are Rain, Reap and Wirex. All three say they take their compliance responsibilities seriously. In January Rain raised $250m from VC funds, including big names like Iconiq and Lightspeed. The crypto providers sign up customers under their own brand. Between April and September the number of identifiable providers grew from around 100 to 250. The involvement of Visa and Mastercard enables the crypto providers to offer, via the issuers, cards carrying the two payments giants' logos (as well as their own) that can be used anywhere in the world.

The spread of stablecoins worries regulators in poor countries, for it could lead to pseudo-dollarisation of their economies. But it should concern their rich-world peers, too. Stablecoins, notes Eswar Prasad of Cornell University, "create additional conduits for illicit financial transactions,

both within and between countries". Illicit crypto flows exceeded $158bn last year, estimates TRM Labs, an analytics firm.

For businesses that want to block certain customers, crypto cards can make it hard to verify who a cardholder is or where they are. Persian- and Russian-language sites, for example, advertise their own or third-party cards to, among others, Iranians and Russians, who are locked out of the Western financial system by sanctions against their countries. One Russian-language site boasts that the cards enable users to "pay for ChatGPT, Claude, Netflix, Spotify...and other international services".

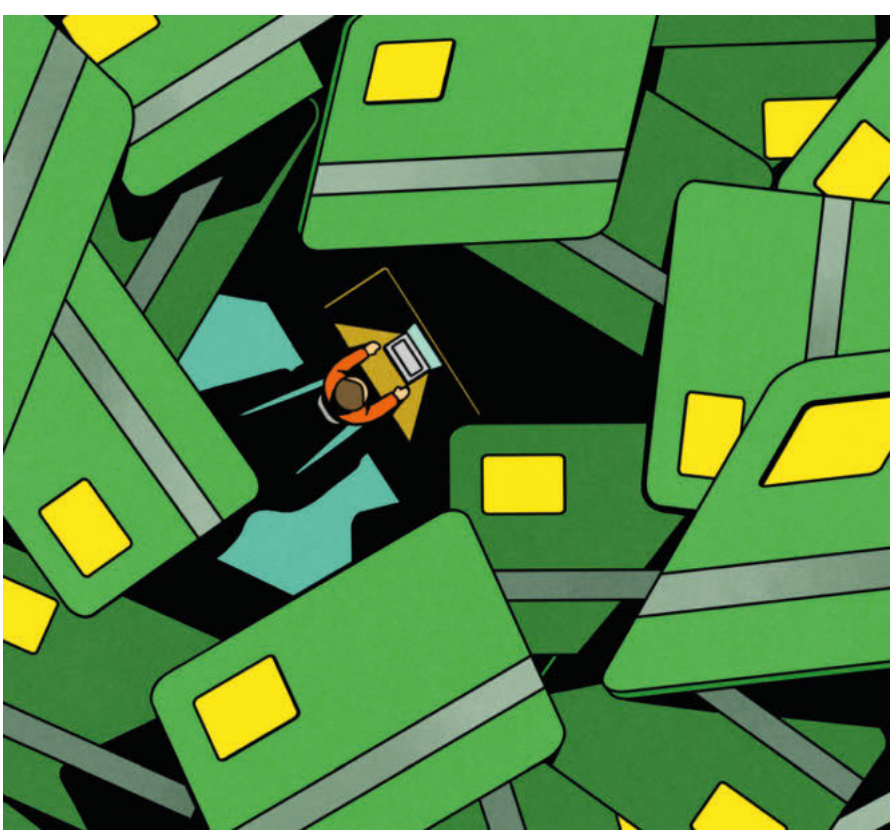
Like payment cards from digital "neobanks" such as Monzo or Revolut, crypto cards can in principle be issued only to verified customers who have gone through the usual know-your-customer (KYC) checks. In practice, however, the providers are not always assiduous. For some of them, weak KYC is a selling point.

A report in April by Crystal Intelligence, another analytics firm, found that nearly seven in ten crypto-card providers had weak KYC requirements or none at all. About half of the no-KYC providers identified by Crystal Intelligence were based in Hong Kong and a fifth in America. Others sell virtual cards, in some cases entirely legally, while being licensed in the Caucasus or Gulf countries, where KYC rules are weaker and less vigorously enforced.

It is possible that some of the firms advertise weak- or no-KYC cards but demand identity verification after a cardholder begins using one. Some can verify identities without asking for documentation. Many crypto-card providers enforce rules on their products and are not dodgy. A few even cap payments to vendors like Starlink, a satellite-internet firm, whose products are illicitly used by Russia's armed forces and its shadow oil-tanker fleet.

But in some instances, Crystal Intelligence found, providers did not ask for identity information. In others no attempt was made to verify the information given. *The Economist* applied for a Visa card through a Russian-language channel on Telegram, a messaging app, that promised "no documents, no waiting and no long forms" and had it linked to an Apple Pay wallet in minutes. The provider is called CinCin, and did not clearly identify the card issuer it worked with. The card advertised monthly payment limits of $500,000. Visa says it follows the law everywhere and prohibits illegal activity on its network.

One growing use of crypto cards is to let Chinese users pay for access to American AI models and chatbots such as Claude and ChatGPT. Chinese social media are full of guides on how to circumvent American restrictions and a domestic ban on foreign AI. Mostly users pay money to a third-party "transfer station", which purchases AI tokens (the chunks of data processed by models) in an unrestricted country and resells them to users in China. Other tutorials teach Chinese users to buy subscriptions directly from the labs, by purchasing fake identities or obscuring their whereabouts with crypto cards.



At least one online Chinese-language guide for dodging the American AI labs' country restrictions advises users to apply for a Mastercard, from a provider in Dubai. Users in mainland China can employ a virtual private network to pretend to be in another country and pick an address in that country on Google Maps to get their card. Providers like the one in Dubai claim to do KYC. Yet in the Chinese tutorial video, the host reassures viewers that the provider does not attempt to verify if the address is real, as a bank might by asking for a utility bill. Mastercard says that it takes governance seriously and prohibits reloadable "no-KYC" cards on its network.

La Paz hold 'em

A source at an American crypto provider whose cards are issued by Rain says that each month he sees hundreds of users signing up for his firm's Visa cards with Chinese-issued IDs with addresses outside China. In one instance, his firm observed a user with an IP address in Shenzhen, a city in southern China, and a Chinese ID apply for a card with a non-Chinese address; the user then spent $50,000 on ChatGPT subscriptions in the Bolivian Apple app store. Rain says it is investigating these reports.

Rain accepts Chinese IDs, since their holders could be residing abroad, but rejects users who apply with a Chinese address. Yet a number of providers which offer cards via Rain allow users who fail KYC checks by entering a Chinese address to give it another go, with one explicitly encouraging them to "try again with another address". Rain says it maintains high standards for its compliance programme, takes seriously attempts to circumvent its controls and abides by all regulations.

Since late 2024 one Hong Kong-based crypto provider serving Chinese-speaking users, and tracked by Crystal Intelligence, has seen more than $355m in dollar stablecoins flow into its card-linked wallets and has processed 4,000 transactions. Some observers suspect that many of these transfers are used for wholesale purchases of AI tokens. "I would hazard a guess", says the source at the American crypto provider, "that a substantial amount of the $1bn-a-month stablecoin card spend is funding the grey token market."

On September 10th Anthropic accused Alibaba, a Chinese internet giant, of illicitly gaining access to the Claude model 151m times between May and July to train its own models in a process known as "distillation". Anthropic alleges that to do so Alibaba used a pool of 5,000 Claude accounts, some paid for with virtual cards to hide its actions. Alibaba did not respond to a request for comment. ■

Financial astrology

Star-struck

MUMBAI

Financial astrology is a serious business in India

DURING DIWALI, a Hindu festival that falls in November this year, India's stockbrokers and their families gather for an especially auspicious one-hour trading session in which they place symbolic trades to usher in prosperity. They perform "puja" rituals to Lakshmi, the goddess of wealth, whose likeness hangs in the lobby of the Bombay Stock Exchange. Business-TV channels host astrologers who chart the movement of planets alongside fund managers speculating about earnings and interest rates.

Traders everywhere are a superstitious lot—no wonder, given how random markets can seem. Americans have lucky ties; Japanese stock-market investors look for omens in the first auction of the year; some Asian businesses pay premiums for things containing auspicious numbers. John Pierpont Morgan, the eponymous founder of a big American bank, is said to have subscribed to an astrology newsletter (even if he probably did not really say that "millionaires don't use astrology; billionaires do"). "Technical analysis", reading the runes of stock charts and searching for patterns such as "heads and shoulders" or a "rising wedge", has a cult-like following on many Western trading floors.

In India, this professional propensity meets an astrology market with $7bn in annual revenues. According to a survey by Pew Research Centre, 44% of Indian adults believe in astrology. Horoscopes are drawn up the moment newborns tumble forth. Doctors try to perform C-sections in a priest-appointed lucky time-window. Cars and motorbikes are delivered to buyers at propitious moments. Why should finance be any different?

One asset manager at a large Indian provider of mutual funds uses astrology to manage his personal wealth. He describes fundamentals as the "engine" and astrology as the "steering" which offers direction. It is not uncommon for money managers to be part of WhatsApp groups run by astrologers. Or take Purvesh Shelatkar, who ran two alternative-investment funds for high net-worth individuals until his sudden death on September 27th. A self-taught practitioner, he gave astrology roughly half the weight in his investment decisions. He once bought stocks in Jindal, a steelmaker, because Mars, a planet associated with red metal, was strong in his birth chart. The stock rocketed. He likened

The altar of capitalism

astrology to a "fog light": unnecessary when visibility is good, but useful when the road disappears into mist.

Retail traders have plenty of ways to brush up on the aspects and ingresses of celestial finance. Online courses promising to predict stock-market movements using astrology cost 35,000-45,000 rupees ($365-470). Online platforms bring traders, investors and astrologers together. In August one of these, InstaAstro, which has 12m registered users, raised $12m from investors. (Coincidence?) AstroTalk charges between 14 and 200 rupees a minute for consultations. It also sells merchandise: a "Dhan Yog" (financial prosperity) bracelet will set you back 699 rupees.

That is much more fun than reading the academic literature. Studies of Chinese and Taiwanese investors have shown that betting on firms linked to the auspicious number 8 and avoiding those associated with number 4 (which in Chinese is similar to the word for "death") produces significant underperformance: investors were willing to overpay for 8-laden stocks.

Anecdote confirms academia. Hersh, a financial analyst in India (who by day uses Claude, an artificial-intelligence chatbot, to crunch numbers), tried with a chum to trade options and futures using astrology during the covid-19 pandemic. "We failed miserably," he says.

Indian investors today have more money and more ways to put it to work than their elders, who relied on property, gold and bank deposits. But mutual funds, equities and derivatives bring greater uncertainty. AI can sift through filings and earnings calls. Quantitative models can discern patterns invisible to the human eye. Neither, though, can say for sure what happens next. Astrology cannot either. Just try telling believers that. ■

The economics of graft

Civil disobedience

SINGAPORE

Research on grasping bureaucrats startles Singapore

YOU WOULD not expect an 85-page equation-packed economics treatise to stir public feeling. Yet in Singapore, a new working paper by four economists, Tomasz Piskorski, Amit Seru, Jian Zhang and Chun Zhao, has. It finds that between 2007 and 2019 civil servants in the squeaky-clean city-state were 60% more likely to buy private flats within a kilometre of planned but unannounced Mass Rapid Transit (MRT) metro stations, compared with a control group. Those who subsequently sold got S$80,000 ($62,000) or so more than owners of similar flats in other parts of town for every year they held on to their purchases. The paper's publication within a week of Singapore's generously remunerated ministers being awarded a pay rise of up to 9% added to the furore. "Tip of the iceberg?" groused one netizen.

On September 25th the government said it took the unearthed "statistical patterns" seriously and launched a probe into potential cases of "actual misconduct". The paper has also caused a scholarly stink. Among the 20 academics named in the acknowledgments was Shengwu Li of Harvard University. Mr Li is the grandson of Lee Kuan Yew, Singapore's late founding father. He is also a vociferous critic of his homeland's recent governments. In a highly unusual step, two professors from the National University of Singapore (NUS), Ivan Png and Jessica Pan, asked for their names to be removed from the thank-yous, claiming they were there by mistake.

The intrigue is surprising because, if you look closely, the paper is not actually that damning. Its starting point is that a government cannot tell precisely how corrupt or honest its society is. It gets signals from things like corruption-attitude surveys, or the odd scandal. All these, though, are noisy data: both could reflect broader rot or be an outlier.

The government's response to sleaze, by contrast, is a clean signal: it can choose to tighten enforcement or not. This in turn causes bureaucrats to recalculate their risk-reward function for mischief. Collectively, their choices influence the strength ▸▸

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of the honesty norm. If its strength increases, and signals of corruption diminish, the government can afford to relax enforcement, and vice versa.

To see how this plays out, the paper examines a period around 2011-12, when Singapore tightened enforcement amid scandals unrelated to suspicious property purchases (a particularly tawdry one involved an official swapping confidential procurement information for sex). The researchers find that evidence of civil servants' savvy homebuying vanishes after 2011. In other words, where norms alone failed, tougher enforcement appears to have persistently boosted honesty. "We do not see the paper as an indictment of Singapore," says Mr Seru. "Quite the opposite."

The palaver over the study does, however, cast a shadow over Singapore's academic culture. In 2014 a critic of Singapore's ruling party, Cherian George, faced a self-described "forced exit" from Nanyang Technological University, after being twice denied tenure, allegedly against the advice of his tenure committee.

The reaction of Mr Png and Ms Pan suggests they felt that any association with the paper was costly. Ms Pan did not respond to a request for comment. Mr Png says, "The news media (and laypersons) seemed to conflate being acknowledged with endorsing or concurring with the conclusions." Nervous professors are a better problem to have than corrupt bureaucrats. But this, too, is a form of rot. ■

BUTTONWOOD
Nurturing capital

Investing for babies involves knotty trade-offs

PATIENCE, CONSISTENCY and frugality are fine habits for an investor. They are not, however, common virtues among newborn babies. The little bundles of joy are impulsive short-term thinkers. They are also some of nature's biggest freeloaders, living entirely beyond their means.

All the same, for new parents with a soft spot for the magic of compounding, like your columnist, a baby presents a tantalising investment opportunity. Many decades of life and potential returns stretch out into the future: $1,000 invested today with a real annual return of 5.2% (the average for global stocks over the past 125 years) would be worth $58,000 by the time the child is 80.

Yet these juicy financial returns must be considered alongside another sort of investment: in the children themselves. Spending on a child's health, education and skills can yield even higher returns (on top of helping them lead a fulfilling life). So how should a new parent think about their budding little portfolio?

Potential financial investments are manifold. Many countries offer tax-advantaged ways for parents to save for their children. American tykes are now eligible for Trump accounts, which start with a $1,000 freebie from the federal government that can be put into mutual funds or index trackers until the child turns 18, while taxes are deferred. Earlier options include the even more tax-friendly 529 accounts, but these may be used solely for specified educational purposes. In Britain, the Junior Individual Savings Account, into which up to £9,000 ($11,900) a year can be invested, is tax-friendlier still.

Investing for babies is a joy even without the tax perks. Long horizons smooth out volatility, allowing for more risk. Ian Ayres and Barry Nalebuff, two economists, propose that young workers should borrow to buy stocks, since the long-term returns to equities are (hopefully) higher than the interest on borrowing. If the logic is true for 20-somethings, it is truer still for infants with a horizon a couple of decades longer.

Some parents fear that creating a large trust fund will turn their offspring into layabouts. Andrew Carnegie, a 19th-century American steel magnate, gave away most of his wealth out of fear that it would dull the work ethic of his descendants. A more recent study of Norwegian inheritances between 2000 and 2004 found that those bequeathed NKr300,000 (some $40,000 at the time) or more reduced their working hours by 7-10% in the six years that followed.

This makes investments in human capital more appealing. In the 1970s and 1980s Gary Becker and Nigel Tomes, another pair of economists, theorised that parents were better off investing in education and skills early on. For example, teaching a child to read is not hugely expensive, but opens up huge opportunities for future earnings. Investing in financial assets starts to make sense later, once those human-capital investments encounter diminishing returns. A wealth of empirical evidence has since backed up the theory.

Beyond basic skills, however, investing in human capital gets fraught. Such investments are sunk costs that cannot be liquidated. They are also infungible: tutoring in advanced maths is not the same as Chinese lessons. Children react differently (and unpredictably) to each. The extraordinarily long timeframes that make investing for babies attractive also raise uncertainty. Perhaps the skills that are valuable today will be obsolete in an AI future?

So what is a new parent to do? The most obvious financial options involve accepting free money. The $1,000 in a Trump account is a no-brainer. Even rabid anti-Trumpers would be mad not to take it. All parents should also accept similar grants from several American states. Those who expect to be on the hook for university tuition would be wise to take advantage of 529 accounts or similar vehicles as early as possible, so that compounding can work its magic to the full. Any money that children are likely eventually to inherit is better off in tax-friendly shelters than in their parents' brokerage accounts.

Investments can misfire: violin lessons may generate little more than screeching and even a modest trust fund could encourage laziness. But with a bit of luck, investing in a child's future and on their behalf will pay off handsomely. Fortunate parents with the ability to do both should approach this like the rest of parenting: try different things and don't agonise too much over your decisions.

FREE EXCHANGE

Governments' new grab-bag

Today's governments want to hold America's shares more than its debts

AFTER THE pain of the Asian financial crisis in 1997 most of South Korea's foreign assets were deployed for one purpose—to defend the won. By 2016 the central bank had gathered a pile of dollars, euros, other currencies and gold worth $400bn. These net reserves, which could likewise be sold to prop up the local currency, made up roughly 80% of the country's total foreign assets. Financial policymaking was laser-focused on the next crisis.

The crisis never came, and foreign assets kept mounting—but for different reasons. Although South Korean foreign-exchange reserves are unchanged from a decade ago, the central bank has been overtaken by the National Pension Service (NPS), a government-owned nest egg, as the country's biggest international investor. NPS owns some $650bn-worth of foreign assets, up from $80bn in 2012.

South Korea is one of several countries where foreign reserves are no longer the government's only big source of financial firepower. Others include Japan, China and Saudi Arabia. In the past decade their governments have accumulated foreign assets in the portfolios of public pension systems, sovereign-wealth funds and state-owned banks. Some, like Saudi Arabia's Public Investment Fund (PIF), have attracted attention owing to flashy stakes in skyscrapers and sports tournaments. But their rise also has less visible implications for the financial system, because the new players take more risks than central bankers, hoping for better returns. This exposes governments' large stockpiles to volatility.

Like South Korea, most countries used to hoard foreign reserves defensively: they might need them to prop up the exchange rate in a crisis. Then, in economies with trade surpluses, policymakers found themselves selling the local currency to prevent its immediate appreciation, which could hurt exports. The result was a mounting pile of foreign assets, so large that it would never need to be liquidated all at once.

As such, they did not need all to be cashlike instruments or gold. Reserve managers could chase not just peace of mind but also returns, by buying riskier securities. Research published in June by Antonin Chenard, Barry Eichengreen, Eric Monnet and Florian Morvillier, four economists, assessed 109 central banks from 1950 to 2022. They found that at the end of that period almost two-thirds of their reserves were held in securities on average, up from one-third around 2000.

This suited policymakers fine. Income from foreign assets is handy while oil wells are depleted, populations age and demands on government spending rise. But returns were still limited because even the most audacious reserve manager would probably not stray beyond investment-grade credit.

Other types of sovereign investors, though, have had no such qualms. State-run banks happily make risky loans. Pension funds dabble in junk bonds. National rainy-day funds buy shares and even take over companies. Between 2014 and 2025, foreign assets held by pension and sovereign-wealth funds in China, Japan, Saudi Arabia and South Korea rose by at least $3.5trn. In the same period the world's total central-bank reserves grew by less than half as much, having ballooned from $2trn in 2001 to $11.6trn in 2014.

According to research by Brad Setser of the Council for Foreign Relations, a think-tank, in the past decade or so seven of China's biggest banks, all state-owned, have acquired foreign assets faster than the People's Bank of China. Their holdings amount to $3.5trn, roughly the same as the PBOC's foreign reserves. Japan's foreign reserves have hovered around $1trn for the past decade, but the foreign holdings of the Government Pension Investment Fund have risen from $400m in 2014 to $1trn in 2026. And Saudi Arabia's PIF boasts overseas holdings with a value close to that of the central bank's foreign reserves.

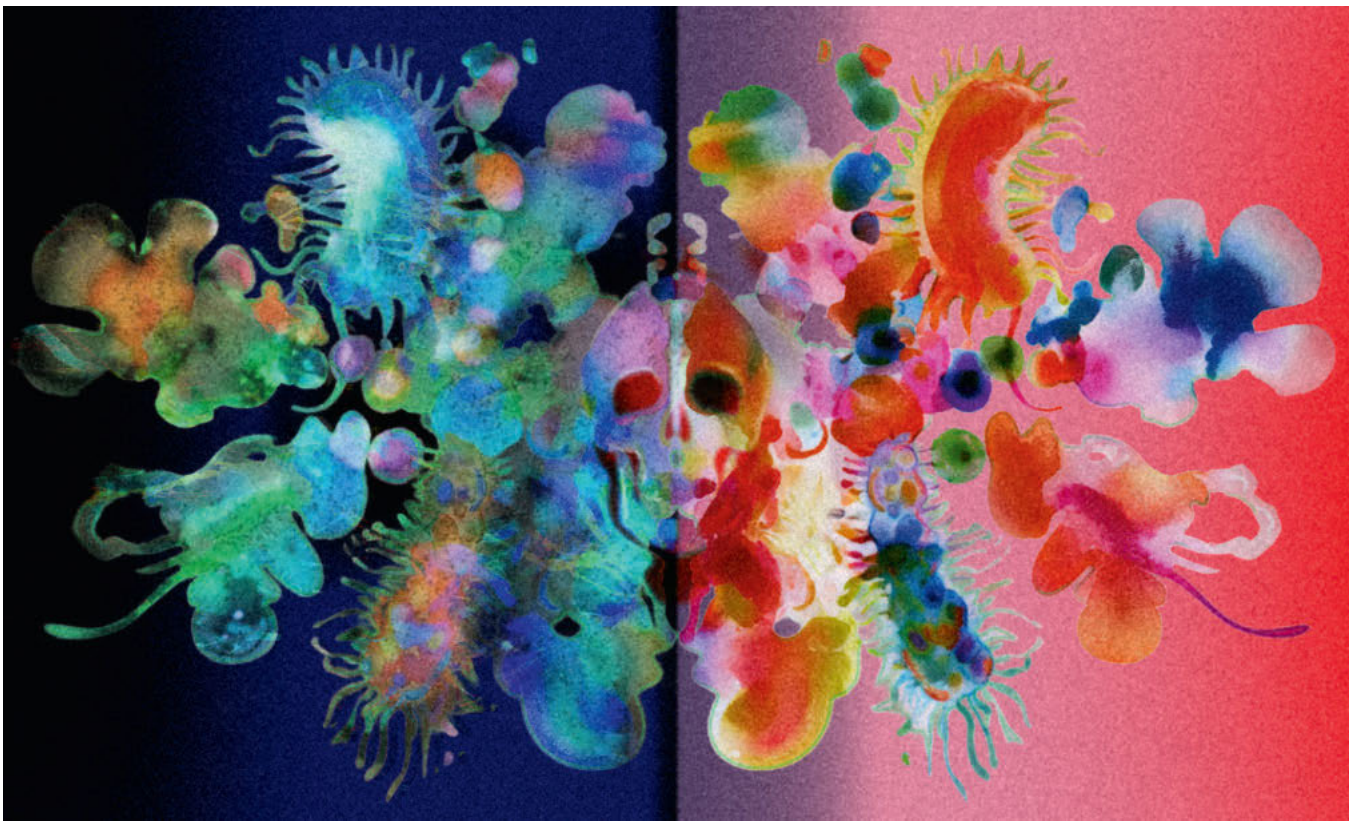
One result of the rise of alternative foreign assets concerns the countries that sell them, mainly America. Economists who argue that the dollar's days as the global reserve currency are numbered point to its dwindling share of the world's foreign-exchange reserves. This peaked at 71% in 2001 and has since fallen to 57%—lower than the greenback's share of foreign-exchange transactions. Add governments' other dollar assets, though, and the dollar continues to look formidable. Mr Setser estimates that nearly all the new funds and banks are more exposed to America than their countries' central banks are. Some 80% of PIF's foreign portfolio and 70% of holdings of China's top five banks, for instance, are parked in dollar assets.

What has changed is the part of America's economy that benefits from foreigners' interest. Reserve managers liked America chiefly for its ultra-safe government bonds. Profit-seeking funds and banks are drawn to the vast profits of American firms, particularly in technology. In the 12 months to July, demand for Treasuries softened but foreigners bought a record value of American equities and fund units. In South Korea NPS owns shares in Apple and Microsoft. This raises the cost of capital for America's government and reduces it for its companies.

Reserve judgment

The other consequence concerns the assets' holders. The new bets promise newly juicy returns but bring new exposures. Old-fashioned reserve assets are liquid, stable and mildly countercyclical: the dollar typically rallies in global downturns. Stocks, start-ups and skyscrapers do not. So care must be taken to ensure that new-look foreign portfolios are not too correlated with the domestic economy. South Korea has every right to hold more foreign assets. But since its economic fate is already bound up with the AI boom through its chipmaking champions, it may want to steer clear of American AI stocks. ■

Science & technology



Mirror life

Through a glass, darkly

It may soon be possible to create organisms that rely on a mirror image of natural biochemistry. That is scary

BIOLOGY IS HANDED. In other words, many molecules essential to living organisms come in two distinct forms which, like human hands, are non-identical mirror images of one another. For a lot of those molecules, though, life on Earth employs only one of the two possibilities.

Why this should be so is lost in evolutionary history. And, until recently, only evolutionary biologists cared. But that is changing. Although biotechnologists are not there yet, they can see on the horizon the possibility of synthesising biological cells from scratch. This has led some to wonder what might happen if someone tried to reverse life's handedness and create a cell—probably a bacterium—that held a mirror to normal biochemistry.

The answer, discussed this week by a conclave of concerned biologists meeting at the Singapore Conference on Mirror Life, is potentially terrifying. If such an organism were to escape the confines of the lab it might create an unstoppable plague or devastate an entire ecosystem. Scientists are, therefore, keen to see what can be done to nip such research in the bud.

Life's handedness—or chirality, to use the technical term—arises in part from molecules called amino acids, the building blocks of proteins (themselves the building blocks of bodies). They consist of one carbon atom attached to four other chemical entities: three are the same across all amino acids, but the fourth varies. The carbon atom's bonds to these four add-ons form a roughly tetrahedral structure. That allows for each amino acid to be constructed in two mirror-symmetrical ways (unless the fourth entity is hydrogen, duplicating one of the permanent three).

A complicated and somewhat arbitrary chemical convention defines one of these arrangements as left-handed and the other as right-handed. By this definition life's amino acids are generally left-handed. The opposite is true for sugar molecules (which, besides the many other jobs they do, form the backbones of the gene-carrying nucleic acids DNA and RNA). They are right-handed.

Fears of mirror life came to public attention in a paper published in *Science* at the end of 2024. At first sight, the idea of through-the-looking-glass creatures seemed like a curiosity rather than a threat. The paper's authors begged to differ. The handedness of biomolecules is so deeply embedded into biological systems that their chiral opposites go largely unnoticed. A mirror version of a bacterium would thus, those authors posited, evade immune systems, human and non-human alike, and could reproduce unchecked.

Mirror bacteria could also, they noted, be indigestible to single-celled creatures, such as amoebae, that are the principal predators of bacteria, and resistant to the viruses, called bacteriophages, which also keep bacterial populations under control. It could, in other words, multiply indefi-

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nitely. Hence the risk to ecosystems.

Nature would push back a little. Mirror life would be unable to metabolise all the left-handed amino acids and right-handed sugars of the natural organisms it encountered, restricting its food supply. But lots of biomolecules have no handedness, and it could feed on them instead.

All this assumes mirror life could be made in the laboratory. With existing technology, that possibility is remote. But synthetic biology is advancing by leaps and bounds, and is accelerating as AI is applied to the process. Novel viruses of certain types can already be designed and synthesised from scratch. What in 2024 looked at least 15 years away may thus arrive sooner than expected.

Mirror organisms might be made in one of two ways: big bang or piecemeal. The big-bang arrival—the creation in a laboratory setting of a completely synthetic mirror cell—would first require a breakthrough in the synthesis of "ordinary" cells. The piecemeal approach would convert cells a bit at a time, perhaps running both biochemistries in parallel.

George Church of Harvard Medical School, one of the authors of the 2024 paper, pointed to two significant developments in these directions. One was a so-called spud cell (it looks like a potato) announced to the world in July by researchers at the University of Minnesota. It had seven tiny chromosomes (called plasmids), encoding 36 enzymes (a small bacterium has hundreds), inside a fatty membrane. Even so, it was able to feed, grow and undergo cell division. That is a step towards a proper synthetic cell.

The other development, at Dr Church's own lab, was a modification of the way normal cells gather amino acids and assemble them into proteins. This can now be made to include non-standard (such as, in principle, right-handed) amino acids.

As to the reasons why anyone might want to make mirror life, there are two. One is scientific curiosity—the possibility of investigating why biology became handed in the first place. The other is that chiral molecules can be useful. Because bodies often do not recognise and destroy molecules with opposite chirality, they have longer periods of action as drugs. D-peptides and spiegelmers are, respectively, opposite-handed short-chain proteins and nucleic acids. D-peptide drugs already exist, and several spiegelmers are under test. But, as those very facts show, there are other ways to manufacture such molecules than invoking mirror life to do it.

As to curiosity, even though it proverbially killed the cat, letting it kill a proportion of the human race or harm some important ecosystem would be a bad bargain. Since its publication, therefore, the 2024 paper's authors have been beavering away, spreading the message. They have had some success. Authorities in Britain and Germany have shown concern and in March the UN Secretary-General's Scientific Advisory Board issued a briefing outlining the risks, though the governments of the field's big beasts, America and China, have yet to opine. The purpose of the meeting in Singapore was to move things along by coming up with recommendations for concrete action. These will be promulgated in a report to be issued in a few months' time.

Dr Church foreshadowed some possible recommendations in his talk. One was extending to mirror life the scrutiny already applied to the business of DNA synthesis in order to stop people trying to make dangerous pathogens. Another was incentivising whistle-blowers. A third was to impose serious penalties, including prison time, on malefactors—as happened in China to a researcher who edited the DNA of human beings when they were embryos. But he also had some more controversial ideas, such as psychological evaluations for researchers in the field to weed out those likely to go off-piste.

Meanwhile, lest anyone have lingering doubts about the nature of the threat, Deepa Agashe, another of the 2024 paper's authors, who is an ecologist and evolutionary biologist at the National Centre for Biological Sciences in Bengaluru, India, was there to disabuse them. She showed data on what was previously an unquantified point: the amount of achiral nutrients in soil, blood and the contents of the gut. Each of these habitats, she opined, would be able to support the well-studied bacterium *E. coli* without its needing to resort to chiral foodstuffs. And if *E. coli* could manage it, a mirror bacterium would probably be able to do likewise.

Dr Agashe also offered a telling analogy. She likened the lack of predators and pathogens a mirror organism might encounter to the experience of a successful invasive species, such as the neotropical cane toads that are running riot in Australia for lack of predators adapted to consume their toxic bodies. Such invaders can sometimes be controlled. They have never yet been eliminated.

Better, then, to act early, before the threat becomes tangible. At the moment, as was pointed out at the meeting, there are no vested economic interests involved to sway business lobbies, nor strategic ones to sway governments. Moreover, anyone sane can see that a bioweapon based on mirror life would be uncontrollable, and therefore unusable. Not everyone is sane, of course—a point that gives weight to Dr Church's thoughts about psychological evaluation. All the same, the warnings from Singapore present an opportunity to avert catastrophe before it is too late. ■

Navigation

One way or another

Robust alternatives to GPS are being developed

THE FIRST casualty of war is truth. But a reliable GPS signal comes a close second. Jamming signals from a global navigation satellite system (GNSS), such as America's GPS, is so easy it has become common when the shooting starts. Stronger signals from newer kit can make jamming harder, but the potential scale of the threat now looks much greater than previously thought. A paper posted in June by a team led by Todd Humphreys of the University of Texas, Austin, traced scores of broad GPS disruptions across Europe and elsewhere since 2019 to a constellation of Russian satellites. Jamming or even inadvertent interference from space, the research suggests, could affect an entire continent.

One alternative to GNSS is simply to mount cameras on a craft and use software to keep track of the terrain below. But that works only over land, and can be stymied by cloud cover, haze or darkness. So two other, quite different, methods are under investigation. One employs quantum phenomena—the angular momentum of photons and the spin of electrons—to follow magnetic maps of Earth's surface. The other establishes an observer's position by tapping ambient radio waves broadcast for other purposes. Neither method can yet match the precision of GNSS, but both are becoming credible backups.

Many rocks are weakly magnetic, and their distribution in Earth's crust results in

Points of contention

irregularities in the planet's magnetic field. The resulting landscape of anomalies is detectable under almost any conditions, and equally well over land and sea. Fit a plane with a magnetometer capable of spotting these anomalies, and a magnetic navigation (MagNav) system that can match them to existing maps, and its position can be accurately plotted. Last year Q-CTRL, an Australian firm, demonstrated a MagNav system that was more accurate than a leading GPS alternative in a trial flight over New South Wales.

Q-CTRL's sensor measures the field's local strength using a device called an optically pumped magnetometer (OPM). This works by firing a laser into a cloud of rubidium atoms. If the laser beam is tuned correctly, the atoms' electrons enter a state in which they are extremely sensitive to external magnetic fields. Another laser can then be used to track the behaviour of the electrons and work out the ambient magnetic field, allowing the system's software to work out where on the map it is.

Other firms are betting on a different type of sensor capable of showing not only the field's strength, as Q-CTRL's sensors already do, but also its direction. They believe these sensors, if used to navigate against the vector maps they help produce, could allow an observer to more accurately pinpoint their position.

One such firm is SBQuantum, a Canadian outfit. Its device is called an NV-diamond magnetometer which, as the name suggests, has a diamond at its heart. Diamonds are not normally magnetic, but can be made so by creating what are known as nitrogen-vacancy (NV) centres in their crystal lattices. These are places where two of the lattice's carbon atoms are replaced by a single atom of nitrogen—opening a gap around which electrons gather. Once the diamond is placed in a magnetic field, its lattice provides a 3D grid against which the electrons' behaviour can be measured. Accurate readings of field strength and direction can then be taken.

For now, the main drawback of both these approaches is a dearth of good maps of the crustal field. Those that are available tend to be mutually inconsistent and often lack the accuracy needed for navigation. Some firms, SBQuantum among them, therefore fly survey missions to create their own maps, while others, such as SandboxAQ, a Californian company, propose improving those that already exist. In June the firm submitted a paper suggesting that America's government agencies, armed forces and researchers work together to do just that on a global scale. They may soon get their wish. That same month Donald Trump called for more research into quantum sensing and ordered the Department of War to explore applications.

Meanwhile, others are developing ways to derive position from existing satellite radio signals. America's Federal Communications Commission has pressed for more work on such "signals of opportunity", and a few teams are giving it a shot. One technique is to measure Doppler shifts. As a satellite approaches and then recedes, the frequency of its transmissions appears to rise and then drop, much like the whistle of a passing train. The closer an observer is to a satellite's path, the more rapid the shift. Information about satellite orbits can then be combined with frequency measurements to estimate position.

One pioneer is ASPIN Lab, a research outfit at Ohio State University. In a demo on a cruise ship off western Greenland on August 24th 2024, its system, called Matrix, processed transmissions from 21 Starlink and OneWeb broadband satellites passing overhead. Given a starting position and speed, Matrix placed the ship just 27 metres from its true position.

Positioning by Doppler, however, has a notable shortcoming: satellites can drift off course or be manoeuvred to avoid collisions. This, in turn, can throw off estimates of a receiver's position. Stefano Tomasin, an engineer at the University of Padua, in Italy, says errors of between 100 metres and more than 1km are common.

Precision, though, should improve. SpaceX, the owner of Starlink, now publishes predictions for the orbits of its more than 11,000 satellites. Unlike tracking data produced by America's Space Force, these take into account planned manoeuvres. Dr Humphreys says SpaceX's forecasts tend to be accurate to within just a couple metres of the satellites' true positions. This should allow receivers to better interpret the Doppler shifts they observe.

Another route to greater accuracy is being pursued by Mark Psiaki of Virginia Tech. The trick, he says, is to place a master receiver at a known location to compare the Doppler shifts expected from a satellite's forecast trajectory with those it actually observes, and broadcast corrections to receivers at unknown locations. With funding from America's Department of Transportation, he is developing one such receiver that he hopes will yield positioning accurate to within about ten metres.

Dr Psiaki's master receiver could also facilitate another way of exploiting signals of opportunity: working out how long it takes for a signal to travel from source to recipient. The idea, he says, is to use the master receiver's known position and the position of a Starlink satellite as predicted by SpaceX to determine when a signal must have been transmitted. That signal's departure time can then be broadcast to receivers at unknown locations, allowing them to determine how long the same signal took to reach them, and, therefore, their distance from the satellite. By combining distance measurements from four Starlink satellites with Doppler data, a receiver should be able to determine its position to within just two metres.

The quest has even expanded to transmissions from mobile-phone towers. In tests with America's air force, ASPIN Lab has used Matrix to navigate an aircraft by tracking tiny shifts in such signals. Matrix has been licensed to companies and military users, says Zak Kassas, the lab's director, who expects the kit to be ready for market within two years. More reliable navigation, then, may be at hand in signals that already exist. ■

Thanatophobia

Cold comfort

Simulations of dying can help reduce people's anxiety

DEATH HAS always had a marketing problem. The debilitating terror it induces has been linked to everything from money hoarding to workaholism. And worrying about death—thanatophobia for the pub quizzers—can even be self-fulfilling. Anxiety can double the risk of death in patients with heart disease and cause others to avoid screening tests most likely to catch cancer early.

So all credit to the brave folk in Australia who signed up for a quick practice kick of the bucket—experiencing a cardiac arrest and out-of-body journey in virtual reality (VR). The ten-minute VR experience, "Passing Electrical Storms" by Shaun Gladwell, was shown at the National Gal-

This death may contain flashing images

lery of Victoria in 2023. Visitors lay on a bed that initially vibrated with their heartbeat, while wearing a VR headset showing them on a table in the gallery. The simulation then takes over: as their heart rate seemingly crashes, the volunteers watch doctors try and fail to resuscitate them. After brain death, they then see themselves leave their body and appear to float through the universe before shrinking inside cells and atoms. Six large fans provide the associated cooling effects.

Participants' reactions to the experience were analysed by researchers before being published in a journal in June. Most participants found it a pleasant trip, and many emerged less anxious about death. "If death is as simple as that, I'd be very happy," a volunteer told the research team, led by Stefan Greuter, a design technologist at Australia's RMIT University. Another compared the experience to yoga. One participant, brought up to believe in hell, admitted to being relieved.

A separate study in Texas, published in *Frontiers in Virtual Reality*, tested an 11-minute VR simulation of a near-death experience. In this case the student volunteers survived, but only after watching themselves undergo a car crash, coma, tunnels of light, serene landscapes and imagined encounters with friends and family. Reported death anxiety among participants dropped by an average 18%. "Very relaxing. I wish I could try it more than once. Maybe every night before going to bed," one participant said. Running the test again with those who are frail, elderly or terminally ill could produce a stronger response, the researchers say. A glimpse of mortality did not, however, help everyone: a quarter of the student volunteers said the VR experience made them more anxious about death, and a third emerged more stressed.

Most people fear their own deaths. Surgeons and doctors also worry about others dying under their care. This, too, can be simulated: a report in the *Journal of Surgical Education* from January describes a "protocol deviation" during a clinical simulation exercise with medical students at the University of Wisconsin, in which a rogue supervisor went off-script and simulated killing the practice patient. The monitor flatlined. The students, who had been calmly preparing the patient for bowel surgery, were not forewarned—as their reactions attest. "Oh my God. I'm going to kill someone. They should not graduate me. I am literally having a panic attack," one said during a debrief session.

Interventions such as these could have some benefit, the researchers say. Many doctors will lose patients. Learning to process the emotional weight in a controlled situation, before it happens in a real operating theatre, may be something that more medical schools should offer students. ■

Well Informed

Are weighted blankets really beneficial?

Measurable gains are modest, but many users report feeling better

ENTHUSIASTS LIKEN sleeping under a weighted blanket to being cocooned, gently hugged or soothed by a light massage. Heavy quilts and sofa throws filled with glass beads or other dense materials have grown popular despite prices that can reach several hundred dollars. Weighted blankets have a curious pedigree: they were inspired in part by padded "squeeze chutes" that restrain and often calm cattle during handling. Might people derive similar measurable benefits from pressure?

In 2023 the results of a small so-called crossover trial (in which participants receive the treatment as well as the control) from a sleep laboratory in Sweden were published. Twenty-six healthy young adults had spent a night under a normal blanket and another beneath one weighing about 12% of their respective body weights. Each night, for an hour before lights-out, saliva samples were taken. Under the weighted blanket, the rise in salivary levels of melatonin, a hormone that promotes sleep, was on average 32% greater.

That sounds promising. These participants, though, did not sleep for longer when under the heavier blanket. And a single night's trial says little about sustained use. A larger trial, published in *BMC Psychiatry* in 2024, therefore followed 95 Chinese adults with diagnosed insomnia for a month. Each was randomly assigned a blanket that weighed either 5.5kg or 3.5kg. A device tallied their awakenings during the first week. Those using the heavier blanket woke less often—but only slightly less. At the end of the trial, both groups scored better on a questionnaire measuring insomnia severity, but by statistically indistinguishable amounts. On other questionnaires covering sleep quality and related problems, though, the two groups diverged. Participants using the heavier blanket reported roughly twice the improvement in overall sleep quality, as well as greater reductions in fatigue, anxiety, stress and pain.

Another randomised trial was even more striking. For four weeks, 119 Swedish adults with a combination of insomnia and one of four psychiatric disorders slept under either a 1.5kg blanket containing plastic chains or a 6kg or 8kg one containing metal ones. (Participants were told only that they would receive a blanket containing chains, to preserve some blinding.) Nearly 60% of those given a heavy blanket saw their insomnia-severity score fall by at least half, against 5% in the control group. Those given heavier blankets also reported reductions in fatigue, anxiety and depression. Writing in the *Journal of Clinical Sleep Medicine* in 2020, the researchers speculated that by mimicking touch and acupressure, a therapy whose practitioners stimulate pressure points with their fingers, the blankets calmed and reassured users while increasing levels of oxytocin, a hormone associated with affectionate touch.

Such conjecture reflects the difficulty of pinning down what weighted blankets actually do. Consider two last studies. In a British crossover trial with 67 autistic children, reported in 2014, a weighted blanket did not help them doze off faster, wake less often or sleep longer. The children and their parents still preferred it. In a trial involving 94 American adults with chronic pain, published in 2022, a heavier blanket did not reduce participants' daily ratings of pain intensity. Its severity, however, as well as the disruption it posed to daily life, did seem to improve. The most decisive experiment, then, may be to try one.

Culture

Hollywood's novel obsession

Hit lit

Why today's new films and TV shows resemble a highbrow reading list

A CATERPILLAR CRAWLS on a finger in the opening shot of a new adaptation of "Sense and Sensibility". It is a neat symbol of Jane Austen's technique. In this and other novels, she describes what, in the sweep of history, are tiny things: stolen looks, crossed wires, unspoken longings, the heart's hidden rhythms. Yet she evokes them so exquisitely as to have transfixed generations of readers—and viewers.

"Pride and Prejudice" is up next. It is a truth universally acknowledged among producers that you are allowed to remake it every decade or so; the latest version is due on Netflix in December. As you may have noticed, Austen's are not the only classics currently transferring to the screen. "Wuthering Heights" was a hit; "The Odyssey" was a sensation. Other recent adaptations include "Frankenstein" and "Little House on the Prairie". Among those impending are "East of Eden", "A Tale of Two Cities" and a version of "A Christmas Carol" starring Johnny Depp.

This shelf-to-screen stampede raises questions about films, TV and books. Why, in a world of apps and AI, is so much airtime devoted to women in bonnets and men in breeches? What distinguishes a soaring reprise from a plodding retread? As for books: if literature is in decline, as it is often said to be, how come the listings resemble a highbrow reading list? The underlying question is about the state of culture: is the costume-drama surge a mark of classiness—or of senescence?

A new version of Austen or Charles Dickens is an adaptation not only of a novel but of previous adaptations; that is part of their fun. This "Sense and Sensibility", directed by Georgia Oakley (and out in America soon), follows the much-loved version of 1995 with Emma Thompson and Kate Winslet. After wrestling with her feelings in "Normal People", Daisy Edgar-Jones does so as sensible but secretly lovelorn Elinor Dashwood, minus the sex but with some impressively frenetic dancing. Esmé Creed-Miles is her unsecretly passionate sister, Marianne.

The setting is rugged moorland rather than pastoral idyll. Again the Dashwoods juggle bland but noble, hot yet caddish and unavailable love interests, but now the tone is more earnest than satirical. The crucial bits in Austen's dramas are often silent, reckons Andrew Davies, author of screenplays including the wet-shirted "Pride and Prejudice" of 1995 (and of "An Unsuitable Boy", a wry memoir). They are moments when "You think, 'God yes! He loves her—and she loves him'." Only fine acting can convey the interior depths plumbed in great novels. This film has it.

The joys and woes of archaic characters

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may (like caterpillars) seem trifling. The narrowness, it turns out, is not a drawback for producers but a plus. These books strip away modern ephemera to focus on elemental themes like love and money. Period settings, notes Jinny Howe, a bigwig at Netflix, feel "exotic" in "a high-speed, high-tech digital world". Like fairy tales, old stories are consoling (as are nice frocks). "When we think we're going to be wiped out by a terrible climate event or AI robots," says Mr Davies, people "reach for something comforting and familiar."

There is a commercial logic, too. Shows and movies based on books are a surer bet than original ideas. So far this year they had 3.4 times the global demand of the average title, calculates Christofer Hamilton of Parrot Analytics, a research firm. Literary titans have established followings that hype up adaptations in online ding-dongs over their merits or anachronisms.

To modish critics, a glut of frock-coated men and women saying "I declare" suggests a failure of imagination. Crafting knockout narratives is hard; reviving old ones seems easier. To which insiders say: humbug! The pressure of expectations can be "daunting", says Ms Howe. The challenges are different but real. Slavish adherence to a text can be sterile; disfiguring it (à la "Wuthering Heights") can look absurd. Louise Willder, author of "The Book Was Better", a witty survey of adaptations, calls out the film of "The Scarlet Letter", which puts in skinny-dipping and a man dancing with a deer carcass on his head.

As when bards retold myths, she argues, every screen version imparts a trace of its age. The go-to moves in dramatising modern novels are to make everyone good-looking, tack on a happy ending and move the story to America. Some of that is trickier with classics (though "Clueless" transplants "Emma" to Beverly Hills). Often sex and violence are added. "The Odyssey" includes plenty of gore, but also omits some Homeric episodes evidently deemed off-putting, like the hero's rampant adultery and mass execution of slave girls. The main job, says Ms Howe, is to identify "what about [the book] cuts through and feels relevant to audiences today."

The new "Sense and Sensibility" does that by emphasising the relationship between the sisters—fittingly in an era disinclined to see men as women's sole concern. Mr Davies is writing a version of "Mansfield Park" which will visit the slave plantation whence a family in Austen's novel derive their wealth. Both tweaks extrapolate from the books without violating them.

Or consider "East of Eden". Featuring a lithe James Dean (and his mesmerising quiff), the movie of 1955, directed by Elia Kazan, tackled only a slice of John Steinbeck's novel of familial rivalry and rejection, destiny and free will. A seven-part version out on Netflix on October 1st—scripted by Zoe Kazan, Elia's grand-daughter—takes on the whole sprawling saga. The opening scenery in Connecticut is weathered but beautiful, as are the men. When the action moves to big-skied California, Ms Kazan reinstates Lee, a Chinese-American character cut in the film.

And she reimagines Cathy (Florence Pugh), a prostitute and murderer who in the novel embodies evil. Here a background of abuse makes her pitiably human; a killer with a hint of avenging angel. Such reboots, says Robert Douglas-Fairhurst, a professor at Oxford, "aren't just adapting old stories; they're showing how these stories can always adapt themselves to changing circumstances".

The book of the film

There is a paradox here. Reading for pleasure is declining. Some teachers assign passages from novels, not whole books, to screen-addled pupils. Many American college students struggled to understand the opening page of Dickens's "Bleak House". Serious fiction is said to be on the skids; romance, fantasy and their mutant offspring are the booming genres.

So you might expect the appeal of the classics for film-makers to wane. You might also suspect that, for many viewers, watching an adaptation of a chewy text is a substitute for reading it. This is true of some students of "Romeo and Juliet", who, noted an examiner's report in England, betrayed their reliance on the film starring Leonardo DiCaprio by "making specific references to guns and petrol stations".

Yet lots of readers are indeed picking up the books of the films. Penguin Classics, a publishing imprint, says that in 2026 sales of "The Odyssey" are up 265% in Britain and 599% in India; sales have quadrupled in America, reports Circana BookScan, a market tracker. "Wuthering Heights" topped Britain's bestseller list, a feat for a bizarre novel published 180 years ago. These boosts augment a trend in which young readers are discovering classic authors on social media, and, as Jessica Harrison of Penguin Classics puts it, "having their minds blown by Dostoyevsky".

Sales perk up months before buzzy releases, says Tom Rowley, a former *Economist* writer now running Backstory, a chic London bookshop. Some diligent people still read the novel in advance. Readers are not seeking refunds on finding there are no scenes inside the Trojan horse in Homer, nor sex in a barn in "Wuthering Heights".

Thus, says Ms Willder, classic novels, like genes, are surviving through adaptations. They are also helping a pinched film industry. Don't be snooty about yet another "Pride and Prejudice" or "A Tale of Two Cities". It is a far, far better thing that they do than letting the art forms wither. ■

A history of pens

The write way

The Pen. By Simon Garfield. *Crown; 304 pages; $33. Canongate Books; £20*

IT WAS NOVEMBER 1944. German soldiers were blocking the entranceways to the charming medieval town of Metz, in northern France. Meanwhile, just outside Metz, and less charmingly, a piece of German mortar was blocking the airway of an American soldier who was, his companion recalled, "thrashing around" and "gasping for breath". The companion acted swiftly. He took the soldier's fountain pen and said "I don't like to do this." Then he slit the man's throat open with a knife, popped the fountain pen in—and the soldier started to breathe. And a pen legend began.

There is, as Simon Garfield, a journalist, points out in an engaging new book, a pen paradox. Pens have changed the course of history, competing with "the wheel, steam engine and microchip" as "our most important tool of progress". Pens make history in the literal sense (the second world war was postponed by pen in Munich; Hermann Göring, a Nazi, committed suicide using poison—possibly—smuggled into his cell in a pen). And pens make history in the literary sense too, since "to write history" is, as the philosopher Michael Oakeshott pointed out, "the only way of making it".

Yet the pen has left surprisingly little mark on the page about itself. The earliest image of a quill does not appear until 547AD: an Italian mosaic in a church in Ravenna depicts St Matthew at his writing

Let that ink in

desk. Most artists and authors ignore their pens entirely except to grouse about them: Virginia Woolf hated her "disgusting, slippery, false" fountain pen; Jane Austen felt "sick of myself & my bad pens". King Charles III harrumphed at his "bloody" pen. The pen is the handmaid of history: it writes history, then is written out of it.

This history more than makes up for that, as it follows the story of the pen (briefly) from pre-history then onwards, and in closer detail, as the pen becomes first a "necessity" and later an industry with billions of dollars in annual sales. But the arrival of the pen proper (in the sense of a hollow thing with a pointy metal bit at the end) was surprisingly late. People describe writers as "penning" their works—but for most of history they didn't. Tang-dynasty poets used brushes; Cicero wrote his letters with a reed pen; Julius Caesar used a stylus, long and pointed as a nail, to write (and, ever the pragmatist, also used it to stab an assassin in the arm).

The pen proper only really began with the quill, from which the modern pen got its name (*penna* is Latin for "feather"). The word's ubiquity reflects the quill's: Shakespeare wrote the "words, words, words" of "Hamlet" with quills; Magna Carta was signed with one; the da-da-da-dum of Beethoven's Fifth was drafted by one. Almost everyone was irked by them: the ink leaked; the nibs split (and had to be reshaped with a "penknife"); a busy writer could get through five in a day. This was annoying for them and not much fun for geese: in 1832 England imported 33.7m goose quills, mostly from the Netherlands and Germany.

It all changed in France in 1827 when someone had the idea—the "inkling"—of manufacturing something a bit more sturdy, with a well of ink inside and a metal nib. Or this may have happened in Iowa in 1867. Or Egypt in 970AD. Or possibly with Leonardo da Vinci in the 15th century. Whatever. Eventually everyone had one. Those who manufactured the most popular pens became—like Messrs Waterman, Parker and (later, with a different nib) Biro—household names. (Around 150bn Bics had been made by 2024.)

They also became a status symbol. Rorschach would claim that you could tell a personality from an ink splat; far more telling were the pens that ink came from. James Bond had an exploding pen. Freud allegedly had a huge red pen (prompting talk of "pens envy").

Once omnipresent, the pen is now becoming obsolescent: "to pen" will perhaps soon join words like "scribe" and "manuscript" as a linguistic literary relic. This book—which could admittedly have done with a red pen through some of its longer sections—persuades the reader of what a shame that would be. ■

Wild-game cooking

Tarmac to table

MARLINTON

What a rural American cooking contest reveals about food culture

WEST VIRGINIA'S official motto is *montani semper liberi*: mountaineers are always free. It is the only state whose boundaries fall entirely within the Appalachian mountains, offering plenty of isolated valleys where people can find seclusion from the modern world (and sometimes the law). But with great freedom comes great distances from grocery shops, not to mention a legacy of poverty. Appalachian cuisine prizes thrift and resourcefulness: hunting, drying and pickling all feature heavily.

For 35 years Pocahontas County—one of the state's biggest and least populated—has honoured those traditions with an annual "Roadkill Cook-Off". The festival draws around 14,000 people to a tiny town tucked deep in the mountains, some from as far afield as Rhode Island and Colorado. Don't worry, nobody is served *lapin à la Michelin*. The animals do not have to be physically scraped off a highway: they just have to be the sorts of creatures commonly hit by cars on rural roads.

The event reveals three things about food culture, in America and elsewhere. First, even in historically poor regions like Appalachia, abundance reigns. A few generations ago, Appalachians would have eaten roadkill out of necessity; today it is by choice. Delicious ingredients are everywhere—if you know where to look for them and have a bit of imagination.

In previous years, cooks have served up elk, snakes, snapping turtle and opossum. At the latest edition on September 26th, contestants made ground buck with cabbage and braised deer with pancakes and beetroot-cream sauce. The day's best dish was a rustic and comforting wild-boar stew with apples and rosemary. The most unexpected was barbecued bear over macaroni—which was slightly dry, but since the chef said she "found it in a ditch off Route 42", better overcooked than under.

The festival is also a reminder that wastefulness should be unpalatable to chefs and eaters alike. "Nose-to-tail cooking" is now fashionable; it is no longer unusual to see bone marrow, liver or sweetbread on the menu of an eatery in a big city. (The term was coined in the mid-1990s by Fergus Henderson, co-founder of St John, a meaty restaurant in London.) But as the cook-off attests, people have long been making the most of every bit of animal. When entrails are prepared with care, you'll want to eat the heart out.

The final lesson is that there are good cooks everywhere. The festival leans into stereotypes of West Virginians as hillbillies who eat what they can peel off the tarmac. It also subverts them. The team that cooked the boar stew called themselves the Coal Hollow Brothers, and drank a homemade concoction called "Swamp Water" from a plastic jug. But they are in fact talented wild-game cooks and produced a dish that could grace the menu of any urban restaurant. Even a haughty gourmet would try it and think: "Give me boar." ■

Why singing is note-worthy

Duet for ever

Sing. By Sacha Langton-Gilks. *Headline Home; 288 pages; £20*

SINGING HAS provided a soundtrack to human history. Songs tell people about their past and their values. Ancient epics, which placed societies in mythic time, were sung or chanted. "I sing of arms and the man," begins Virgil's "Aeneid"—not "I speak." Singing has long been a way to bind ideological communities, from the hymn-singing early Protestants to pro-democracy demonstrators in Belarus.

It can help people at all stages of life. Singing seems to hold babies' attention, and prevent them from crying, longer than speech. A study published in the *British Journal of Psychiatry* found that singing groups were better than playgroups at alleviating severe postnatal depression. The act of singing can help rehabilitate people who have suffered horrendous injuries, including Gabrielle Giffords, an American congresswoman who was shot in the head in 2011. (Music therapy helped her recover verbal fluency.)

Many people derive pleasure from singing in choirs and other groups. Some 1.5m Germans belong to choirs that are affiliated with the main singing associations; millions more sing in casual groups. In "Sing" Sacha Langton-Gilks, a teacher who trained as a choral scholar, argues that almost everyone can, and should, sing. But that raises a question.

If singing is so enjoyable and beneficial, why doesn't everyone do it more? One answer is that it is extremely difficult to sing well. Good technique must be learned; bad habits must be unlearned. Ms Langton-Gilks was informed by one fierce-sounding teacher that her voice had been "strangled" by years of singing in a college choir. Many things can set a singer back: germs, nodules, the hormone progesterone (which dries the vocal cords)—and, above all, ageing.

Social difficulties abound too. Ms Langton-Gilks defends the practice of auditioning children for school singing groups, but many children (and adults) fear the exposure and judgment that it entails. A request to sing solo in front of others can be no less terrifying than a request to take off your clothes. It might not help that wonderful singers can be summoned by a few taps on a phone screen. Next to someone like Adele or Caroline Polachek, hardly anybody sounds good.

"Sing" insists that group singing is gratifying regardless of talent, and that it brings people together. True, but it can also be infuriating and disheartening. Your correspondent has sung in choirs where people ostentatiously turn away or cover an ear to muffle the wrong notes being sung by a neighbour. One study of an amateur choir found that the common practice of bringing in professional singers shortly before performances, to "stiffen" the sound, greatly offended the amateurs. The professionals did not enjoy it, either.

If singing is, or ought to be, an essential part of life, it is not a surprise that life intrudes on singing. The strains and irritations of ordinary existence can wreck people's voices—Ms Langton-Gilks writes that this temporarily happened to her—and throw singing groups into disharmony. A better book, with fewer digressions, would have more to say about the bum notes. ■

Raising their voices

What's up with supplements?

The weight and C approach

Make Me Well. By Xand van Tulleken. *Cornerstone Press: 400 pages; £22. To be published in America by Crown in January; $34*

DURING THE Tang dynasty, Emperor Xuanzong became so distracted by a beautiful young woman with luminous skin, Yang Guifei, that he neglected his duties and nearly lost his empire. The secret behind her good looks was *ejiao*, a Chinese collagen supplement made from boiling donkey hides. Or so claim marketers promoting the pricey product, which promises youthful skin and graceful ageing. (Though Guifei was executed, so never experienced the latter.)

Today collagen is one of the most popular supplements in the wellness sector. But after researchers removed studies with industry funding or poor design, they found no clinical evidence that the "miracle" supplement with the supposed power to topple a dynasty truly makes an impact on your forehead wrinkles.

A new book by Xand van Tulleken, an English doctor, argues that the booming wellness industry is "a modern myth factory". When supplements such as collagen are promoted with trumped-up medical "science", it can "blur the line between wellness and health care" with harmful consequences, he argues.

"Make Me Well" is well-timed. Ten years ago "biohacking", or the practice of using science, data and lifestyle changes to optimise your health, was not recognised by dictionaries. But in 2025 it was shortlisted by Collins Dictionary as a word of the year. Wellness is now a $2trn industry, by a McKinsey estimate (which includes supplements and health-tracking devices, but not groceries). American consumers spend $500bn a year on wellness products. Every Tom, Dick and Harry (or rather Kim, Khloé and Kylie) is now an ambassador for one.

Mr van Tulleken argues two things convincingly. First, despite extravagant promises that supplements make you live longer, see better and feel brighter, there is little proof that they have the ability to boost your health at all unless you are severely nutritionally deficient.

He takes a swipe at the hype around menopause supplements, for example. Following wild exaggerations of the supposed dangers of HRT, an oestrogen-based treatment for symptoms of menopause, the alternative-medicine industry started pushing supplements to perimenopausal wom-

en. However, recent investigations show a lack of standardisation across ingredients, limited effectiveness and, in some cases, harmfully high doses of vitamins. In America spending on menopause supplements was $18bn in 2024, over 300 times more than was invested in menopause research. Instead of injecting money into improving care for women, the wellness industry benefits from "the confusion that 20 years of miscommunicated science has created".

Second, the business of personal testing and wearable technology creates a culture of expensive and often unnecessary health interventions. You can now monitor your glucose spikes, sleep and gut microbiome at home. Decentralised monitoring claims to democratise health, empower the patient through self-knowledge and offer "peace of mind". But there is little evidence that doing lots of testing makes most people healthier. Excess screenings can lead to false positives, increased patient anxiety and further unnecessary tests.

"Make Me Well" is a thorough, entertaining investigation into the unintended consequences of the wellness industry. It may well wake consumers up to the marketing fluff that drives their purchases. The author argues for starting a registry of supplements, requiring independent third-party testing and enforcing regulation of bogus health claims. Government officials are already pushing to strengthen oversight of supplements to ensure safety and efficacy. That would be, well, welcome. ■

BACK STORY
Captured by the Kremlin

Prison is still the best way to understand Russia, as Evan Gershkovich's memoir shows

THERE ARE three types of people in Russia, runs an old joke: those who have been in prison, those who are currently in prison—and those who will be. Under capricious rulers, incarceration is an ambient risk, which can be a mark of virtue rather than degeneracy. For Dostoyevsky and others, it incubated moral clarity and spiritual rebirth. In his new memoir Evan Gershkovich, an American journalist, quotes a character of Tolstoy's: "The only suitable place for an honest man in Russia...is prison."

A son of Russian-speaking Soviet emigrants to America, Mr Gershkovich is now in the first of those three carceral categories. He spent 16 months in Russian custody, an ordeal he chronicles in "This Cursed Beautiful Land". Prison, it shows, is still a barred window onto the truths of an inescapable country.

He moved to Russia in 2017 to work for the *Moscow Times,* later joining the *Wall Street Journal*. It seemed, initially, that the old rules of post-Soviet journalism still applied: that foreign reporters enjoyed a licence which their beleaguered Russian counterparts lacked. In bygone days, if an official criticised your coverage, it was generally a bid to extract a bung (or a bottle). "I am sick," a minder once complained to *The Economist,* "and nobody has brought me any medicine."

But the rules were changing. Russia began grabbing Americans to trade as hostages. With the invasion of Ukraine, any residual freedom of speech was squished. Mr Gershkovich saw the danger, but reckoned he was more likely to be expelled than arrested. In March 2023 he flew to Yekaterinburg to look into tank production. Seized by balaclava-wearing agents of the FSB, Russia's almighty security service, he was hooded, punched and accused of espionage.

Vladimir Putin had instructed his goons to catch more spies; they gave him Mr Gershkovich. He spent most of his captivity in Lefortovo, an FSB-run prison in Moscow with a grim reputation, where he was relatively coddled. He evokes the grinding routines and indignities (the deadbolt and peephole, grotty food and smells); the snatched joys, such as the bliss of once-a-week showers; and the slow-quick wooziness of time inside.

Prison culture in Russia, however, is not just a curiosity. It helps explain the mindset of the gangsterish leadership—not least the imperative, explained to Mr Gershkovich, to always "hit first in a fight". Prison is a front in the war on Ukraine: he encountered men who were recruited in jail, fought, survived, came home and got rearrested. Others were pressed to sign up. As in Dostoyevsky's era, this moving prison sketch is a scan of Russia's soul.

It is a bleak picture, but not entirely. Travel around Russia, and you find people who rant against the West, then politely invite you home for tea. Mr Gershkovich experienced an extreme version of this whiplash. Many of his guards, and some FSB agents, were curious about America, friendly, even kind; they suggest the inferiority complex and political apathy behind the Kremlin's bombast.

At the poignant heart of this book is the author's bond with his cellmates (who, for their safety, he presents as a composite figure, Seryoga). Though he initially feared Seryoga was an informant, they became improbably close and mutually supportive, playing backgammon and watching movies on their cell TV. At moments Mr Gershkovich felt he "cared about him more than anyone else in the world". After he was freed, he was served stuffed pigeon at a fancy dinner and "thought of Seryoga and the pigeons that visited our cell window".

That was after complex diplomacy which saw him and others traded for a Russian group including an FSB assassin. (Until his death, it seemed Alexei Navalny, an opposition leader who chose prison over exile, might be part of the deal.) First though, for his sham trial Mr Gershkovich was taken back to Yekaterinburg, where he met inmates tortured with electric shocks. He wrote an obligatory, but sassy, pardon request to Mr Putin. In August 2024 Joe Biden hugged him on the tarmac in Maryland.

So this tale of punishment without a crime turns into an offbeat love story. It also has some unassuming heroes—not the overseas diplomats and politicians who succeed in freeing Mr Gershkovich, but the Russian lawyers who bravely conduct his futile defence. The futility is what makes it heroic. "One day all of this will come crumbling down," one says. For now, the aim is "to preserve some semblance of a foundation from which to rebuild". As another Russian proverb insists: hope dies last.

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	Gross domestic product, % change on year ago				Consumer prices, % change on year ago			Unemployment rate, %		Current-account balance, % of GDP, 2026†	Budget balance, % of GDP, 2026†	Interest rates, 10-yr gov't bonds			Currency units	
	latest		quarter*	2026†	latest		2026†					latest, %		change on year ago, bp	per $ Sep 30th	% change on year ago
United States	2.2	Q2	2.2	2.1	3.4	Aug	3.2	4.1	Aug	-3.5	-6.5	5.3		113	-	
China	4.3	Q2	3.6	4.7	0.8	Aug	0.9	5.3	Aug‡§	4.2	-5.4	1.4	§§	-21	6.70	6.3
Japan	0.9	Q2	1.4	0.8	1.9	Aug	1.8	2.4	Jul	4.0	-1.8	3.1		142	157	-5.5
Britain	1.4	Q2	2.1	1.2	3.1	Aug	3.3	4.9	Jun††	-3.7	-5.2	5.4		64	0.76	-1.3
Canada	1.1	Q2	3.3	0.9	3.0	Aug	2.8	6.4	Aug	-0.3	-2.1	4.0		81	1.42	-2.1
Euro area	1.2	Q2	2.6	0.9	3.2	Aug	2.9	6.4	Jul	2.4	-3.6	3.6		87	0.88	-3.4
Austria	0.4	Q2	-0.6‡	0.4	2.9	Aug	3.1	6.2	Jul	1.2	-4.1	3.8		79	0.88	-3.4
Belgium	0.5	Q2	0.1	0.6	4.6	Sep	3.7	6.0	Jul	0.2	-5.2	4.3		102	0.88	-3.4
France	0.5	Q2	0.1	0.6	3.4	Sep	2.4	8.3	Jul	-0.4	-5.3	4.9		133	0.88	-3.4
Germany	1.0	Q2	1.3	1.0	3.3	Sep	2.7	4.0	Jul	4.5	-4.2	3.6		87	0.88	-3.4
Greece	1.8	Q2	1.4	2.0	3.7	Aug	3.8	7.9	Jul	-5.6	0.6	4.5		106	0.88	-3.4
Italy	1.0	Q2	0.7	0.9	4.1	Sep	2.7	5.8	Jul	0.7	-3.0	4.6		106	0.88	-3.4
Netherlands	1.6	Q2	2.6	1.3	2.8	Aug	2.9	4.0	Aug	8.3	-3.0	3.7		80	0.88	-3.4
Spain	2.6	Q2	2.8	2.6	5.0	Sep	3.6	10.0	Jul	2.4	-2.5	4.1		82	0.88	-3.4
Czech Republic	1.8	Q2	1.1	2.0	1.9	Aug	2.0	3.0	Q2‡	0.1	-2.8	5.4		84	21.5	-3.8
Denmark	5.8	Q2	3.6	2.7	2.1	Aug	1.9	3.1	Aug	11.3	1.2	3.4		87	6.59	-3.5
Norway	2.0	Q2	2.7	1.5	3.3	Aug	3.0	4.4	Jul‡‡	13.0	10.0	4.6		57	9.60	4.0
Poland	3.9	Q2	4.1	3.4	4.0	Sep	3.0	5.8	Aug§	-2.0	-7.0	6.4		86	3.85	-5.5
Russia	1.3	Q2	8.3	0.7	6.3	Aug	6.2	2.2	Aug§	2.3	-3.2	16.9		195	84.4	-1.9
Sweden	3.2	Q2	6.4	2.1	0.3	Aug	1.6	8.5	Aug§	5.3	-2.3	3.2		62	9.99	-5.7
Switzerland	2.8	Q2	7.8	1.7	0.8	Aug	0.6	3.1	Aug	8.2	0.2	0.6		38	0.83	-4.8
Turkey	2.3	Q2	4.6	2.7	31.5	Aug	31.0	7.7	Aug§	-2.4	-3.5	32.8		362	49.0	-15.1
Australia	2.1	Q2	1.7	1.6	4.0	Aug	3.7	4.6	Aug	-3.7	-1.5	5.2		97	1.43	6.3
Hong Kong	4.3	Q2	-2.5	4.1	1.7	Aug	1.9	3.8	Aug‡‡	8.1	-2.6	4.1		104	7.85	-0.8
India	7.8	Q2	6.8	7.2	4.8	Aug	4.6	6.1	Aug	-1.5	-4.3	7.2		61	95.9	-7.5
Indonesia	5.3	Q2	4.4	5.3	3.2	Aug	3.4	4.7	Feb§	-2.1	-2.9	7.1		78	17,954	-7.3
Malaysia	6.0	Q2	9.3	5.6	1.9	Aug	1.9	3.0	Jul§	3.1	-3.6	4.0		57	4.08	3.2
Pakistan	4.8	2026**	na	3.0	11.2	Aug	9.0	6.9	2025	-1.1	-4.7	12.7	†††	73	277	2.3
Philippines	2.3	Q2	2.4	2.9	6.1	Aug	5.6	6.0	Q3§	-4.6	-6.4	7.7		166	62.6	-7.3
Singapore	5.9	Q2	5.7	5.7	2.3	Aug	2.1	1.9	Q2	15.8	1.3	2.5		56	1.28	0.8
South Korea	3.8	Q2	2.6	3.5	3.1	Aug	2.7	2.0	Aug§	13.6	-0.6	4.4		146	1,352	3.5
Taiwan	12.9	Q2	5.7	12.0	2.0	Aug	2.0	3.3	Aug	25.2	1.6	2.0		58	31.8	-4.2
Thailand	1.9	Q2	-0.7	2.1	2.5	Aug	1.9	1.2	Aug§	-0.3	-5.5	2.3		92	33.6	-4.0
Argentina	2.0	Q2	-2.3	1.7	33.5	Aug	32.5	7.9	Q2§	0.1	0.2	na		na	1,525	-10.9
Brazil	2.0	Q2	1.9	2.0	4.2	Aug	4.4	5.3	Aug§‡‡	-2.6	-7.3	14.1		41	5.20	2.3
Chile	-0.2	Q2	-0.1	0.4	4.1	Aug	3.8	9.6	Aug§‡‡	-1.1	-1.8	6.2		62	973	-0.8
Colombia	3.4	Q2	5.2	2.5	6.2	Aug	6.0	9.4	Aug§	-2.0	-7.3	12.8		153	3,341	16.8
Mexico	2.1	Q2	5.8	1.3	3.3	Aug	3.8	2.7	Aug	-0.4	-4.0	9.6		81	18.0	1.8
Peru	2.6	Q2	-1.0	2.8	4.4	Aug	3.8	4.9	Aug§	3.2	-2.1	6.7		80	3.44	1.2
Egypt	4.7	Q2	4.0	4.6	14.5	Aug	13.7	5.8	Q2§	-5.1	-5.1	25.6		50	52.1	-7.7
Israel	7.0	Q2	14.9	4.3	1.5	Aug	1.9	2.8	Aug	1.0	-3.3	4.1		nil	3.07	7.8
Saudi Arabia	-4.7	Q2	-17.9	-4.8	1.8	Aug	1.9	3.0	Q2	-3.8	-7.2	na		na	3.76	nil
South Africa	0.9	Q2	-1.0	1.1	4.2	Aug	4.1	33.6	Q2§	-0.3	-4.4	9.0		-21	16.4	5.4

Source: Haver Analytics *% change on previous quarter, annual rate. †The Economist Intelligence Unit estimate/forecast. §Not seasonally adjusted. ‡New series. **Year ending June. ††Latest 3 months. ‡‡3-month moving average. §§5-year yield. †††Dollar-denominated bonds. Note: Euro-area consumer prices are harmonised

Markets

In local currency	Index Sep 30th	% change on: one week	Dec 31st 2025
United States S&P 500	7,651.5	-0.7	11.8
United States NAS Comp	26,861.1	-0.3	15.6
China Shanghai Comp	3,842.2	-2.4	-3.2
China Shenzhen Comp	2,404.5	-4.8	-5.0
Japan Nikkei 225	66,753.7	2.7	32.6
Japan Topix	4,108.6	0.4	20.5
Britain FTSE 100	10,606.0	-0.9	6.8
Canada S&P TSX	35,235.9	-1.4	11.1
Euro area EURO STOXX 50	6,269.0	-0.5	8.2
France CAC 40	7,964.5	-2.0	-2.3
Germany DAX*	25,399.2	nil	3.7
Italy FTSE/MIB	51,372.0	-1.2	14.3
Netherlands AEX	1,114.9	0.6	17.2
Spain IBEX 35	19,426.2	-1.0	12.2
Poland WIG	156,882.0	-0.4	33.8
Russia RTS, $ terms	854.5	-1.0	-22.9
Switzerland SMI	13,912.2	-0.1	4.9
Turkey BIST	11,947.2	-9.8	6.1
Australia All Ord.	8,969.2	0.1	-0.5
Hong Kong Hang Seng	24,613.3	-0.9	-4.0
India BSE	72,480.3	-3.1	-14.9
Indonesia IDX	6,071.1	-4.8	-29.8
Malaysia KLSE	1,651.2	-1.5	-1.7

	Index Sep 30th	% change on: one week	Dec 31st 2025
Pakistan KSE	169,969.3	-1.3	-2.3
Singapore STI	5,675.9	-0.6	22.2
South Korea KOSPI	6,838.0	-3.4	62.3
Taiwan TWI	47,940.1	-0.5	65.5
Thailand SET	1,559.0	-3.2	23.8
Argentina MERV	2,819,323.3	-5.1	-7.6
Brazil BVSP*	186,340.5	0.3	15.6
Mexico IPC	64,214.4	-0.1	-0.1
Egypt EGX 30	51,894.8	-4.3	24.1
Israel TA-125	4,075.6	-2.0	11.2
Saudi Arabia Tadawul	10,440.6	-2.2	-0.5
South Africa JSE AS	108,506.9	-2.7	-6.3
World, dev'd MSCI	4,909.2	-0.6	10.8
Emerging markets MSCI	1,701.8	-2.7	21.2

US corporate bonds, spread over Treasuries

Basis points	latest	Dec 31st 2025
Investment grade	93	93
High-yield	336	354

Sources: FactSet; Moscow Exchange; Standard & Poor's Global Fixed Income Research *Total return index

Commodities

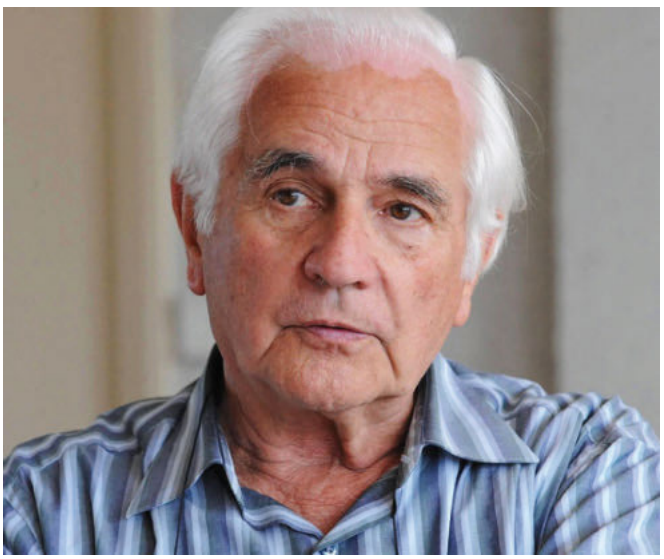
***The Economist* commodity-price index, 2020=100**	Sep 22nd	Sep 29th*	% change on month	% change on year
Dollar Index				
All items	153.9	148.5	-5.9	9.0
Food	152.1	143.8	-9.9	-0.7
Industrials				
All	155.3	152.4	-2.4	18.1
Non-food agriculturals	150.8	152.7	-2.5	20.9
Metals	156.5	152.3	-2.4	17.4
Sterling Index				
All items	148.1	144.4	-3.6	11.1
Euro Index				
All items	153.7	149.6	-3.7	13.0
Gold				
$ per oz	4,329.6	4,163.4	-4.4	8.8
Brent				
$ per barrel	99.2	102.6	8.4	53.1

Sources: CME Group; FactSet; LME; NOREXECO; NZ Wool Services; S&P; Thompson Lloyd & Ewart; USDA *Provisional. Note: new series for gold, oil and cotton

For historical indicators data, visit economist.com/economic-and-financial-indicators

OBITUARY
Stanislas Hutin

The exposer of French war crimes in Algeria died on September 2nd, aged 95



AT FIRST HE thought it was jackals he heard, on that cold January night. But the cries went on and on, and they were close, coming from the lieutenant's tent not far away. Sometimes, shrill and terrible, they sounded like a pig having its throat slit. Surely, Stanislas Hutin thought, it could not be the boy. But it was.

The boy was one of those barefoot child-shepherds, thin and filthy as their flocks, who were a common sight in Constantine, in north-east Algeria. This one was 14, though he looked 12, and rather than meet a French army patrol he had fled for the woods. In case he was going to alert the *fellaghas*, the fighters for Algerian independence, he was pursued and caught. Now he was wired up by wrist and ear lobe to the magneto, the field generator. Shocks coursed through his small body.

Mr Hutin saw much else in 1955-56, when he served as a recalled reservist in Algeria. Especially vivid was the savage beating of the company's first captive at Christmas, as if in celebration, and the curled-up prisoner afterwards, barely conscious. He recorded these things day by day, secretly, in a red notebook, and also wrote down the testimonies of his comrades. They had seen more: men strung up naked by their iron shackles, with their heads dunked and held down in a bucket of water; men interrogated with their hands bound to their feet behind their backs; the magneto applied, as it usually was, to the testicles; and the *corvée de bois*, "wood-chopping duty", where a supposedly released prisoner would be felled, as he raced away, with a burst of machinegun fire.

The whole company might be called in to watch the "spectacle", and this in itself revolted him: a gang of 19- to 20-year-old soldiers, the supposed flower of French youth, cracking crude jokes and laughing at the men they abused. Most of them, as he had travelled with them across France, had had no wish to go; "Algeria for Algerians" was the motto they scrawled on the trains. They hated being asked to stifle independence. Once deep in the country, however, hardened by fear, they learned fast to see Algerians of all sorts as the enemy and nothing more.

Of course the French army was fighting a war; one that lasted from 1954 to 1962, and which France was losing, as it had already lost Indochina. Not that it was called a war. This was a "police operation", or "pacification" of France's most precious colony, part of the patrimony, administered as three French departments. The political and military powers in Paris were so keen to keep it that the vilest conduct went undisclosed; not only then, but well into the 21st century. Yet in 1957 Mr Hutin had already published "*Des rappelés témoignent*" (Reservists Bear Witness), and several famous writers had added their voices to his. He had also detailed the abuse in long letters to his family. He freed himself from trauma by telling that story, and he told it for the rest of his life.

From the start he had been an oddity in the army, a Jesuit seminarian with real doubts about whether he should do his compulsory military service. Perhaps he should refuse on principle. But he came from a family of journalists, and when he asked his father, the founder of the newspaper *Ouest-France*, for advice, it was definite: if you go, you can bear witness. So in 1954, at 24, he went to Madagascar, where he got his first taste of French colonialism on a coffee plantation where the land had been seized from peasants and the peasants, in effect, enslaved. He co-wrote a report exposing this, and his path was set.

Called back *sous les drapeaux* in Algeria, he deliberately refused to go on patrol or fight, like his comrades. He was meant to be a machine-gunner, but never fired one; his role was more on the pacification side, re-establishing a school that the *fellaghas* had destroyed. In a windowless building where the only light, summer or freezing winter, came through the open door, he taught around 50 grubby, ragged ten-year-olds the basics of sums and Algerian history. He did not deal much with French history, though a picture he showed them from the Nazi occupation of brave *maquis* about to ambush a German troop lorry reminded him forcefully of the other war he had lived through. His father had been in the Resistance and arrested. If these children's fathers were *fellaghas*, what was the difference between them and him?

His lieutenant, a brute and a sadist, called him a communist. His comrades considered him a priest already, who would obviously be soft on prisoners as he was, taking them bread at night and chatting to them. He fraternised with village men too, not least to persuade them to send their children to school, and squatted with them to drink coffee in their long jobless days. He carried no visible arms, though his lieutenant insisted that a helmeted soldier went with him to school and that he had an automatic pistol in his pocket. Originally he was told to prop a rifle by the blackboard. Who could possibly teach like that?

It was a short stint, yet it marked him so deeply that when he returned the priesthood no longer drew him. He had to make restitution for the crimes he had witnessed. For decades he immersed himself in development projects in Africa, especially in Algeria. In 2004 he immediately joined a new group, The Association of Former Conscripts in Algeria and Their Friends Against the War, a gathering of veterans who gave their war pensions, around 800 euros monthly, to help 24 NGOs operating there.

He had also brought back photographs from the Algerian war. Some were of his pupils, some of prisoners after beatings. One became the face of the conflict, reproduced on the critical histories that eventually followed. It was a portrait of the 14-year-old shepherd, Said Boutout, whose screams he had heard in the night. It was taken the next day, under a blue sky. The boy, with curly hair and a ripped sweater, did not look distressed. He was a tough kid. Rather, he looked puzzled; perplexed, perhaps, by the barbarity of men who called themselves more civilised than he was. ■

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